Agricultural and Forest Entomology
Royal Entomological Society

Research Article

Testing the wind disruption and disturbance refugia hypothesis for mountain pine beetle outbreaks (Dendroctonus ponderosae)

Submission ID	8e11b3ef-8095-4dfa-8794-93a0857cee26
Submission Version	Initial Submission
PDF Generation	12 Oct 2026 14:06:16 EST by Atypon ReX

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Testing the wind disruption and disturbance refugia hypothesis for mountain pine beetle outbreaks (*Dendroctonus ponderosae*)

Terrain, stand density and flight-period wind as controls on attack in the Selkirk Mountains of British Columbia

Running title. Terrain, wind and mountain pine beetle refugia

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Keywords. Mountain pine beetle, Disturbance refugia, Curculionidae, Scolytinae, British Columbia, thinning, dispersal, topography

Data availability statement. Landsat Collection 2 surface reflectance, the British Columbia Vegetation Resources Inventory, the Natural Resources Canada High Resolution Digital Elevation Model (Natural Resources Canada, 2017) and Environment and Climate Change Canada hourly station records are public. The derived data and the analysis code that reproduce every number, table and figure are at *https://github.com/seamusrobertmurphy/beetle-topography-and-wind-study* and will be deposited in the Dryad Digital Repository with the revised manuscript. The 28 field plots of Murphy et al. (2026) were used with the permission of its co-authors, Adrian Leslie, John Wilson and Lauren K. Banks.

Conflict of interest statement. The author declares no conflict of interest. The field plots used to train the classifier were collected under an earlier study the author led (Murphy et al., 2026) and are used with its co-authors' agreement.

Author contributions. Seamus Murphy conceived and designed the study, assembled the data, wrote the code, performed the analysis and wrote the manuscript. The field plots and the Landsat

classifications of red-stage mortality came from Murphy et al. (2026), whose co-authors took no part in the present article.

Acknowledgements. This work used no external funding. The author thanks the United States Geological Survey, the British Columbia Data Catalogue, Natural Resources Canada and Environment and Climate Change Canada for the open archives the study used.

Abstract

1. Disturbance refugia from mountain pine beetle outbreaks have been proposed in thin stands, on shaded ground and where wind disrupts the aggregation pheromone, but few studies have tested these mechanisms together with attack nearby and earlier in the model.
2. This study mapped red-stage attack across 5,573 ha of the Selkirk Mountains, British Columbia, in nine outbreak years and 47 sixteen-day Landsat periods, and modelled it on forest inventory, terrain and wind, entering the previous year's attack first.
3. Attack was clustered in every year, with positive spatial autocorrelation to a median of 2,640 m, and attack in and around a cell the year before dominated every model.
4. Attack then rose with stand basal area (+0.167 log-odds per standard deviation, $p < 0.001$), so thin stands had less attack, as the refugia hypothesis predicted, and ground shaded by terrain had less attack while north-facing ground had more; attack peaked in stands of 25 to 30 cm mean diameter and fell among fewer larger pine.
5. Wind during flight lowered attack where canopy was open and raised it where closed ($p < 0.001$), as plume disruption predicted, and ground facing the afternoon wind had least attack, as deposition predicted, although that effect was small and reversed under a latent spatial field.
6. Refugia on this landscape were thin stands with few host and terrain-shaded ground, wind and shelter added small modifiers to an outbreak spread mostly by contagion, and opening a stand protected it only during flight.

1. Introduction

Mountain pine beetle (*Dendroctonus ponderosae* Hopkins [Coleoptera: Curculionidae: Scolytinae]) is a bark beetle native to the pine forests of western North America, where lodgepole pine (*Pinus contorta* var. *latifolia* Engelm. ex S. Watson) is its main host through most of its range (Cooke et al., 2026; Safranyik & Carroll, 2006). Endemic populations persist in weakened and suppressed trees, and outbreaks occur infrequently followed by sharp population declines (Raffa et al., 2008; Safranyik & Carroll, 2006). After 1976, when the Pacific Decadal Oscillation of Ocean temperatures entered its warm phase, extreme winter cold that kills the brood under the bark became rarer across the province (Macias Fauria & Johnson, 2009), and carried by wind, the outbreak then crossed the northern Rocky Mountains (Giroday et al., 2012).

In north-central Colorado and southern Wyoming, mapped from Landsat images for 1996 to 2011, tree mortality rarely reached 100 per cent even in a major outbreak (Meddens & Hicke, 2014). Places within stands where more trees survive are considered disturbance refugia, which Krawchuk et al. (2020) defined as locations disturbed less severely or less frequently than the landscape around them, noting that “few studies have explicitly identified refugia from insect outbreaks” (p. 235). They proposed that refugia from this beetle may occur “in areas with cooler temperatures (eg from topographic shading) that protect trees from water stress; in areas with lower host density, allowing for greater wind disruption of beetle pheromone communication and more vigorous tree growth and chemical defenses; and in areas with fewer large-diameter host trees” (Krawchuk et al., 2020, p. 239). In southern Oregon, refugia mapped from Landsat imagery during a severe outbreak in 2009 were “associated with topographically shaded slopes, convergent environments such as valleys, areas of relatively low soil bulk density, and in thinner forest stands” (Cartwright, 2018, p. 1).

In experimental stands of lodgepole pine, trees contained a few attacking beetles but died once the bark held more than about 40 beetle galleries per square metre (Raffa & Berryman, 1983), and over three to six years tree defences made no difference once beetle numbers in a stand passed a critical threshold (Boone et al., 2011). Once an outbreak is under way, whether a tree is killed therefore depends more on how many beetles are nearby than on its own defences. Populations become outbreaks once they can kill the average large tree in a stand (Shore et al., 2006), and lodgepole pines larger than 25 cm in diameter, whose thicker phloem feeds more larvae, produce on average more beetles than attack them (Carroll & Safranyik, 2004; Safranyik et al., 2010). When marked beetles were released in a mature lodgepole pine stand, most were caught 3 m above the ground and only 0.2 per cent rose above the canopy (Safranyik et al., 1992). Within a stand beetles fly under the crowns, downwind until they meet the scent of an attacked tree and then upwind towards it (Carroll & Safranyik, 2004; Safranyik et al., 1989), and new clusters of attack appeared 30 to 50 m from those of the year before (Robertson et al., 2007). On the Chilcotin Plateau an outbreak in a 12 km cell was best predicted by outbreaks in the neighbouring cells that year or in the same cell in the two years before (Aukema et al., 2008), and tree by tree in the Cypress Hills of Saskatchewan, infestations nearby in the previous year were the strongest single predictor of attack in every phase of the outbreak (Kunegel-Lion & Lewis, 2020).

Cold is the largest single effect of weather on the beetle's numbers, and where summers are too cool for the brood to develop in one year, as at high elevations, mortality is severe and mass attack less successful (Logan & Powell, 2001; Safranyik & Carroll, 2006). In Colorado and southern Wyoming the outbreak began in several separate places, which suggested drought as a regional trigger (Chapman et al., 2012), and in Oregon and Washington, once beetle numbers nearby were accounted for, winter minimum temperature and drought had the largest effects on whether an outbreak grew large (Preisler et al., 2012). Within a landscape temperature follows

the sun on each slope, and in the Morice Timber Supply Area of north-central British Columbia the most intense infestations were found early on warmer slopes facing south and west (Nelson et al., 2007). Elsewhere terrain entered as elevation, slope and aspect, whose effects followed the phase of the outbreak and the host. In Colorado the best predictors of red attack changed "from forest susceptibility to dispersal to host availability" as the outbreak grew (Walter & Platt, 2013, p. 317), and in western Montana the odds of red stage attack rose with elevation because lodgepole pine grew higher than the Douglas-fir around it (Wulder et al., 2006).

Thinned lodgepole and ponderosa pine stands across the western United States lost fewer trees to the beetle (Amman et al., 1988; Fettig et al., 2007; Mitchell et al., 1983). Mitchell et al. (1983) attributed this to the vigour of the remaining trees, while Bartos & Amman (1989), in the Uinta Mountains of Utah, proposed microclimate, having measured more light, warmth and wind in a thinned stand than in the unthinned stand beside it, a wind difference that doubled to at least 3.2 km/h between 4 and 6 p.m., when most beetles flew. In crowded stands, however, trees grow slowly and have thin phloem, so adding trees can raise attack in an open stand and lower it in a crowded one (Shore et al., 2006).

Wind can lower attack by breaking up the pheromone plume, the trail of scent downwind of an attacked tree, which Krawchuk et al. (2020) called "wind disruption of beetle pheromone communication" (p. 239), most of all in open stands, where more wind reaches the trunks (Bartos & Amman, 1989). Wind can also concentrate attack, because "alterations in wind speed may cause increased settlement in areas where wind speed is reduced" (de la Giroday et al., 2011, p. 1098), and the first beetles to arrive were "unable to distinguish between hosts, dead hosts and nonhosts during landing" (Hynum & Berryman, 1980) and judged a tree only after landing (Safranyik et al., 2010), so ground sheltered from the wind should receive more attack whatever the stand. Both effects act only during flight, which peaks in the early to mid afternoon (Safranyik & Carroll, 2006) and stops in winds above about 2 metres per second (Carroll & Safranyik, 2004),

and studies of flight above the canopy resolved wind over terrain but not within the stand (Ainslie & Jackson, 2010; Jackson et al., 2008).

Few studies had tested these mechanisms together while accounting for attack nearby. This study mapped red-stage attack, trees whose needles had turned red in the year after they were attacked, across part of the Selkirk Mountains of southeastern British Columbia from the 28 field plots of an earlier study on the same ground (Murphy et al., 2026), once a year to test the refugia mechanisms and every sixteen days to test wind. It asked three questions, the first whether stand density, shading by terrain and the scarcity of large trees predicted attack once attack nearby was in the model, the second whether wind during flight lowered attack more in open stands than in closed ones, as disruption of the pheromone plume predicted, and the third whether ground sheltered from the afternoon wind, or facing away from it, had more attack whatever the stand, as deposition predicted.

2. Methods

2.1 Site terrain

The study area covered 5,573 ha on the eastern face of the Selkirk Mountains in southeastern British Columbia, above the Creston valley where the Kootenay River enters the south arm of Kootenay Lake (*Figure 1*), from 830 to 1,744 m in elevation. It was centred on the 480 ha burned by the 2015 Mt Midgeley fire (Murphy et al., 2026) and extended 5 km beyond the burn within the same band of elevation, to take in the range of stand density the pheromone mechanism needed. Along the latitude of its centre the ground fell from 2,072 m to 531 m within 7.6 km (Figure S3). The area held 61,923 cells of 30 m on the grid of Murphy et al. (2026). Attack was mapped once a year for the questions on refugia and every sixteen days for the wind test, the inventory once a year, the station winds every hour and the terrain once overall (Table S1).

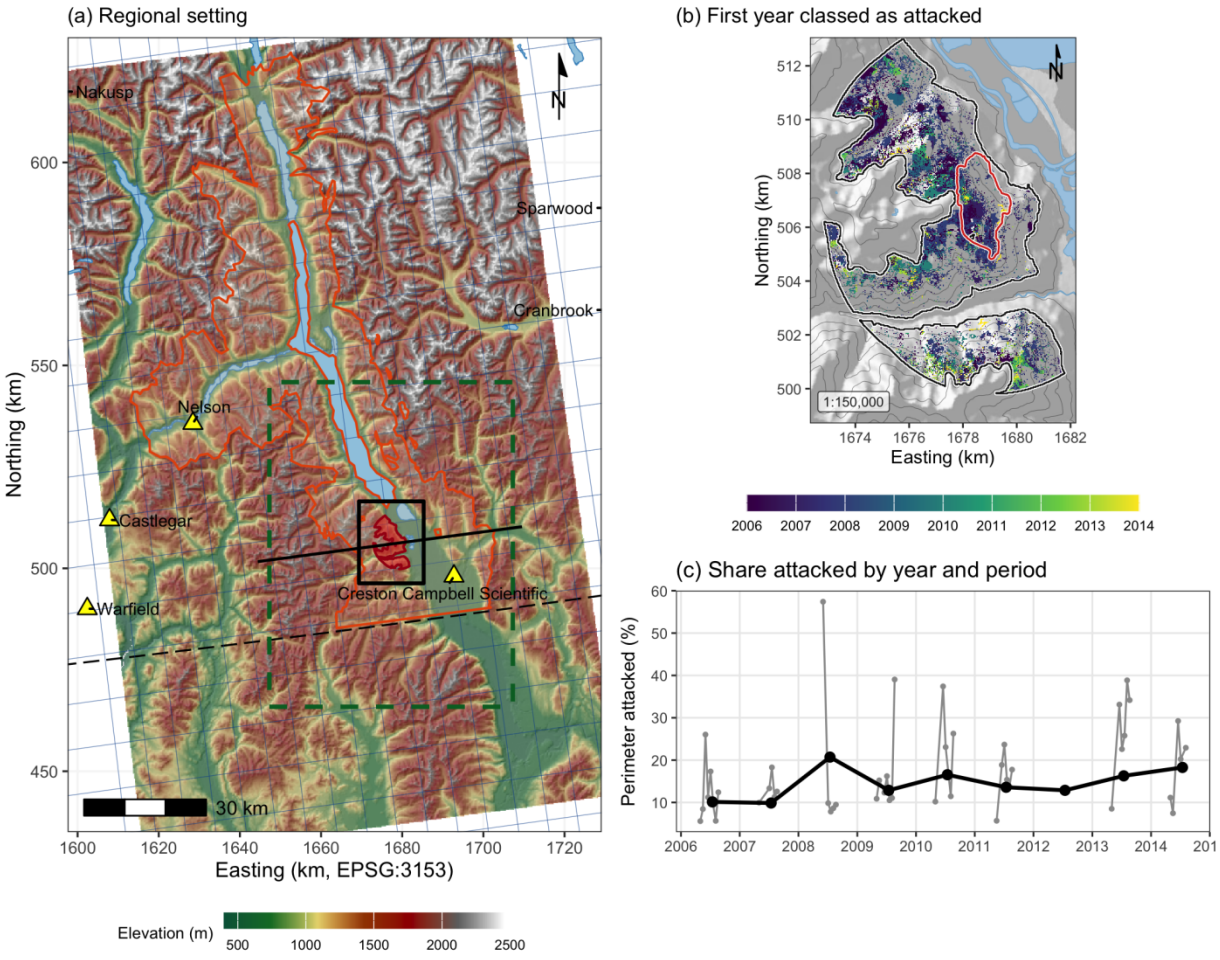


Figure 1: The study area and its red-stage attack. (a) Regional setting over the Copernicus 30 m elevation model. The red fill is the study perimeter, 5,573 ha, the black box the window of the 1 m lidar, the orange outline the 2017 lidar project, the green dashed box the regional wind domain, the blue grid the ERA5-Land reanalysis at 0.1 degree, the yellow triangles the seven hourly stations within 150 km, the black line the profile of Figure S3, and the long dash the Canada and United States border. (b) The first year each cell was attacked, with the 2015 Mt Midgeley burn in red. (c) The share of the perimeter attacked each year, black, and in each sixteen-day period, grey, the periods of each summer joined.

143 Stand structure came from British Columbia's Vegetation Resources Inventory, the snapshot
144 published in each study year, rasterised to the 30 m grid (*Table 1*). The attributes were basal area,
145 crown closure, live stems per hectare, quadratic mean diameter of stems of 12.5 cm and larger,
146 stand age, stand height, standing volume and susceptible pine basal area, the product of basal
147 area and pine cover. The 2007 snapshot omitted basal area and live stems, so the 2006 snapshot

stood in for it. The inventory is a projection rather than a census, and polygons interpreted from photographs taken late in the outbreak described stands the beetle had already attacked.

Terrain was computed in SAGA GIS from an elevation model extending beyond the perimeter, so that search radii near the edge fell on measured ground, and split terrain ruggedness into the properties of the ground the beetle's biology concerns, with ruggedness kept as a candidate. Exposure was measured by the windward-leeward index, effective air flow height, the wind exposition index, topographic openness, sky view, the wind shelter index of Plattner et al. (2004) and geomorphon class, which assigns each cell a landform such as ridge, slope or valley. Landform was measured by wetness, valley depth and height above the valley floor, because groups of infested trees are "frequently associated with draws and gullies" (Safranyik & Carroll, 2006, p. 41). Aspect entered as northness, eastness and a heat load index. Radiation was computed for the flight window, 1 July to 15 August from 12:00 to 17:00, and for the growing season, 1 May to 30 September, over which shading can protect trees from water stress (Krawchuk et al., 2020). Table S3 lists the variables that entered the models with the mechanism each represents.

2.2 Beetle outbreak

The response for the first and third questions was red-stage attack in each outbreak year from 2006 to 2014, measured, following Murphy et al. (2026), as the fall in the normalised difference moisture index (NDMI) from its value in 2005, when the outbreak began. NDMI tracks the moisture of the canopy and is used to detect forest disturbance in Landsat time series (Jin & Sader, 2005), and Murphy et al. (2026) validated their map of the burn site against 28 field plots of 20 by 20 m, in which beetle-killed pine was confirmed from pitch tubes, frass and the pattern of the egg galleries.

Each year was mapped from the median of cloud-masked Collection 2 Level-2 Landsat 5 and 8 scenes of 1 June to 31 August, differenced against 2005 and resampled by nearest neighbour onto the 30 m grid of Murphy et al. (2026), with Landsat 8 converted to the Landsat 5 scale by the coefficients of Roy et al. (2016) and matched to the Landsat 5 range over undisturbed forest. With lakes and rivers masked, attack covered from 9.9 to 20.7 per cent of the perimeter by year (Table S2, *Figure 1*).

No Landsat 5 or Landsat 8 images were taken in the summer of 2012, so its map was drawn from the median of the summer's Landsat 7 scenes, which filled most of the strips that the failed scan-line corrector left in each one (Wulder et al., 2011). The index was put on the Landsat 5 scale by a line fitted on stable forest in 2010, when both satellites flew.

The response for the second question was red-stage attack in each sixteen-day period, the Landsat revisit interval, from 1 May to 22 September of the same years, with NDMI as the median of the period's scenes differenced against the same period of 2005, so that the seasonal course of leaf moisture did not enter the difference. Periods in which the imagery saw less than a tenth of the perimeter were dropped, leaving 47 periods over eight years (Table S5).

The fall in NDMI that counted as attack was set by a classifier trained on the four pixels nearest the centre of each field plot, 112 pixels, against the 84 points of undisturbed forest that Murphy et al. (2026) digitised on the 2020 Landsat 8 scene, which fell in 68 distinct cells, each pixel entering with its deepest annual fall in NDMI against 2005. Random forest, a radial support vector machine and gradient boosting were scored over 100 splits, each holding out a quarter of the field plots and of the 100 m blocks of undisturbed pixels. The radial support vector machine was the most accurate, with kappa 0.859 ± 0.081 and overall accuracy 0.934 (Table S10; Table S6), and its cut, a fall in NDMI of 0.0616 against 2005, was applied to the annual and sixteen-day differences. The undisturbed pixels came from one patch of about 230 by 455 m, so the cross-validation

measured how well the plots were separated from that patch rather than from undisturbed forest across the landscape.

2.3 Wind and lift

Wind was summarised over the afternoon flight hours, 12:00 to 17:00, the flight period Safranyik & Carroll (2006) identified for this region (p. 18), which overlaps the emergence peak of 11:00 to 14:00 that Gray et al. (1972) recorded in eastern Washington. Across 236,079 hourly station records from May to September, 89.5 per cent of afternoon hours between 1 July and 15 August fell within the 19 to 41 degrees C in which beetles fly (McCambridge, 1971), against 51.4 per cent in the rest of the season (Figure S18).

Hourly speed and direction from Environment and Climate Change Canada stations within 150 km were combined as vector components, adjusted for terrain with the MicroMet model of Liston & Elder (2006), which weights each observation by the slope in the wind direction and the curvature of the ground, and interpolated to the grid by inverse distance weighting. Each sixteen-day period was summarised by its mean flight-hour wind, which ran from 4.0 to 8.6 km/h between periods, and each year by its mean wind of June and of July.

Wind over the terrain was modelled with WindNinja 4.0.0 (Forthofer et al., 2014), driven by the hourly ERA5-Land reanalysis (Muñoz-Sabater et al., 2021) for every flight hour of June to August 2005 to 2014 with its slope winds on, at 90 m over the study area, at 30 m and 10 m for each flight window, and at 500 m and 1 km over the surrounding valleys for checking against stations. Each cell received the mean speed, corrected by the model's loss of speed on flat ground, a factor of 1.63 at 90 m, the share of hours in which the air moved upslope, and the slope updraft. At the valley stations the modelled direction erred by about 60 degrees (Table S10), so these terms were read as each slope's exposure to the regional flow, and the valley wind was taken from 28 provincial stations (Figure S4).

Convective lift from the reanalysis heat flux, back-trajectories of the air over the site, the sun on each slope under each hour's cloud, the modelled afternoon air temperature, the Landsat land surface temperature of the flight window and the roughness, north-facing share and spread of position of the 1 m lidar terrain within each cell entered as further terms, described in Methods S2.

These terms were read at the scales the beetle's flight allows. Beetles that rise above the canopy travel on the wind, an estimated 30 to 110 km in a day (Jackson et al., 2008), so the terrain wind, the lift and the trajectories described which slopes the above-canopy stream reached. Within a stand a surrogate pheromone fell to about a tenth of its concentration within 10 m (Brush & Lewis, 2024; Thistle et al., 2004), an effect that entered the models only through attack in and around the cell. The wind terms were therefore read as predictors of where beetles arrived and not of where they settled, and each entered the models from the same period or flight window of the previous summer, one term to a specification (Table S3).

2.4 Model design

Before any model was fitted, the spatial pattern of attack was described on each annual map, with attack present or absent in each cell. The join count (Moran, 1948) counted pairs of neighbouring attacked cells among the eight around each cell, and Moran's I (Moran, 1950) was computed in distance classes of 60 m out to 6 km as a correlogram (Legendre & Fortin, 1989), both against 999 random permutations of the attacked cells among the cells mapped that year. The range of positive autocorrelation was the first distance class at which Moran's I no longer exceeded the permuted values at $p \leq 0.05$.

The annual models were built in the order Aukema et al. (2008) used, in which they first "determined an appropriate spatial neighborhood structure(s) and time lag(s) to account for spatial and temporal dependencies" (p. 351) and only then entered environmental variables. The first

243 dependence term was the cell's own state in the previous outbreak year, and the second the
244 share of cells attacked around it in that year within 42, 90, 150, 210, 510 or 1,050 m, excluding
245 the cell itself, with the radius chosen on AIC (Table S7). Each year contributed up to 4,000
246 attacked and 4,000 unattacked cells, because at a landscape prevalence near one cell in ten an
247 unbalanced fit would report its intercept rather than its covariates.

248 The 52 environmental variables were grouped by mechanism (*Table 1*) and entered four logistic
249 regressions in sequence, each adding one mechanism to the one before, and each was scored
250 on its full fit, under ten-fold cross-validation on 1 km blocks and on a blind test that fitted the years
251 to 2012 and predicted 2013 and 2014 (Table S9). Every model included the geomorphon class of
252 the cell, and every continuous term was standardised, so that each coefficient was a change in
253 log-odds per standard deviation. The sequence was fitted on all nine years without the
254 dependence terms and on the eight years with a previous map without and with them, so that the
255 change in each environmental term once attack nearby entered could be read directly. Before the
256 final model was read, each candidate was screened on the years with a previous map by a
257 likelihood ratio test of the variable added alone to the two dependence terms, by its variance
258 inflation against the main effects of M3, and by its coefficient in an elastic net (Zou & Hastie, 2005)
259 that entered every candidate together with the dependence terms unpenalised, and its distribution
260 was tested by the Shapiro-Wilk test on 5,000 cell-years (*Table 1*).

261 Each annual model took the form

262 $$\operatorname{logit}\Pr(y_{it}=1)=\alpha+\mathbf{x}_{it}^{\top}\boldsymbol{\beta}+\gamma_{g(i)}+\sum_{k}\delta_{k}\,z_{k,it}+\rho_{1}\,y_{i,t-1}+\rho_{2}\,\bar{y}_{i,t-1}\qquad(1)$$

263 where y_{it} was attack in cell i in year t, $\mathbf{x}_{it}$ the standardised environmental terms of the model with
264 coefficients $\boldsymbol{\beta}$, $\gamma_{g(i)}$ the effect of the geomorphon class of the cell, $z_{k,it}$ the interactions of basal
265 area with the wind shelter index, the July wind and the flight-window radiation that entered in M3,

$y_{i,t-1}$ the cell's own attack in the previous outbreak year and $\bar{y}_{i,t-1}$ the share of cells attacked within 150 m of it in that year. The last two terms were the dependence terms, present only in the fits over the eight years with a previous map.

The first and third questions were answered with these annual models, the third by comparing the main effect of the wind shelter index with its interaction with basal area, in M3 with and without flight-window radiation. Each interaction was read as the slope of one term at one standard deviation either side of the mean of the other. Attack was also tabulated by quadratic mean diameter class, with Wilson intervals, a chi-square test across classes and a two-proportion test across 25 cm, the diameter Carroll & Safranyik (2004) gave as the boundary between beetle sinks and sources.

The second question was tested on the sixteen-day maps, on which wind varied between periods within a season as well as between years. Attack in a period was regressed on 15 stand and terrain variables, mean flight-hour wind, the geomorphon class and the interactions of wind with stand structure (E1), and E2 added the cell's own state in the previous period and the share attacked within 90 m around it, chosen on AIC, with each period contributing up to 2,000 cells of each class. Because the red crowns of a period recorded the flight of the previous summer, every wind test was also run with the wind of the same period one year earlier.

The sixteen-day models took the form

$$\operatorname{logit}\Pr\left(y_{ip}=1\right)=\alpha+\mathbf{x}_{ip}^{\top}\boldsymbol{\beta}+\gamma_{g(i)}+\theta\,w_{ip}+\sum_{k}\delta_{k}\;s_{k,ip}\,w_{ip}+\rho_{1}\,y_{i,p-1}+\rho_{2}\,\bar{y}_{i,p-1}\qquad(2)$$

where y_{ip} was attack in cell i in period p, w_{ip} the flight-hour wind of the period, from the stations or from the terrain wind model, or the wind of the same period of the previous summer, $s_{k,ip}$ the stand terms it interacted with, and the last two terms the dependence terms of E2, which E1 lacked.

Residual autocorrelation was measured as Moran's I of the deviance residuals within each year. M3 with the dependence terms was then refitted with a Gaussian process smooth of easting and northing (Wood, 2017), its range set to the median range of positive autocorrelation, 2,640 m, and on cells coarsened to 90, 270 and 990 m with the dependence terms rebuilt at each grain, a coarse cell counting as attacked if any 30 m cell inside it was, as Aukema et al. (2008) defined their 12 km cells.

The refit under the spatial field took the form

$$\operatorname{logit}\Pr(y_{it}=1)=\eta_{it}+f(e_i,n_i) \qquad (3)$$

where η_{it} was the linear predictor of *Equation 1* with the dependence terms and f a Gaussian process smooth of the easting e_i and northing n_i of the cell, with a Matérn covariance at that range and 200 basis functions.

Further terms were tested beside M3 and E2 and are described in the Supporting Information. These were valley and ridge flatness, canopy height from lidar, winter minimum temperature, station pressure, and attack upwind and downwind of each cell weighted by winter survival.

3. Results

Across the 5,573 ha perimeter and the nine outbreak years from 2006 to 2014, attack was clustered in every year, with 3.2 to 5.8 times as many neighbouring pairs of attacked cells as random permutation produced and positive autocorrelation to a median of 2,640 m (Figure S17). Attack in the cell and within 150 m the year before dominated every model it entered. A cell attacked the year before had 9.1 times the odds of attack, and the two terms raised the area under the receiver operating characteristic curve (AUC) of M3 from 0.814 to 0.890. Of the 31 candidate variables, 28 improved the fit by likelihood ratio test once the dependence terms were in the model, the elastic net kept 16, and the largest variance inflation among the main effects of M3

312 was 2.9 (*Table 1*; Table S13). Each estimate below is a change in log-odds per standard deviation
313 with the dependence terms in the model unless stated otherwise.

*Table 1: Descriptive statistics and screening of red-stage attack and the variables of M3 over the 72,000 sampled cell-years before standardisation. Mean, standard deviation (SD) and skewness over the sample, Shapiro-Wilk W on 5,000 cell-years, and the screening on the years with a previous map, β the slope of the variable added alone to the two dependence terms, the deviance it explained and the p-value of the likelihood ratio test. The landform class was screened as a whole by the same likelihood ratio test. Table S3 defines each variable. * p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001, **** p ≤ 0.0001.*

Covariate	Mean	SD	Skew	W	β	Deviance	LR-test
Red-stage attack (%), 2006 to 2014	14.6	3.65	+0.30				
Susceptible pine basal area (m^2 ha^{-1})	8.47	10.3	+1.31	0.804	+0.3329	886.97	0.001****
Stand age (years)	113	25.4	+1.30	0.893	+0.0923	85.93	0.001****
Quadratic mean diameter (cm)	26.5	6.11	+1.10	0.934	-0.0141	2.19	0.139
Sky view factor	88.3	4.86	-0.47	0.984	+0.4119	1,362.54	0.001****
Northness	0.214	0.685	-0.45	0.886	+0.0435	18.26	0.001****
Elevation (m)	1,400	276	-0.54	0.906	+0.7528	3,987.76	0.001****
Stand basal area (m^2 ha^{-1})	35.3	12.4	-0.78	0.915	+0.1961	371.10	0.001****
Wind shelter index	21	29.8	+1.70	0.597	+0.0650	43.04	0.001****
Topographic position index	3.12	16.1	-0.51	0.977	+0.1393	201.48	0.001****
Convergence index	0.538	8.81	-0.07	0.974	+0.0668	41.60	0.001****
Profile curvature	0.0000622	0.00227	-0.06	0.992	+0.0457	21.34	0.001****
Flight-window direct radiation (kWh/m^2)	140	37.6	-0.15	0.993	+0.2147	469.53	0.001****
July mean wind (km/h)	7.17	0.576	+0.25	0.916	+0.2349	501.58	0.001****
June mean wind (km/h)	7.18	0.508	-0.28	0.948	+0.0526	17.97	0.001****
Landform class, seven classes against slope						139.51	0.001****

314 ## 3.1 Refugia mechanisms

315 The first question asked whether stand density, shading by terrain and the scarcity of large trees
316 predicted attack once attack nearby was in the model. The refugia hypothesis predicted less
317 attack in thin stands, on ground shaded by terrain and in stands without large trees. Density and
318 shade moved as predicted and host size did not. Attack rose with stand basal area, +0.167 ($p <$
319 0.001), so the thinnest stands had the least attack. The term was +0.276 without the dependence
320 terms ($p < 0.001$), +0.200 under the spatial field ($p < 0.001$) and +0.094 at a 270 m grain ($p =$

0.026), and basal area lowered AIC by 478 when it entered, more than any other step of the sequence (*Table 2*; *Figure 2*; Table S9). Crown closure gave the same direction in the sixteen-day model, +0.129 ($p < 0.001$), while live stems per hectare, which counted small trees as well as large, ran the other way, -0.043 ($p < 0.001$). The sun of the flight window raised attack in dense stands and not in thin ones, by +0.133 ($p < 0.001$) one standard deviation above the mean basal area against +0.037 ($p = 0.070$) one below it.

Ground shaded by terrain had less attack. Sky view, the share of the sky a cell saw, entered at +0.170 ($p < 0.001$) and the direct sun of the flight window at +0.085 ($p < 0.001$), so attack rose with open sky and with sun, as the hypothesis predicted. Northness ran against it. Ground facing north had more attack, +0.073 ($p < 0.001$), and the term grew under the spatial field to +0.133 ($p < 0.001$), while sky view fell under the field to +0.037 ($p = 0.285$) and was -0.003 at a 270 m grain ($p = 0.945$). Shade by terrain therefore protected ground at 30 m once attack nearby was in the model, and the protection did not survive the field, whereas the warmth of a south face protected nothing at any grain.

The hypothesis predicted less attack where large trees were few, so attack rising with tree size. Attack instead fell with quadratic mean diameter, -0.136 ($p < 0.001$), and -0.069 under the field ($p < 0.001$). By diameter class, attack rose from 48.1 per cent of the balanced sample at 20 to 25 cm to 56.4 per cent at 25 to 30 cm and fell to 46.6 per cent at 30 to 40 cm and 35.4 per cent above 40 cm [$\chi^2(5) = 980.1$, $p < 0.001$, Cramér's $V = 0.117$], a step of 8.2 percentage points across the 25 cm source threshold (95 per cent confidence interval 7.3 to 9.1, $p < 0.001$). The stands of the largest trees held the least pine, 0.4 $m^2\ ha^{-1}$ of susceptible pine basal area against 10.1 at 25 to 30 cm, and susceptible pine basal area itself entered at +0.090 ($p < 0.001$). Host size therefore acted as a threshold at 25 cm and not as the gradient the hypothesis described. Elevation had the largest coefficient of any environmental term, +0.636 ($p < 0.001$).

*Table 2: Regression estimates of red-stage attack by model, in log-odds per standard deviation of each continuous term, with the landform classes in every model as log-odds against slope, the most common class, ridge including the shoulder cells and a class of fewer than 30 cells not estimated. M0 held host size, shading and landform, M1 added stand basal area, M2 terrain exposure, shape and flight-window radiation, and M3 the interactions, over the nine years; M3 was refitted with the dependence terms over the eight years with a previous map, and with a latent spatial field. Performance is the AUC and RMSE of the full fit, the RMSE ratio of the test folds over the training folds under ten-fold cross-validation on 1 km blocks, and Theil's U bias. * p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001.*

Term	M0	M1	M2	M3
Susceptible pine basal area	+0.272***	+0.195***	+0.191***	+0.186***
Stand age	+0.118***	+0.101***	+0.101***	+0.102***
Quadratic mean diameter	-0.097***	-0.249***	-0.250***	-0.250***
Sky view	+0.420***	+0.381***	+0.387***	+0.369***
Northness	+0.128***	+0.077***	+0.100***	+0.105***
Elevation	+0.934***	+0.945***	+0.941***	+0.940***
Stand basal area		+0.279***	+0.273***	+0.276***
Wind shelter index			-0.069***	-0.073***
Topographic position			-0.022	-0.011
Convergence			-0.049***	-0.049***
Profile curvature			-0.074***	-0.071***
Flight-window radiation			+0.062***	+0.078***
July wind				+0.064***
June wind				+0.049***
Attack in the cell, previous year				
Attack within 150 m, previous year				
Basal area × shelter				-0.030**
Basal area × July wind				+0.004
Basal area × radiation				+0.066***
Landform class, against slope				
Slope, 38,898 cells, 14 per cent attacked	reference	reference	reference	reference
Spur, 11,796 cells, 18 per cent attacked	-0.107***	-0.059**	-0.012	-0.011
Hollow, 8,751 cells, 12 per cent attacked	+0.071*	+0.036	-0.024	-0.028
Valley, 1,609 cells, 11 per cent attacked	+0.013	-0.069	-0.200**	-0.210**
Ridge, 665 cells, 24 per cent attacked	-0.626***	-0.528***	-0.453***	-0.447***
Peak, 32 cells, 42 per cent attacked	-0.475	-0.368	-0.284	-0.268
Footslope, 1 cell, 89 per cent attacked	not estimated	not estimated	not estimated	not estimated
AUC	0.804	0.807	0.808	0.809
RMSE	0.423	0.422	0.421	0.421
RMSE ratio	1.003	1.000	1.005	1.008
Theil's U bias	0.0000	0.0001	0.0001	0.0001

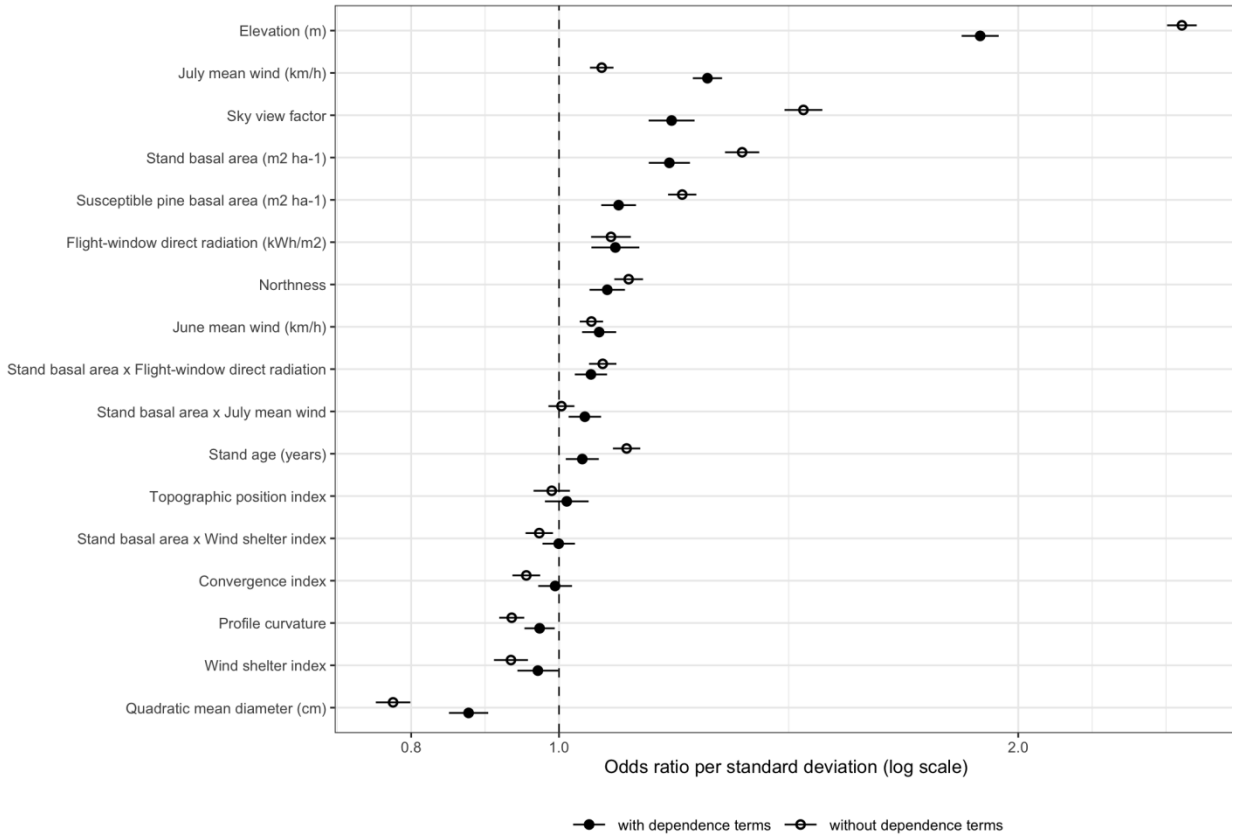


Figure 2: Odds ratios of the environmental terms of M3 per standard deviation, with 95 per cent confidence intervals, without the dependence terms over the nine years, open circles, and with them over the eight years with a previous map, filled circles. Terms are ordered by their odds ratio with the dependence terms, and the dashed line marks no effect.

3.2 Wind and stand density

The second question asked whether wind during flight lowered attack more in open stands than in closed ones, as disruption of the pheromone plume predicted, so that the wind term should be negative in open stands and weaker or positive in closed ones. It was, with the terrain wind of the previous summer in the sixteen-day model. The speed itself was -0.014 at the mean crown closure ($p = 0.271$), its interaction with crown closure was +0.071 ($p < 0.001$), and the wind effect was -0.085 ($p < 0.001$) in stands one standard deviation below the mean crown closure and +0.056 ($p = 0.003$) one above it (Figure S16; Table S4). The wind at 10 m gave the same pattern, an interaction of +0.070 ($p < 0.001$) with a wind effect of -0.030 ($p = 0.073$) in open stands and +0.110

(p < 0.001) in closed ones. Wind also lowered attack in stands of small trees and not in stands of large ones, -0.055 (p = 0.002) one standard deviation below the mean diameter against +0.026 (p = 0.125) above it.

The station wind interpolated from the valleys did not show the effect. The wind of the same period raised attack, +0.116 (p < 0.001), as did the previous summer's, +0.088 (p < 0.001), and its interaction with crown closure could not be distinguished from zero (p = 0.963). In the annual models the July station wind raised attack, +0.224 (p < 0.001), by +0.185 (p < 0.001) in stands one standard deviation below the mean basal area and +0.263 (p < 0.001) one above it, the same order as the sixteen-day result but with no fall in attack at any density.

3.3 Shelter and deposition

The third question asked whether ground sheltered from the afternoon wind, or facing away from it, had more attack whatever the stand, as deposition predicted. The wind shelter index was higher on slopes facing the prevailing wind, so the prediction was a negative coefficient and no interaction with basal area. Both held, and the first was weak. The index entered at -0.032 (p = 0.043), from -0.073 without the dependence terms (p < 0.001), and its interaction with basal area could not be distinguished from zero (p = 0.971), so the effect did not depend on the stand. It was -0.025 at a 90 m grain (p = 0.175) and changed sign under the spatial field, to +0.096 (p < 0.001), with residual Moran's I still at a median of 0.345 under the field. The terrain wind gave the same direction more strongly. The shelter index at each cell's own modelled wind direction entered at -0.051 (p < 0.001) in the annual model and -0.064 (p < 0.001) in the sixteen-day model, the share of flight hours in which the wind blew upslope at -0.057 (p < 0.001) over sixteen days and -0.003 (p = 0.804) annually, and the slope updraft, the wind pushed up the face, at -0.103 (p < 0.001) over sixteen days and -0.029 (p = 0.050) annually, falling most steeply on ridges and spurs, -0.029 (p < 0.001). The faces that met the south-west afternoon wind had the least attack, 10 per cent of cells facing south and 12 per cent facing south-west against 23 per cent facing west and

379 19 per cent facing north-west (*Figure 3*). Ridges and valleys had less attack than slopes without
380 the dependence terms, -0.447 ($p < 0.001$) and -0.210 ($p = 0.003$), and neither differed from slopes
381 once attack nearby was in the model ($p = 0.697$ and $p = 0.978$; *Table 2*).

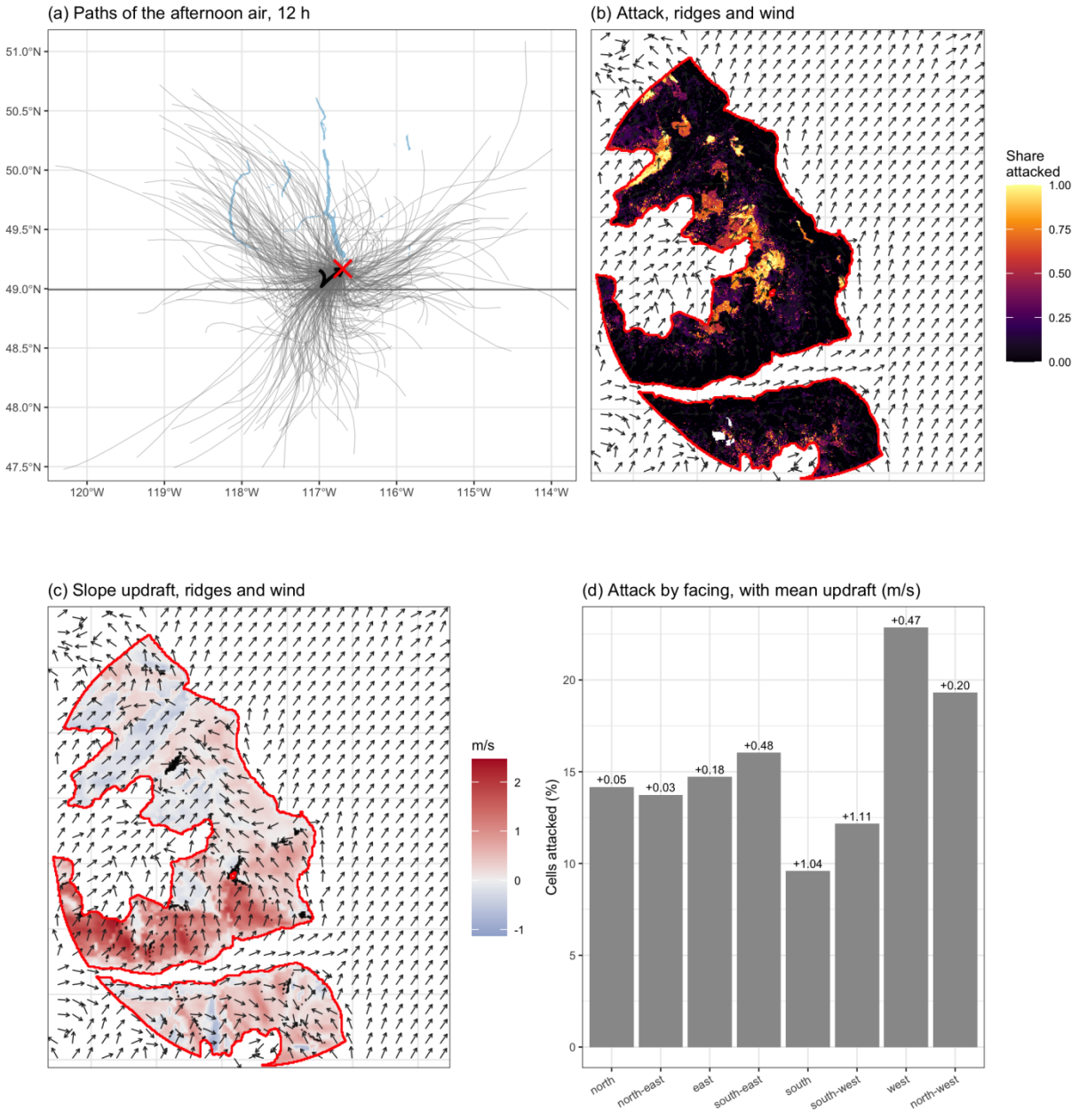


Figure 3: The afternoon air over the site and the attack it met, 1 July to 15 August. (a) HYSPLIT paths of the air over the 12 h before 14:00 on each flight day of 2005 to 2013, with the mean path in black. (b) The share of the annual maps of 2006 to 2014 on which each cell was attacked, with ridges in black

and the modelled flight-hour wind as arrows. (c) The slope updraft of that wind, red where it rose and blue where it sank. (d) The share of cells attacked by the direction the ground faced, with the mean updraft above each bar. The red outline is the study perimeter.

The remaining terms, surface temperature, the lidar terrain, convective lift, the back-trajectories, pressure, valley and ridge flatness, cold-air pooling, canopy height, winter cold, the upwind sources and the maps of 2012, were fitted beside M3 and E2 and none entered the final models (Tables S11 and S12).

4. Discussion

Attack in a 30 m cell was predicted first by attack in and around it the year before, the order Aukema et al. (2008) followed on the Chilcotin Plateau, where temperature still "contributed to explaining outbreak probabilities" once neighbouring and earlier outbreaks were in the model (p. 348). The best neighbourhood here was 150 m, close to the 140 m within which previous attack raised red attack above its background in Colorado (Walter & Platt, 2013), and the outbreak was in the phase in which "almost all new mortality was associated with intensification of existing outbreaks, not expansion" (Meddens & Hicke, 2014, p. 83). The three refugia mechanisms that Krawchuk et al. (2020) proposed were therefore tested as modifiers of an outbreak that spread mostly by contagion, and the novelty of the study lay in testing the three together on one landscape at 30 m with that dependence in the model, which few earlier studies had done. Density and shade by terrain moved as the hypothesis predicted and host size did not, wind lowered attack only in open stands and only over sixteen-day periods, and ground facing the afternoon wind had the least attack.

4.1 Refugia mechanisms

Beside contagion, attack rose with the amount of host, the basis of the stand susceptibility rating used in British Columbia, which is built on "the percentage of susceptible pine by basal area", pine age and stand density (Safranyik et al., 2010, p. 429; Shore et al., 2000), so the refugia this

landscape showed were thin stands with little pine, as in thinning trials in which stands reduced in basal area lost fewer trees (Fettig et al., 2007; Hood et al., 2016; Mitchell et al., 1983). The amount of host is itself a product of forest history, because within climatically suitable areas “forest age-class structure will be the primary factor influencing host susceptibility and outbreak severity” (Taylor & Carroll, 2003, p. 42). Live stems ran the other way because a stand of many stems was a crowded stand, in which trees grow slowly and have thin phloem (Shore et al., 2006), so the count of stems measured the size of the trees more than the amount of host.

Shade by terrain protected ground once attack nearby was in the model, through the sky view and the sun of the flight window, which was the mechanism Krawchuk et al. (2020) proposed, cooler ground that protects trees from water stress, although the effect was lost under the spatial field and at a 270 m grain, so it was a property of the slope at the scale of a stand and not of the landscape. The faces that were cooler by aspect did not share it. North-facing ground had more attack, the reverse of the Morice Timber Supply Area, where the most intense infestations were found early on warmer slopes facing south and west (Nelson et al., 2007), although its surface was 3.4 °C cooler in the late morning. Two observations of the beetle’s flight point to heat during flight rather than water stress. The heaviest attacks on a bole are “usually found on the northern aspect” (Safranyik & Carroll, 2006, p. 18), and in the Black Hills “The decreased number of attacks during midday coincides with the period of highest mean temperatures” (Schmid et al., 1991, p. 1445), so the sunlit afternoon faces may have been too hot for landing beetles while the ground shaded by terrain was too cool for the brood, where summers too cool for development in one year make mass attack less successful (Logan & Powell, 2001; Safranyik & Carroll, 2006).

Attack peaked in stands of 25 to 30 cm quadratic mean diameter, just above the 25 cm at which lodgepole pines change on average from beetle sinks to beetle sources (Carroll & Safranyik, 2004), because thicker phloem feeds more brood, from an average of 16 beetles to 94 as phloem thickened from 2.3 to 5.8 mm (Amman, 1972). Attack fell in stands of larger trees because those

stands held little pine, not because the beetle avoided large trees (Shore et al., 2006), and epidemics also “often deplete the large diameter pine component of stands” (Shore et al., 2006, p. 110). The scarcity of large trees therefore made a refugium only below 25 cm, where the trees could not feed a brood, and above 30 cm the refugium was the absence of pine. Elevation had the largest coefficient in every model, against the historical decline of mortality with elevation (Shore et al., 2006), but on this range the higher ground also held the pine, as in western Montana (Wulder et al., 2006), and milder winters after 1976 had favoured outbreaks at height (Macias Fauria & Johnson, 2009).

4.2 Wind and stand density

Wind lowered attack where the canopy was open and raised it where the canopy was closed, so the stand property through which wind acted was crown closure. That was the mechanism Bartos & Amman (1989) proposed in the Uinta Mountains of Utah, where a thinned stand was windier than the unthinned stand beside it by 3.2 km/h or more in the late afternoon and, in a recently thinned stand, “canopy density usually is insufficient to trap the pheromone and move it horizontally to attract other beetles” (p. 9), and a tracer gas released in place of pheromone was diluted fastest in the most open of three canopies (Thistle et al., 2004). The effect appeared with the terrain wind over sixteen-day periods and not with the station wind interpolated from the valleys, which varied between periods and years but hardly between cells, so the test needed a wind that differed from one slope to the next within a season. In the annual models wind raised attack everywhere, more so where basal area was greater. Wind above the canopy carries beetles into new stands (Chen & Jackson, 2017; Jackson et al., 2008), which predicts more attack with wind where large hosts stand under a closed canopy, and under epidemic pressure pheromones from attacked trees nearby “may override the positive benefits of increased spacing and improved tree growth derived from partial cutting” (Schmid & Mata, 2005, p. 9). For

453 management the two readings agree, since a stand opened by thinning was protected by the wind
454 only during flight and least where attacked trees stood close.

455 ## 4.3 Shelter and deposition

456 The afternoon air reached the site from the south-west, from 193 to 256 degrees, and the least
457 attack fell on the faces that met it, 10 per cent of cells on south-facing and 12 per cent on south-
458 west-facing ground against 23 per cent on west-facing and 19 per cent on north-west-facing
459 ground. This was the reverse of most studies of aspect, in which attack favoured southern slopes
460 (Nelson et al., 2007; Walter & Platt, 2013), and of the Peace River region, where beetles “tend to
461 accumulate on windward sides of barriers” (de la Giroday et al., 2011, p. 1107). Beetles released
462 by Safranyik et al. (1989) “climbed towards the tree tops, and usually tracked to the north or
463 northeast, generally crosswind and downwind” (p. 508), and “bark beetles do not fly in winds that
464 exceed their maximum flight speed” (Carroll & Safranyik, 2004, pp. 22–23). These observations
465 suggest that the afternoon wind lifted beetles up the south and south-west faces, carried them
466 over the ridges and set them down on the north and west faces beyond, where the wind
467 slackened. That would explain the least attack on the windward faces, the fall in attack where the
468 wind was pushed up the slope, steepest on ridges and spurs, and a shelter effect that did not
469 depend on the stand, as deposition predicted and plume disruption did not, and it matches the
470 reading Kunegel-Lion & Lewis (2020) gave of a north-facing onset in the Cypress Hills, where
471 “the dominant winds in Cypress Hills come from the south-west” (p. 9). Deposition also depends
472 on the stand, since in simulations of beetles flying through forest, dispersal downwind shortened
473 as trunks became denser (Byers, 2000). The shelter index taken at one fixed bearing was small,
474 was lost at a 90 m grain and changed sign under the spatial field, so it described where attack
475 clustered as much as a property of the slope, and a refugium defined at 30 m and one defined at
476 1 km were different things, the concern for scale that Aukema et al. (2008) raised. No study had

compared beetles landing on the windward and lee faces of the same ridge, and the test of this hypothesis would be a record of wind and of landing beetles on the slopes themselves.

4.4 Limits and management

The response was classified rather than observed, and the classifier's accuracy measured its separation of the field plots from undisturbed forest rather than agreement with ground mortality across the landscape (Meddens et al., 2013). The inventory postdated part of the outbreak, the terrain wind took its direction from the reanalysis and gave about half the speed measured on the valley floor, and the study covered one mountain range over nine outbreak years. For management, the refugia this landscape showed were stands with little host, the structure that thinning and a younger age structure produce, which terrain maps alone could not locate. Partial cutting substantially lowers the susceptibility of lodgepole pine stands (Mata et al., 2003; Shore et al., 2000), but in ponderosa pine it "may be ineffective for partially cut parcels of <10 acres if the partially cut stands are surrounded by unmanaged susceptible stands" (Schmid & Mata, 2005), which the dominance of attack nearby supported, and Six et al. (2014) found the field evidence that thinning protects stands during an outbreak weaker than policy had assumed.

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Figures

Figure 1

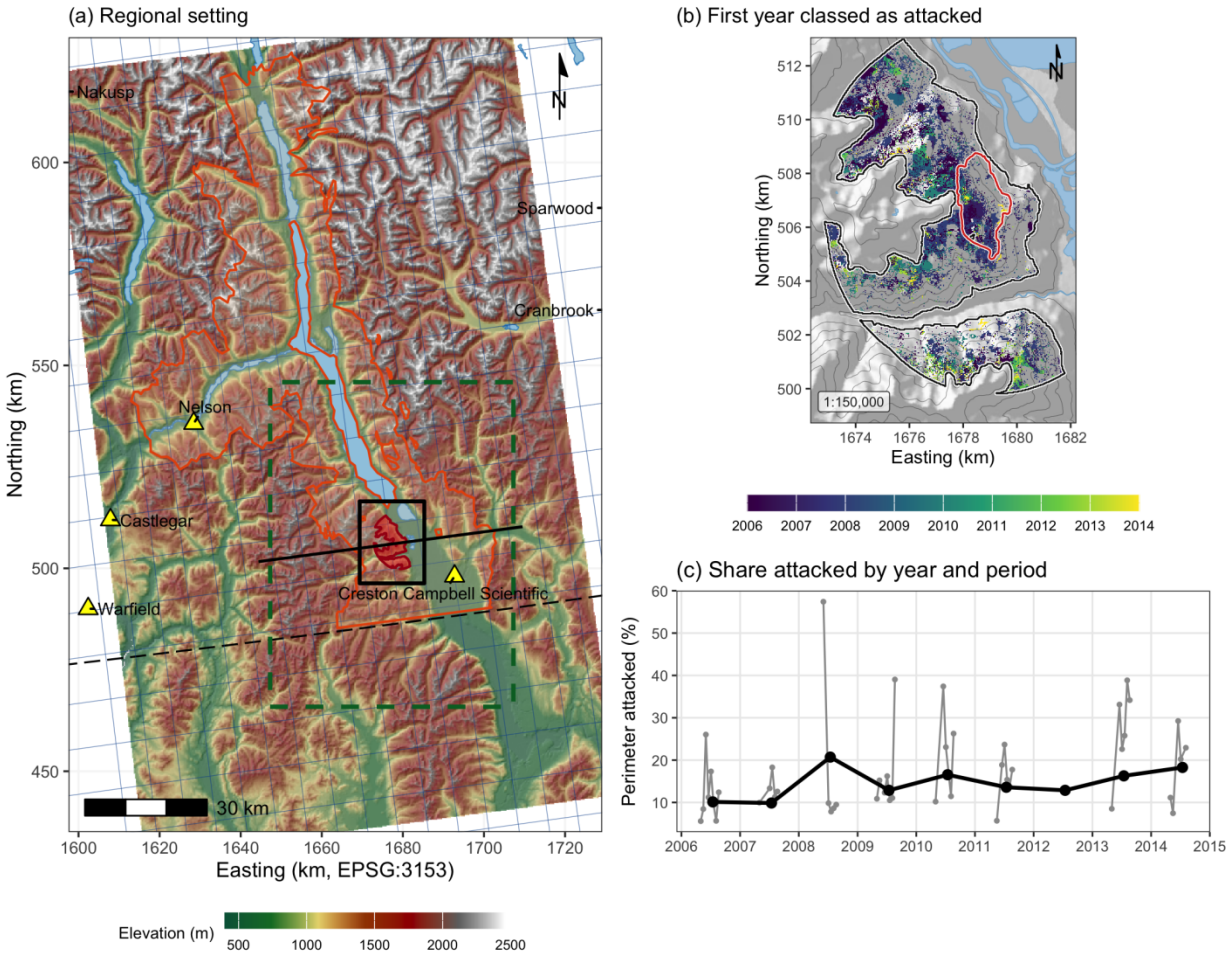
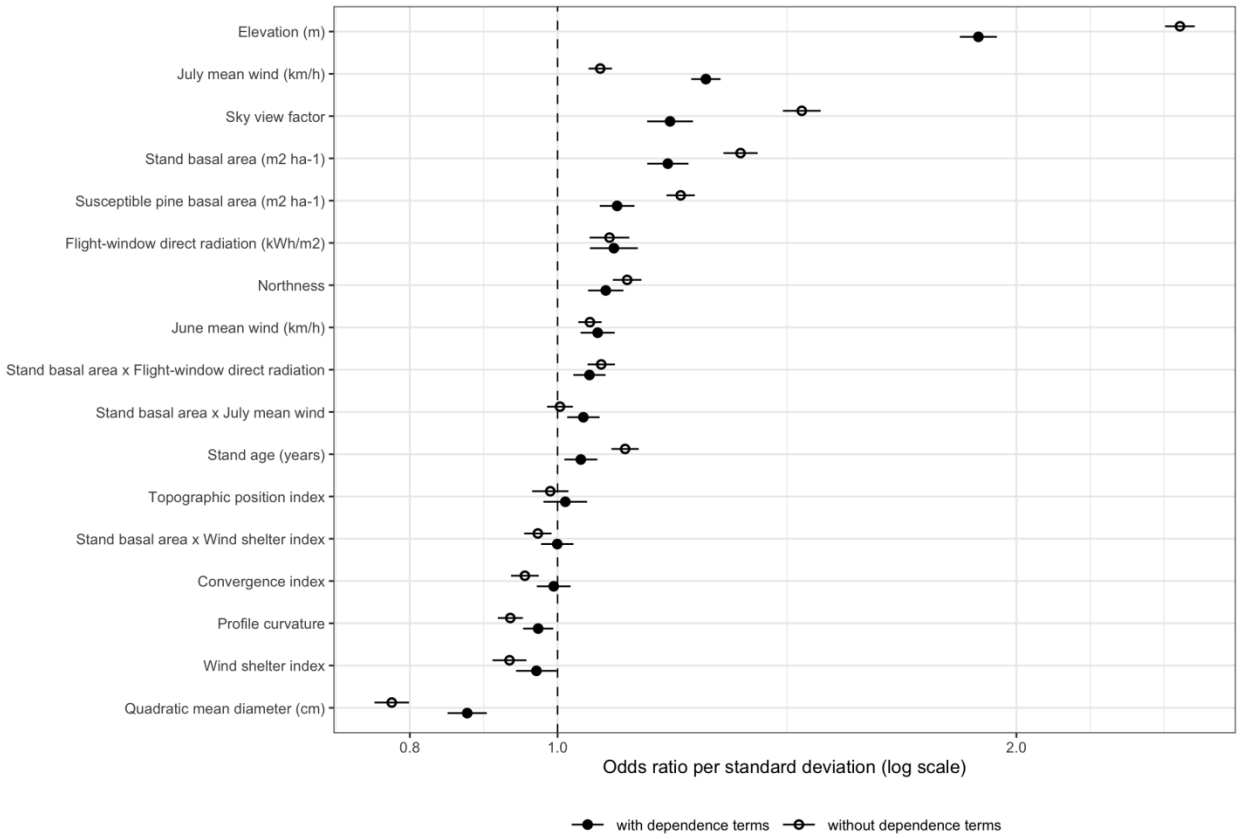


Figure 1. The study area and its red-stage attack. (a) Regional setting over the Copernicus 30 m elevation model. The red fill is the study perimeter, 5,573 ha, the black box the window of the 1 m lidar, the orange outline the 2017 lidar project, the green dashed box the regional wind domain, the blue grid the ERA5-Land reanalysis at 0.1 degree, the yellow triangles the seven hourly stations within 150 km, the black line the profile of Figure S3, and the long dash the Canada and United States border. (b) The first year each cell was attacked, with the 2015 Mt Midgeley burn in red. (c) The share of the perimeter attacked each year, black, and in each sixteen-day period, grey, the periods of each summer joined.

12

13 **Figure 2**



14

15 Figure 2. Odds ratios of the environmental terms of M3 per standard deviation, with 95 per cent
16 confidence intervals, without the dependence terms over the nine years, open circles, and with
17 them over the eight years with a previous map, filled circles. Terms are ordered by their odds ratio
18 with the dependence terms, and the dashed line marks no effect.

19

Figure 3

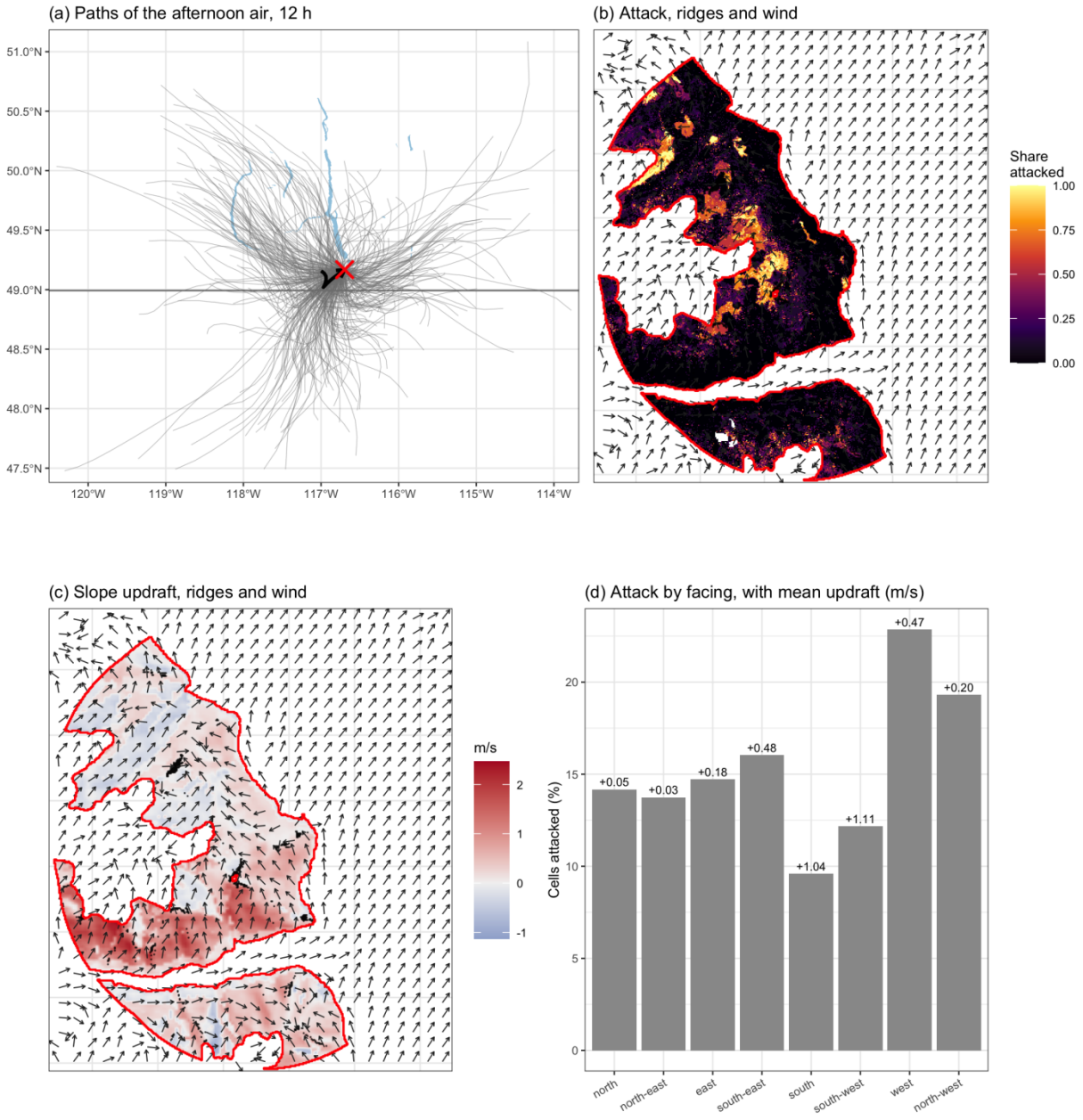


Figure 3. The afternoon air over the site and the attack it met, 1 July to 15 August. (a) HYSPLIT paths of the air over the 12 h before 14:00 on each flight day of 2005 to 2013, with the mean path in black. (b) The share of the annual maps of 2006 to 2014 on which each cell was attacked, with ridges in black and the modelled flight-hour wind as arrows. (c) The slope updraft of that wind,

26 red where it rose and blue where it sank. (d) The share of cells attacked by the direction the
27 ground faced, with the mean updraft above each bar. The red outline is the study perimeter.

Tables

Table 1

Descriptive statistics and screening of red-stage attack and the variables of M3 over the 72,000 sampled cell-years before standardisation. Mean, standard deviation (SD) and skewness over the sample, Shapiro-Wilk W on 5,000 cell-years, and the screening on the years with a previous map, β the slope of the variable added alone to the two dependence terms, the deviance it explained and the p-value of the likelihood ratio test. The landform class was screened as a whole by the same likelihood ratio test. Table S3 defines each variable. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$.

Covariate	Mean	SD	Skew	W	β	Deviance	LR-test
Red-stage attack (%), 2006 to 2014	14.6	3.65e+00	0.30	NA	NA		
Susceptible pine basal area (m² ha⁻¹)	8.47	1.03e+01	1.31	0.804	0.3329	886.97	0.001****
Stand age (years)	113	2.54e+01	1.30	0.893	0.0923	85.93	0.001****
Quadratic mean diameter (cm)	26.5	6.11e+00	1.10	0.934	-0.0141	2.19	0.139
Sky view factor	88.3	4.86e+00	-0.47	0.984	0.4119	1,362.54	0.001****
Northness	0.214	6.85e-01	-0.45	0.886	0.0435	18.26	0.001****
Elevation (m)	1,400	2.76e+02	-0.54	0.906	0.7528	3,987.76	0.001****
Stand basal area (m² ha⁻¹)	35.3	1.24e+01	-0.78	0.915	0.1961	371.10	0.001****
Wind shelter index	21	2.98e+01	1.70	0.597	0.0650	43.04	0.001****
Topographic position index	3.12	1.61e+01	-0.51	0.977	0.1393	201.48	0.001****
Convergence index	0.538	8.81e+00	-0.07	0.974	0.0668	41.60	0.001****
Profile curvature	0.0000622	2.27e-03	-0.06	0.992	0.0457	21.34	0.001****
Flight-window direct radiation (kWh/m²)	140	3.76e+01	-0.15	0.993	0.2147	469.53	0.001****
July mean wind (km/h)	7.17	5.76e-01	0.25	0.916	0.2349	501.58	0.001****
June mean wind (km/h)	7.18	5.08e-01	-0.28	0.948	0.0526	17.97	0.001****
Landform class, seven classes against slope		NA	NA	NA	NA	139.51	0.001****

Table 2

Regression estimates of red-stage attack by model, in log-odds per standard deviation of each continuous term, with the landform classes in every model as log-odds against slope, the most common class, ridge including the shoulder cells and a class of fewer than 30 cells not estimated. M0 held host size, shading and landform, M1 added stand basal area, M2 terrain exposure, shape and flight-window radiation, and M3 the interactions, over the nine years; M3 was refitted with the dependence terms over the eight years with a previous map, and with a latent spatial field. Performance is the AUC and RMSE of the full fit, the RMSE ratio of the test folds over the training folds under ten-fold cross-validation on 1 km blocks, and Theil's U bias. * $p \le 0.05$, ** $p \le 0.01$, *** $p \le 0.001$.

Term	M0	M1	M2	M3	M3 + dependence	M3 + field
Susceptible pine basal area	+0.272***	+0.195***	+0.191***	+0.186***	+0.090***	+0.032
Stand age	+0.118***	+0.101***	+0.101***	+0.102***	+0.035**	+0.029
Quadratic mean diameter	-0.097***	-0.249***	-0.250***	-0.250***	-0.136***	-0.069***
Sky view	+0.420***	+0.381***	+0.387***	+0.369***	+0.170***	+0.037
Northness	+0.128***	+0.077***	+0.100***	+0.105***	+0.073***	+0.133***
Elevation	+0.934***	+0.945***	+0.941***	+0.940***	+0.636***	+1.199***
Stand basal area		+0.279***	+0.273***	+0.276***	+0.167***	+0.200***
Wind shelter index			-0.069***	-0.073***	-0.032*	+0.096***
Topographic position			-0.022	-0.011	+0.012	+0.023
Convergence			-0.049***	-0.049***	-0.006	+0.011
Profile curvature			-0.074***	-0.071***	-0.029*	-0.002
Flight-window radiation			+0.062***	+0.078***	+0.085***	+0.215***
July wind				+0.064***	+0.224***	+0.209***
June wind				+0.049***	+0.061***	+0.064***
Attack in the cell, previous year					+2.208***	+2.282***
Attack within 150 m, previous year					+2.216***	+0.905***
Basal area × shelter				-0.030**	-0.000	+0.009
Basal area × July wind				+0.004	+0.039**	+0.034*
Basal area × radiation				+0.066***	+0.048***	-0.018
Landform class, against slope						
Slope, 38,898 cells, 14 per cent attacked	reference	reference	reference	reference	reference	reference
Spur, 11,796 cells, 18 per cent attacked	-0.107***	-0.059**	-0.012	-0.011	+0.080*	+0.031
Hollow, 8,751 cells, 12 per cent attacked	+0.071*	+0.036	-0.024	-0.028	+0.057	+0.049

Term	M0	M1	M2	M3	M3 + dependence	M3 + field
Valley, 1,609 cells, 11 per cent attacked	+0.013	-0.069	-0.200**	-0.210**	-0.002	+0.054
Ridge, 665 cells, 24 per cent attacked	-0.626***	-0.528***	-0.453***	-0.447***	+0.038	+0.068
Peak, 32 cells, 42 per cent attacked	-0.475	-0.368	-0.284	-0.268	+0.063	+0.456
Footslope, 1 cell, 89 per cent attacked	not estimated	not estimated	not estimated	not estimated	not estimated	not estimated
AUC	0.804	0.807	0.808	0.809	0.890	
RMSE	0.423	0.422	0.421	0.421	0.365	
RMSE ratio	1.003	1.000	1.005	1.008	0.998	
Theil's U bias	0.0000	0.0001	0.0001	0.0001	0.0000	

21

Supporting Information

Testing the wind disruption and disturbance refugia hypothesis for mountain pine beetle outbreaks (*Dendroctonus ponderosae*). Terrain, stand density and flight-period wind as controls on attack in the Selkirk Mountains of British Columbia

Seamus Murphy

Table S1

The datasets the study combined, with the spatial and temporal resolution of each.

Dataset	Source	Spatial	Temporal	Extent
Red-stage attack	Landsat 5 and 8 Collection 2 Level-2	30 m	1 year and 16 days	9 years and 47 periods
Stand structure	Vegetation Resources Inventory, annual releases	30 m, rasterised	1 year	8 snapshots
Terrain	Natural Resources Canada High Resolution Digital Elevation Model	30 m	static	61,923 cells
Station wind	Environment and Climate Change Canada hourly stations	9 stations within 150 km	1 hour	19,157 hours
Terrain-resolved	MicroMet terrain adjustment of station wind	30 m	1 year and 16 days	9 years and 47 periods
Model tables	Rows joined from the above	30 m	1 year and 16 days	72,000 cell-years and 173,086 cell-periods

Table S2

The yearly red-stage maps, with the cells mapped in each year, the share classed as attacked, the ratio of observed to expected joins between neighbouring attacked cells under 999 random permutations, Moran's I in the first distance class of 60 m, and the range of positive autocorrelation.

Year	Cells mapped	Attacked (%)	Joins, observed / expected	Moran's I, 60 m	Range (m)
2006	61,922	10.1	4.45	0.375	3,420
2007	61,923	9.9	4.92	0.411	1,380
2008	61,922	20.7	3.23	0.561	2,760
2009	61,919	12.9	4.78	0.541	1,680
2010	61,761	16.5	4.21	0.619	3,240
2011	61,923	13.6	5.26	0.651	2,700
2012	61,923	12.9	5.80	0.686	2,640

Year	Cells mapped	Attacked (%)	Joins, observed / expected	Moran's I, 60 m	Range (m)
2013	61,920	16.3	4.55	0.669	2,460
2014	61,923	18.3	4.14	0.680	2,520

Table S3

The environmental variables that entered the annual models, grouped by the mechanism each represented. M0 contained host size, shading and landform, M1 added stand density, M2 added terrain exposure to wind, terrain shape and flight-window radiation, and M3 added station wind and the interactions of basal area with the wind shelter index, July wind and flight-window radiation.

Variable	Mechanism
Susceptible pine basal area ($m^2\ ha^{-1}$)	Host size
Stand age (years)	Host size
Quadratic mean diameter (cm)	Host size
Sky view factor	Topographic shading
Northness	Topographic shading
Elevation (m)	Landform
Stand basal area ($m^2\ ha^{-1}$)	Stand density
Wind shelter index	Terrain exposure to wind
Topographic position index	Terrain shape
Convergence index	Terrain shape
Profile curvature	Terrain shape
Flight-window direct radiation (kWh/m^2)	Flight-window radiation
July mean wind (km/h)	Station wind
June mean wind (km/h)	Station wind
July wind, previous summer (km/h)	Station wind, previous summer
June wind, previous summer (km/h)	Station wind, previous summer
Surface temperature, previous flight window (°C)	Measured surface temperature, previous summer
Ground roughness within the cell at 1 m (m)	Terrain at the scale of single trees
Share of the cell facing north at 5 m	Terrain at the scale of single trees
Spread of position at 55 m (m)	Terrain at the scale of single trees
Crown closure (%)	Stand density
Modelled wind at 90 m, previous summer (km/h)	Terrain wind model, previous summer
Upslope share of flight hours, previous summer	Terrain wind model, previous summer

Variable	Mechanism
Consistency of modelled direction, previous summer	Terrain wind model, previous summer
Shelter at the modelled wind direction, previous summer	Terrain wind model, previous summer
Modelled wind at 30 m, previous summer (km/h)	Terrain wind model, previous summer
Modelled wind at 10 m, previous summer (km/h)	Terrain wind model, previous summer
Convective lift, previous summer (m/s)	Terrain wind model, previous summer
Slope updraft of the terrain wind, previous summer (m/s)	Terrain wind model, previous summer
Cloud-adjusted flight-hour sun, previous summer (kWh/m²)	Terrain wind model, previous summer
Slope temperature departure, previous summer (°C)	Terrain wind model, previous summer
Trajectory endpoints over red attack, previous summer	Air trajectory source, previous summer
Reanalysis vertical air speed, previous summer (m/s)	Reanalysis vertical air speed, previous summer
Valley bottom flatness	Valley bottoms, cold air, canopy height and winter cold
Ridge top flatness	Valley bottoms, cold air, canopy height and winter cold
Cold-air pooling (°C)	Valley bottoms, cold air, canopy height and winter cold
Canopy height, 2017 lidar (m)	Valley bottoms, cold air, canopy height and winter cold
Canopy height variation, 2017 lidar (m)	Valley bottoms, cold air, canopy height and winter cold
Canopy over 5 m, 2017 lidar	Valley bottoms, cold air, canopy height and winter cold
Lowest winter minimum before the flight (°C)	Valley bottoms, cold air, canopy height and winter cold
Surviving sources within 150 m, all directions	Previous year's sources by wind direction
Surviving sources 150 to 1,050 m, all directions	Previous year's sources by wind direction
Surviving sources within 150 m, upwind	Previous year's sources by wind direction
Surviving sources within 150 m, downwind	Previous year's sources by wind direction
Surviving sources 150 to 1,050 m, upwind	Previous year's sources by wind direction
Surviving sources 150 to 1,050 m, downwind	Previous year's sources by wind direction
Red crowns within 150 m, all directions	Previous year's sources by wind direction
Red crowns 150 to 1,050 m, all directions	Previous year's sources by wind direction
Red crowns within 150 m, upwind	Previous year's sources by wind direction
Red crowns within 150 m, downwind	Previous year's sources by wind direction
Red crowns 150 to 1,050 m, upwind	Previous year's sources by wind direction
Red crowns 150 to 1,050 m, downwind	Previous year's sources by wind direction

Table S4

Coefficients of the sixteen-day models. E1 had no dependence terms, E2 added the cell's own state in the previous period and the share attacked around it, and E3 added the same two terms for the same period of the previous year. Estimates are changes in log-odds per standard

23 deviation, with standard errors, the number of cells and the AUC of each model; intercepts and
24 geomorphon classes are not shown. * p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001, **** p ≤ 0.0001.

Model	Term	Estimate	SE	n	AUC
E1	Elevation (m)	+0.393****	0.0	146 39	0.6
E1	Susceptible pine basal area (m² ha⁻¹)	+0.202****	0.0	146 39	0.6
E1	Standing volume (m³ ha⁻¹)	+0.106****	0.0	146 39	0.6
E1	Live stems (n/ha)	-0.104*	0.0	146 39	0.6
E1	Quadratic mean diameter (cm)	-0.078*	0.0	146 39	0.6
E1	Crown closure (%)	+0.098****	0.0	146 39	0.6
E1	Terrain ruggedness index	-0.073*	0.0	146 39	0.6
E1	Vector ruggedness measure	-0.017*	0.0	146 39	0.6
E1	Topographic position index	+0.030***	0.0	146 39	0.6
E1	Mid-slope position	-0.075*	0.0	146 39	0.6
E1	Valley depth (m)	-0.032*	0.0	146 39	0.6
E1	Profile curvature	-0.002	0.0	146 39	0.6
E1	Growing-season direct radiation (kWh/m²)	-0.092*	0.0	146 39	0.6
E1	Flight-window direct radiation (kWh/m²)	+0.234****	0.0	146 39	0.6
E1	Windward-leeward index	+0.163****	0.0	146 39	0.6
E1	Flight-hour wind (km/h)	+0.014*	0.0	146 39	0.6
E1	Live stems x Flight-hour wind	-0.012*	0.0	146 39	0.6
E1	Standing volume x Flight-hour wind	-0.003	0.0	146 39	0.6
E1 all hours	Elevation (m)	+0.394****	0.0	146 39	0.6
E1 all hours	Susceptible pine basal area (m² ha⁻¹)	+0.209****	0.0	146 39	0.6
E1 all hours	Standing volume (m³ ha⁻¹)	+0.099****	0.0	146 39	0.6
E1 all hours	Live stems (n/ha)	-0.093*	0.0	146 39	0.6
E1 all hours	Quadratic mean diameter (cm)	-0.067*	0.0	146 39	0.6
E1 all hours	Crown closure (%)	+0.101****	0.0	146 39	0.6
E1 all hours	Terrain ruggedness index	-0.076*	0.0	146 39	0.6
E1 all hours	Vector ruggedness measure	-0.019*	0.0	146 39	0.6
E1 all hours	Topographic position index	+0.025**	0.0	146 39	0.6
E1 all hours	Mid-slope position	-0.077*	0.0	146 39	0.6
E1 all hours	Valley depth (m)	-0.028*	0.0	146 39	0.6
E1 all hours	Profile curvature	-0.001	0.0	146 39	0.6
E1 all hours	Growing-season direct radiation (kWh/m²)	-0.086*	0.0	146 39	0.6
E1 all hours	Flight-window direct radiation (kWh/m²)	+0.226****	0.0	146 39	0.6
E1 all hours	Windward-leeward index	+0.159****	0.0	146 39	0.6

Model	Term	Estimate	SE	n	AUC
E1 all hours	All-hours wind (km/h)	+0.025****	0.0	146,396	0.6
E1 all hours	Live stems x All-hours wind	-0.043*	0.0	146,396	0.6
E1 all hours	Standing volume x All-hours wind	-0.040*	0.0	146,396	0.6
E2	Elevation (m)	+0.326****	0.0	68,813	0.7
E2	Susceptible pine basal area (m² ha⁻¹)	-0.016	0.0	68,813	0.7
E2	Standing volume (m³ ha⁻¹)	+0.081****	0.0	68,813	0.7
E2	Live stems (n/ha)	-0.043*	0.0	68,813	0.7
E2	Quadratic mean diameter (cm)	-0.110*	0.0	68,813	0.7
E2	Crown closure (%)	+0.129****	0.0	68,813	0.7
E2	Terrain ruggedness index	-0.125*	0.0	68,813	0.7
E2	Vector ruggedness measure	-0.003	0.0	68,813	0.7
E2	Topographic position index	-0.020	0.0	68,813	0.7
E2	Mid-slope position	-0.069*	0.0	68,813	0.7
E2	Valley depth (m)	-0.069*	0.0	68,813	0.7
E2	Profile curvature	-0.003	0.0	68,813	0.7
E2	Growing-season direct radiation (kWh/m²)	-0.193*	0.0	68,813	0.7
E2	Flight-window direct radiation (kWh/m²)	+0.192****	0.0	68,813	0.7
E2	Windward-leeward index	+0.114****	0.0	68,813	0.7
E2	Flight-hour wind (km/h)	+0.116****	0.0	68,813	0.7
E2	Attack in the same cell, previous period	+0.884****	0.0	68,813	0.7
E2	Attack within 90 m, previous period	+1.983****	0.0	68,813	0.7
E2	Live stems x Flight-hour wind	+0.051****	0.0	68,813	0.7
E2	Standing volume x Flight-hour wind	+0.010	0.0	68,813	0.7
E3	Elevation (m)	+0.231****	0.0	38,453	0.8
E3	Susceptible pine basal area (m² ha⁻¹)	+0.015	0.0	38,453	0.8
E3	Standing volume (m³ ha⁻¹)	+0.029	0.0	38,453	0.8
E3	Live stems (n/ha)	-0.060*	0.0	38,453	0.8
E3	Quadratic mean diameter (cm)	-0.087*	0.0	38,453	0.8
E3	Crown closure (%)	+0.144****	0.0	38,453	0.8
E3	Terrain ruggedness index	-0.083*	0.0	38,453	0.8
E3	Vector ruggedness measure	-0.010	0.0	38,453	0.8
E3	Topographic position index	+0.040*	0.0	38,453	0.8
E3	Mid-slope position	-0.054*	0.0	38,453	0.8
E3	Valley depth (m)	-0.003	0.0	38,453	0.8
E3	Profile curvature	-0.012	0.0	38,453	0.8
E3	Growing-season direct radiation (kWh/m²)	-0.102*	0.0	38,453	0.8

Model	Term	Estimate	SE	n	AUC
E3	Flight-window direct radiation (kWh/m²)	+0.069*	0.0	38,453	0.8
E3	Windward-leeward index	+0.063**	0.0	38,453	0.8
E3	Flight-hour wind (km/h)	+0.115****	0.0	38,453	0.8
E3	Attack in the same cell, previous period	+0.751****	0.0	38,453	0.8
E3	Attack within 90 m, previous period	+1.847****	0.0	38,453	0.8
E3	Attack in the same cell, previous year	+1.702****	0.0	38,453	0.8
E3	Attack within 90 m, previous year	+0.764****	0.0	38,453	0.8
E3	Live stems x Flight-hour wind	+0.009	0.0	38,453	0.8
E3	Standing volume x Flight-hour wind	+0.008	0.0	38,453	0.8
E1 lagged	Elevation (m)	+0.393****	0.0	146,39	0.6
E1 lagged	Susceptible pine basal area (m² ha⁻¹)	+0.203****	0.0	146,39	0.6
E1 lagged	Standing volume (m³ ha⁻¹)	+0.104****	0.0	146,39	0.6
E1 lagged	Live stems (n/ha)	-0.102*	0.0	146,39	0.6
E1 lagged	Quadratic mean diameter (cm)	-0.077*	0.0	146,39	0.6
E1 lagged	Crown closure (%)	+0.099****	0.0	146,39	0.6
E1 lagged	Terrain ruggedness index	-0.074*	0.0	146,39	0.6
E1 lagged	Vector ruggedness measure	-0.017*	0.0	146,39	0.6
E1 lagged	Topographic position index	+0.030***	0.0	146,39	0.6
E1 lagged	Mid-slope position	-0.075*	0.0	146,39	0.6
E1 lagged	Valley depth (m)	-0.035*	0.0	146,39	0.6
E1 lagged	Profile curvature	-0.002	0.0	146,39	0.6
E1 lagged	Growing-season direct radiation (kWh/m²)	-0.088*	0.0	146,39	0.6
E1 lagged	Flight-window direct radiation (kWh/m²)	+0.234****	0.0	146,39	0.6
E1 lagged	Windward-leeward index	+0.165****	0.0	146,39	0.6
E1 lagged	Flight-hour wind, previous summer (km/h)	+0.002	0.0	146,39	0.6
E1 lagged	Live stems x Flight-hour wind, previous summer	-0.011	0.0	146,39	0.6
E1 lagged	Standing volume x Flight-hour wind, previous summer	-0.023*	0.0	146,39	0.6
E2 lagged	Elevation (m)	+0.326****	0.0	68,813	0.7
E2 lagged	Susceptible pine basal area (m² ha⁻¹)	-0.007	0.0	68,813	0.7
E2 lagged	Standing volume (m³ ha⁻¹)	+0.068****	0.0	68,813	0.7
E2 lagged	Live stems (n/ha)	-0.027*	0.0	68,813	0.7
E2 lagged	Quadratic mean diameter (cm)	-0.096*	0.0	68,813	0.7
E2 lagged	Crown closure (%)	+0.139****	0.0	68,813	0.7
E2 lagged	Terrain ruggedness index	-0.126*	0.0	68,813	0.7
E2 lagged	Vector ruggedness measure	-0.005	0.0	68,813	0.7
E2 lagged	Topographic position index	-0.020	0.0	68,813	0.7

Model	Term	Estimate	SE	n	AUC
E2 lagged	Mid-slope position	-0.068*	0.0	68,813	0.7
E2 lagged	Valley depth (m)	-0.066*	0.0	68,813	0.7
E2 lagged	Profile curvature	-0.003	0.0	68,813	0.7
E2 lagged	Growing-season direct radiation (kWh/m²)	-0.186*	0.0	68,813	0.7
E2 lagged	Flight-window direct radiation (kWh/m²)	+0.193****	0.0	68,813	0.7
E2 lagged	Windward-leeward index	+0.117****	0.0	68,813	0.7
E2 lagged	Flight-hour wind, previous summer (km/h)	+0.087****	0.0	68,813	0.7
E2 lagged	Attack in the same cell, previous period	+0.882****	0.0	68,813	0.7
E2 lagged	Attack within 90 m, previous period	+1.949****	0.0	68,813	0.7
E2 lagged	Live stems x Flight-hour wind, previous summer	+0.008	0.0	68,813	0.7
E2 lagged	Standing volume x Flight-hour wind, previous summer	-0.036*	0.0	68,813	0.7
E2 lagged with pressure	Elevation (m)	+0.329****	0.0	68,813	0.7
E2 lagged with pressure	Susceptible pine basal area (m² ha⁻¹)	-0.013	0.0	68,813	0.7
E2 lagged with pressure	Standing volume (m³ ha⁻¹)	+0.081****	0.0	68,813	0.7
E2 lagged with pressure	Live stems (n/ha)	-0.032*	0.0	68,813	0.7
E2 lagged with pressure	Quadratic mean diameter (cm)	-0.114*	0.0	68,813	0.7
E2 lagged with pressure	Crown closure (%)	+0.137****	0.0	68,813	0.7
E2 lagged with pressure	Terrain ruggedness index	-0.118*	0.0	68,813	0.7
E2 lagged with pressure	Vector ruggedness measure	-0.004	0.0	68,813	0.7
E2 lagged with pressure	Topographic position index	-0.015	0.0	68,813	0.7
E2 lagged with pressure	Mid-slope position	-0.069*	0.0	68,813	0.7
E2 lagged with pressure	Valley depth (m)	-0.074*	0.0	68,813	0.7
E2 lagged with pressure	Profile curvature	-0.003	0.0	68,813	0.7
E2 lagged with pressure	Growing-season direct radiation (kWh/m²)	-0.181*	0.0	68,813	0.7
E2 lagged with pressure	Flight-window direct radiation (kWh/m²)	+0.204****	0.0	68,813	0.7
E2 lagged with pressure	Windward-leeward index	+0.119****	0.0	68,813	0.7
E2 lagged with pressure	Flight-hour wind, previous summer (km/h)	+0.041****	0.0	68,813	0.7
E2 lagged with pressure	Attack in the same cell, previous period	+0.879****	0.0	68,813	0.7
E2 lagged with pressure	Attack within 90 m, previous period	+2.034****	0.0	68,813	0.7
E2 lagged with pressure	Pressure departure, previous summer (kPa)	+0.011	0.0	68,813	0.7
E2 lagged with pressure	Three-hour pressure change, previous summer (kPa)	+0.149****	0.0	68,813	0.7
E2 lagged with pressure	Live stems x Flight-hour wind, previous summer	+0.001	0.0	68,813	0.7
E2 lagged with pressure	Standing volume x Flight-hour wind, previous summer	-0.035*	0.0	68,813	0.7
E2 lagged, diameter and closure	Elevation (m)	+0.325****	0.0	68,813	0.7
E2 lagged, diameter and closure	Susceptible pine basal area (m² ha⁻¹)	-0.010	0.0	68,813	0.7
E2 lagged, diameter and closure	Standing volume (m³ ha⁻¹)	+0.067****	0.0	68,813	0.7

Model	Term	Estimate	SE	n	AUC
E2 lagged, diameter and closure	Live stems (n/ha)	-0.027*	0.0	68,813	0.7
E2 lagged, diameter and closure	Quadratic mean diameter (cm)	-0.094*	0.0	68,813	0.7
E2 lagged, diameter and closure	Crown closure (%)	+0.139****	0.0	68,813	0.7
E2 lagged, diameter and closure	Terrain ruggedness index	-0.125*	0.0	68,813	0.7
E2 lagged, diameter and closure	Vector ruggedness measure	-0.006	0.0	68,813	0.7
E2 lagged, diameter and closure	Topographic position index	-0.020	0.0	68,813	0.7
E2 lagged, diameter and closure	Mid-slope position	-0.068*	0.0	68,813	0.7
E2 lagged, diameter and closure	Valley depth (m)	-0.066*	0.0	68,813	0.7
E2 lagged, diameter and closure	Profile curvature	-0.003	0.0	68,813	0.7
E2 lagged, diameter and closure	Growing-season direct radiation (kWh/m²)	-0.187*	0.0	68,813	0.7
E2 lagged, diameter and closure	Flight-window direct radiation (kWh/m²)	+0.197****	0.0	68,813	0.7
E2 lagged, diameter and closure	Windward-leeward index	+0.122****	0.0	68,813	0.7
E2 lagged, diameter and closure	Flight-hour wind, previous summer (km/h)	+0.088****	0.0	68,813	0.7
E2 lagged, diameter and closure	Attack in the same cell, previous period	+0.883****	0.0	68,813	0.7
E2 lagged, diameter and closure	Attack within 90 m, previous period	+1.938****	0.0	68,813	0.7
E2 lagged, diameter and closure	Quadratic mean diameter x Flight-hour wind, previous summer	-0.064*	0.0	68,813	0.7
E2 lagged, diameter and closure	Crown closure x Flight-hour wind, previous summer	-0.000	0.0	68,813	0.7
E2 modelled wind	Elevation (m)	+0.327****	0.0	67,488	0.7
E2 modelled wind	Susceptible pine basal area (m² ha⁻¹)	-0.013	0.0	67,488	0.7
E2 modelled wind	Standing volume (m³ ha⁻¹)	+0.057*	0.0	67,488	0.7
E2 modelled wind	Live stems (n/ha)	-0.053*	0.0	67,488	0.7
E2 modelled wind	Quadratic mean diameter (cm)	-0.121*	0.0	67,488	0.7
E2 modelled wind	Crown closure (%)	+0.139****	0.0	67,488	0.7
E2 modelled wind	Terrain ruggedness index	-0.105*	0.0	67,488	0.7
E2 modelled wind	Vector ruggedness measure	-0.001	0.0	67,488	0.7
E2 modelled wind	Topographic position index	-0.001	0.0	67,488	0.7
E2 modelled wind	Mid-slope position	-0.062*	0.0	67,488	0.7
E2 modelled wind	Valley depth (m)	-0.077*	0.0	67,488	0.7
E2 modelled wind	Profile curvature	-0.003	0.0	67,488	0.7
E2 modelled wind	Growing-season direct radiation (kWh/m²)	-0.180*	0.0	67,488	0.7
E2 modelled wind	Flight-window direct radiation (kWh/m²)	+0.215****	0.0	67,488	0.7
E2 modelled wind	Windward-leeward index	+0.103****	0.0	67,488	0.7
E2 modelled wind	Modelled wind at 90 m, previous summer (km/h)	-0.014	0.0	67,488	0.7
E2 modelled wind	Attack in the same cell, previous period	+0.878****	0.0	67,488	0.7
E2 modelled wind	Attack within 90 m, previous period	+1.905****	0.0	67,488	0.7
E2 modelled wind	Stand basal area (m² ha⁻¹)	+0.042	0.0	67,488	0.7

Model	Term	Estimate	SE	n	AUC
E2 modelled wind	Modelled wind at 90 m, previous summer x Stand basal area	-0.047	0.0	67,488	0.7
E2 modelled wind	Quadratic mean diameter x Modelled wind at 90 m, previous summer	+0.041***	0.0	67,488	0.7
E2 modelled wind	Crown closure x Modelled wind at 90 m, previous summer	+0.071****	0.0	67,488	0.7
E2 modelled wind	Live stems x Modelled wind at 90 m, previous summer	-0.045*	0.0	67,488	0.7
E2 modelled wind	Standing volume x Modelled wind at 90 m, previous summer	+0.016	0.0	67,488	0.7
E2 upslope share	Elevation (m)	+0.304****	0.0	67,488	0.7
E2 upslope share	Susceptible pine basal area (m² ha⁻¹)	-0.014	0.0	67,488	0.7
E2 upslope share	Standing volume (m³ ha⁻¹)	+0.054*	0.0	67,488	0.7
E2 upslope share	Live stems (n/ha)	-0.047*	0.0	67,488	0.7
E2 upslope share	Quadratic mean diameter (cm)	-0.133*	0.0	67,488	0.7
E2 upslope share	Crown closure (%)	+0.110****	0.0	67,488	0.7
E2 upslope share	Terrain ruggedness index	-0.097*	0.0	67,488	0.7
E2 upslope share	Vector ruggedness measure	-0.001	0.0	67,488	0.7
E2 upslope share	Topographic position index	-0.013	0.0	67,488	0.7
E2 upslope share	Mid-slope position	-0.064*	0.0	67,488	0.7
E2 upslope share	Valley depth (m)	-0.098*	0.0	67,488	0.7
E2 upslope share	Profile curvature	-0.000	0.0	67,488	0.7
E2 upslope share	Growing-season direct radiation (kWh/m²)	-0.163*	0.0	67,488	0.7
E2 upslope share	Flight-window direct radiation (kWh/m²)	+0.227****	0.0	67,488	0.7
E2 upslope share	Windward-leeward index	+0.085****	0.0	67,488	0.7
E2 upslope share	Upslope share of flight hours, previous summer	-0.057*	0.0	67,488	0.7
E2 upslope share	Attack in the same cell, previous period	+0.879****	0.0	67,488	0.7
E2 upslope share	Attack within 90 m, previous period	+1.907****	0.0	67,488	0.7
E2 upslope share	Stand basal area (m² ha⁻¹)	+0.061*	0.0	67,488	0.7
E2 upslope share	Upslope share of flight hours, previous summer x Stand basal area	-0.026	0.0	67,488	0.7
E2 upslope share	Quadratic mean diameter x Upslope share of flight hours, previous summer	+0.077****	0.0	67,488	0.7
E2 upslope share	Crown closure x Upslope share of flight hours, previous summer	+0.048***	0.0	67,488	0.7
E2 upslope share	Live stems x Upslope share of flight hours, previous summer	-0.025	0.0	67,488	0.7
E2 upslope share	Standing volume x Upslope share of flight hours, previous summer	-0.007	0.0	67,488	0.7
E2 shelter at the modelled wind	Elevation (m)	+0.307****	0.0	67,488	0.7
E2 shelter at the modelled wind	Susceptible pine basal area (m² ha⁻¹)	-0.018	0.0	67,488	0.7
E2 shelter at the modelled wind	Standing volume (m³ ha⁻¹)	+0.048*	0.0	67,488	0.7
E2 shelter at the modelled wind	Live stems (n/ha)	-0.051*	0.0	67,488	0.7
E2 shelter at the modelled wind	Quadratic mean diameter (cm)	-0.122*	0.0	67,488	0.7
E2 shelter at the modelled wind	Crown closure (%)	+0.107****	0.0	67,488	0.7
E2 shelter at the modelled wind	Terrain ruggedness index	-0.107*	0.0	67,488	0.7

Model	Term	Estimate	SE	n	AUC
E2 shelter at the modelled wind	Vector ruggedness measure	-0.001	0.0	67, 488	0.7
E2 shelter at the modelled wind	Topographic position index	-0.017	0.0	67, 488	0.7
E2 shelter at the modelled wind	Mid-slope position	-0.068*	0.0	67, 488	0.7
E2 shelter at the modelled wind	Valley depth (m)	-0.092*	0.0	67, 488	0.7
E2 shelter at the modelled wind	Profile curvature	-0.002	0.0	67, 488	0.7
E2 shelter at the modelled wind	Growing-season direct radiation (kWh/m²)	-0.163*	0.0	67, 488	0.7
E2 shelter at the modelled wind	Flight-window direct radiation (kWh/m²)	+0.214****	0.0	67, 488	0.7
E2 shelter at the modelled wind	Windward-leeward index	+0.085****	0.0	67, 488	0.7
E2 shelter at the modelled wind	Shelter at the modelled wind direction, previous summer	-0.064*	0.0	67, 488	0.7
E2 shelter at the modelled wind	Attack in the same cell, previous period	+0.880****	0.0	67, 488	0.7
E2 shelter at the modelled wind	Attack within 90 m, previous period	+1.918****	0.0	67, 488	0.7
E2 shelter at the modelled wind	Stand basal area (m² ha⁻¹)	+0.062*	0.0	67, 488	0.7
E2 shelter at the modelled wind	Shelter at the modelled wind direction, previous summer x Stand basal area	+0.002	0.0	67, 488	0.7
E2 shelter at the modelled wind	Quadratic mean diameter x Shelter at the modelled wind direction, previous summer	+0.060****	0.0	67, 488	0.7
E2 shelter at the modelled wind	Crown closure x Shelter at the modelled wind direction, previous summer	+0.046***	0.0	67, 488	0.7
E2 shelter at the modelled wind	Live stems x Shelter at the modelled wind direction, previous summer	-0.030*	0.0	67, 488	0.7
E2 shelter at the modelled wind	Standing volume x Shelter at the modelled wind direction, previous summer	-0.007	0.0	67, 488	0.7
E2 modelled wind, 30 m	Elevation (m)	+0.325****	0.0	67, 488	0.7
E2 modelled wind, 30 m	Susceptible pine basal area (m² ha⁻¹)	-0.012	0.0	67, 488	0.7
E2 modelled wind, 30 m	Standing volume (m³ ha⁻¹)	+0.063**	0.0	67, 488	0.7
E2 modelled wind, 30 m	Live stems (n/ha)	-0.047*	0.0	67, 488	0.7
E2 modelled wind, 30 m	Quadratic mean diameter (cm)	-0.120*	0.0	67, 488	0.7
E2 modelled wind, 30 m	Crown closure (%)	+0.132****	0.0	67, 488	0.7
E2 modelled wind, 30 m	Terrain ruggedness index	-0.116*	0.0	67, 488	0.7
E2 modelled wind, 30 m	Vector ruggedness measure	-0.003	0.0	67, 488	0.7
E2 modelled wind, 30 m	Topographic position index	-0.006	0.0	67, 488	0.7
E2 modelled wind, 30 m	Mid-slope position	-0.066*	0.0	67, 488	0.7
E2 modelled wind, 30 m	Valley depth (m)	-0.072*	0.0	67, 488	0.7
E2 modelled wind, 30 m	Profile curvature	-0.001	0.0	67, 488	0.7
E2 modelled wind, 30 m	Growing-season direct radiation (kWh/m²)	-0.190*	0.0	67, 488	0.7
E2 modelled wind, 30 m	Flight-window direct radiation (kWh/m²)	+0.202****	0.0	67, 488	0.7
E2 modelled wind, 30 m	Windward-leeward index	+0.098****	0.0	67, 488	0.7
E2 modelled wind, 30 m	Modelled wind at 30 m, previous summer (km/h)	+0.015	0.0	67, 488	0.7
E2 modelled wind, 30 m	Attack in the same cell, previous period	+0.879****	0.0	67, 488	0.7
E2 modelled wind, 30 m	Attack within 90 m, previous period	+1.913****	0.0	67, 488	0.7
E2 modelled wind, 30 m	Stand basal area (m² ha⁻¹)	+0.037	0.0	67, 488	0.7

Model	Term	Estimate	SE	n	AUC
E2 modelled wind, 30 m	Modelled wind at 30 m, previous summer x Stand basal area	+0.002	0.0	67,488	0.7
E2 modelled wind, 30 m	Quadratic mean diameter x Modelled wind at 30 m, previous summer	+0.031**	0.0	67,488	0.7
E2 modelled wind, 30 m	Crown closure x Modelled wind at 30 m, previous summer	+0.060****	0.0	67,488	0.7
E2 modelled wind, 30 m	Live stems x Modelled wind at 30 m, previous summer	-0.063**	0.0	67,488	0.7
E2 modelled wind, 30 m	Standing volume x Modelled wind at 30 m, previous summer	-0.009	0.0	67,488	0.7
E2 modelled wind, 10 m	Elevation (m)	+0.325****	0.0	67,488	0.7
E2 modelled wind, 10 m	Susceptible pine basal area (m² ha⁻¹)	-0.010	0.0	67,488	0.7
E2 modelled wind, 10 m	Standing volume (m³ ha⁻¹)	+0.057*	0.0	67,488	0.7
E2 modelled wind, 10 m	Live stems (n/ha)	-0.049**	0.0	67,488	0.7
E2 modelled wind, 10 m	Quadratic mean diameter (cm)	-0.117**	0.0	67,488	0.7
E2 modelled wind, 10 m	Crown closure (%)	+0.130****	0.0	67,488	0.7
E2 modelled wind, 10 m	Terrain ruggedness index	-0.123**	0.0	67,488	0.7
E2 modelled wind, 10 m	Vector ruggedness measure	-0.002	0.0	67,488	0.7
E2 modelled wind, 10 m	Topographic position index	-0.011	0.0	67,488	0.7
E2 modelled wind, 10 m	Mid-slope position	-0.067**	0.0	67,488	0.7
E2 modelled wind, 10 m	Valley depth (m)	-0.068**	0.0	67,488	0.7
E2 modelled wind, 10 m	Profile curvature	-0.001	0.0	67,488	0.7
E2 modelled wind, 10 m	Growing-season direct radiation (kWh/m²)	-0.197**	0.0	67,488	0.7
E2 modelled wind, 10 m	Flight-window direct radiation (kWh/m²)	+0.195****	0.0	67,488	0.7
E2 modelled wind, 10 m	Windward-leeward index	+0.100****	0.0	67,488	0.7
E2 modelled wind, 10 m	Modelled wind at 10 m, previous summer (km/h)	+0.040***	0.0	67,488	0.7
E2 modelled wind, 10 m	Attack in the same cell, previous period	+0.878****	0.0	67,488	0.7
E2 modelled wind, 10 m	Attack within 90 m, previous period	+1.919****	0.0	67,488	0.7
E2 modelled wind, 10 m	Stand basal area (m² ha⁻¹)	+0.040	0.0	67,488	0.7
E2 modelled wind, 10 m	Modelled wind at 10 m, previous summer x Stand basal area	-0.019	0.0	67,488	0.7
E2 modelled wind, 10 m	Quadratic mean diameter x Modelled wind at 10 m, previous summer	+0.028*	0.0	67,488	0.7
E2 modelled wind, 10 m	Crown closure x Modelled wind at 10 m, previous summer	+0.070****	0.0	67,488	0.7
E2 modelled wind, 10 m	Live stems x Modelled wind at 10 m, previous summer	-0.042**	0.0	67,488	0.7
E2 modelled wind, 10 m	Standing volume x Modelled wind at 10 m, previous summer	-0.009	0.0	67,488	0.7
E2 lift	Elevation (m)	+0.339****	0.0	67,488	0.7
E2 lift	Susceptible pine basal area (m² ha⁻¹)	-0.015	0.0	67,488	0.7
E2 lift	Standing volume (m³ ha⁻¹)	+0.011	0.0	67,488	0.7
E2 lift	Live stems (n/ha)	-0.032**	0.0	67,488	0.7
E2 lift	Quadratic mean diameter (cm)	-0.104**	0.0	67,488	0.7
E2 lift	Crown closure (%)	+0.154****	0.0	67,488	0.7
E2 lift	Terrain ruggedness index	-0.123**	0.0	67,488	0.7

Model	Term	Estimate	SE	n	AUC
E2 lift	Vector ruggedness measure	-0.007	0.0	67, 488	0.7
E2 lift	Topographic position index	-0.004	0.0	67, 488	0.7
E2 lift	Mid-slope position	-0.057*	0.0	67, 488	0.7
E2 lift	Valley depth (m)	-0.052*	0.0	67, 488	0.7
E2 lift	Profile curvature	-0.002	0.0	67, 488	0.7
E2 lift	Growing-season direct radiation (kWh/m²)	-0.184*	0.0	67, 488	0.7
E2 lift	Flight-window direct radiation (kWh/m²)	+0.334****	0.0	67, 488	0.7
E2 lift	Windward-leeward index	+0.104****	0.0	67, 488	0.7
E2 lift	Convective lift, previous summer (m/s)	-0.239*	0.0	67, 488	0.7
E2 lift	Attack in the same cell, previous period	+0.872****	0.0	67, 488	0.7
E2 lift	Attack within 90 m, previous period	+2.044****	0.0	67, 488	0.7
E2 lift	Stand basal area (m² ha⁻¹)	+0.065*	0.0	67, 488	0.7
E2 lift	Convective lift, previous summer x Stand basal area	-0.045	0.0	67, 488	0.7
E2 lift	Quadratic mean diameter x Convective lift, previous summer	-0.023	0.0	67, 488	0.7
E2 lift	Crown closure x Convective lift, previous summer	-0.002	0.0	67, 488	0.7
E2 lift	Live stems x Convective lift, previous summer	-0.071*	0.0	67, 488	0.7
E2 lift	Standing volume x Convective lift, previous summer	+0.016	0.0	67, 488	0.7
E2 lift with season	Elevation (m)	+0.316****	0.0	67, 488	0.7
E2 lift with season	Susceptible pine basal area (m² ha⁻¹)	-0.013	0.0	67, 488	0.7
E2 lift with season	Standing volume (m³ ha⁻¹)	+0.010	0.0	67, 488	0.7
E2 lift with season	Live stems (n/ha)	-0.028	0.0	67, 488	0.7
E2 lift with season	Quadratic mean diameter (cm)	-0.092*	0.0	67, 488	0.7
E2 lift with season	Crown closure (%)	+0.163****	0.0	67, 488	0.7
E2 lift with season	Terrain ruggedness index	-0.123*	0.0	67, 488	0.7
E2 lift with season	Vector ruggedness measure	-0.008	0.0	67, 488	0.7
E2 lift with season	Topographic position index	-0.001	0.0	67, 488	0.7
E2 lift with season	Mid-slope position	-0.063*	0.0	67, 488	0.7
E2 lift with season	Valley depth (m)	-0.070*	0.0	67, 488	0.7
E2 lift with season	Profile curvature	-0.004	0.0	67, 488	0.7
E2 lift with season	Growing-season direct radiation (kWh/m²)	-0.194*	0.0	67, 488	0.7
E2 lift with season	Flight-window direct radiation (kWh/m²)	+0.275****	0.0	67, 488	0.7
E2 lift with season	Windward-leeward index	+0.108****	0.0	67, 488	0.7
E2 lift with season	Convective lift, previous summer (m/s)	-0.121*	0.0	67, 488	0.7
E2 lift with season	Attack in the same cell, previous period	+0.884****	0.0	67, 488	0.7
E2 lift with season	Attack within 90 m, previous period	+2.131****	0.0	67, 488	0.7
E2 lift with season	Stand basal area (m² ha⁻¹)	+0.065*	0.0	67, 488	0.7

Model	Term	Estimate	SE	n	AUC
E2 lift with season	Convective lift, previous summer x Stand basal area	-0.023	0.0	67, 488	0.7
E2 lift with season	Quadratic mean diameter x Convective lift, previous summer	-0.030*	0.0	67, 488	0.7
E2 lift with season	Crown closure x Convective lift, previous summer	-0.003	0.0	67, 488	0.7
E2 lift with season	Live stems x Convective lift, previous summer	-0.089*	0.0	67, 488	0.7
E2 lift with season	Standing volume x Convective lift, previous summer	+0.004	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Elevation (m)	+0.295****	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Susceptible pine basal area (m² ha⁻¹)	+0.016	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Standing volume (m³ ha⁻¹)	-0.230*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Live stems (n/ha)	-0.040*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Quadratic mean diameter (cm)	-0.004	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Crown closure (%)	+0.107****	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Terrain ruggedness index	-0.083*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Vector ruggedness measure	-0.002	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Topographic position index	+0.009	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Mid-slope position	-0.052*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Valley depth (m)	-0.066*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Profile curvature	-0.005	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Growing-season direct radiation (kWh/m²)	-0.133*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Flight-window direct radiation (kWh/m²)	+0.372****	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Windward-leeward index	+0.069****	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Convective lift, previous summer (m/s)	-0.360*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Attack in the same cell, previous period	+0.896****	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Attack within 90 m, previous period	+2.671****	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Stand basal area (m² ha⁻¹)	+0.302****	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Modelled wind at 10 m, previous summer (km/h)	-0.003	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Upslope share of flight hours, previous summer	-0.040*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Convective lift, previous summer x Stand basal area	-0.070*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Quadratic mean diameter x Convective lift, previous summer	-0.046*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Crown closure x Convective lift, previous summer	-0.009	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Live stems x Convective lift, previous summer	-0.075*	0.0	67, 488	0.7
E2 lift, 10 m speed and upslope	Standing volume x Convective lift, previous summer	+0.085**	0.0	67, 488	0.7
E2 slope updraft	Elevation (m)	+0.299****	0.0	67, 488	0.7
E2 slope updraft	Susceptible pine basal area (m² ha⁻¹)	-0.023*	0.0	67, 488	0.7
E2 slope updraft	Standing volume (m³ ha⁻¹)	+0.048*	0.0	67, 488	0.7
E2 slope updraft	Live stems (n/ha)	-0.045*	0.0	67, 488	0.7
E2 slope updraft	Quadratic mean diameter (cm)	-0.127*	0.0	67, 488	0.7

Model	Term	Estimate	SE	n	AUC
E2 slope updraft	Crown closure (%)	+0.129****	0.0	67,488	0.7
E2 slope updraft	Terrain ruggedness index	-0.061*	0.0	67,488	0.7
E2 slope updraft	Vector ruggedness measure	+0.003	0.0	67,488	0.7
E2 slope updraft	Topographic position index	-0.005	0.0	67,488	0.7
E2 slope updraft	Mid-slope position	-0.061*	0.0	67,488	0.7
E2 slope updraft	Valley depth (m)	-0.111*	0.0	67,488	0.7
E2 slope updraft	Profile curvature	-0.003	0.0	67,488	0.7
E2 slope updraft	Growing-season direct radiation (kWh/m²)	-0.134*	0.0	67,488	0.7
E2 slope updraft	Flight-window direct radiation (kWh/m²)	+0.248****	0.0	67,488	0.7
E2 slope updraft	Windward-leeward index	+0.081****	0.0	67,488	0.7
E2 slope updraft	Slope updraft of the terrain wind, previous summer (m/s)	-0.103*	0.0	67,488	0.7
E2 slope updraft	Attack in the same cell, previous period	+0.881****	0.0	67,488	0.7
E2 slope updraft	Attack within 90 m, previous period	+1.897****	0.0	67,488	0.7
E2 slope updraft	Stand basal area (m² ha⁻¹)	+0.064*	0.0	67,488	0.7
E2 slope updraft	Slope updraft of the terrain wind, previous summer x Stand basal area	-0.064*	0.0	67,488	0.7
E2 slope updraft	Quadratic mean diameter x Slope updraft of the terrain wind, previous summer	+0.045***	0.0	67,488	0.7
E2 slope updraft	Crown closure x Slope updraft of the terrain wind, previous summer	+0.041**	0.0	67,488	0.7
E2 slope updraft	Live stems x Slope updraft of the terrain wind, previous summer	-0.045*	0.0	67,488	0.7
E2 slope updraft	Standing volume x Slope updraft of the terrain wind, previous summer	+0.049	0.0	67,488	0.7
E2 slope updraft at ridges	Elevation (m)	+0.261****	0.0	67,488	0.7
E2 slope updraft at ridges	Susceptible pine basal area (m² ha⁻¹)	+0.005	0.0	67,488	0.7
E2 slope updraft at ridges	Standing volume (m³ ha⁻¹)	-0.152*	0.0	67,488	0.7
E2 slope updraft at ridges	Live stems (n/ha)	-0.060*	0.0	67,488	0.7
E2 slope updraft at ridges	Quadratic mean diameter (cm)	-0.030	0.0	67,488	0.7
E2 slope updraft at ridges	Crown closure (%)	+0.073****	0.0	67,488	0.7
E2 slope updraft at ridges	Terrain ruggedness index	-0.047*	0.0	67,488	0.7
E2 slope updraft at ridges	Vector ruggedness measure	+0.004	0.0	67,488	0.7
E2 slope updraft at ridges	Topographic position index	+0.026	0.0	67,488	0.7
E2 slope updraft at ridges	Mid-slope position	-0.051*	0.0	67,488	0.7
E2 slope updraft at ridges	Valley depth (m)	-0.109*	0.0	67,488	0.7
E2 slope updraft at ridges	Profile curvature	-0.006	0.0	67,488	0.7
E2 slope updraft at ridges	Growing-season direct radiation (kWh/m²)	-0.130*	0.0	67,488	0.7
E2 slope updraft at ridges	Flight-window direct radiation (kWh/m²)	+0.213****	0.0	67,488	0.7
E2 slope updraft at ridges	Windward-leeward index	+0.071****	0.0	67,488	0.7
E2 slope updraft at ridges	Slope updraft of the terrain wind, previous summer (m/s)	-0.077*	0.0	67,488	0.7
E2 slope updraft at ridges	Attack in the same cell, previous period	+0.900****	0.0	67,488	0.7

Model	Term	Estimate	SE	n	AUC
E2 slope updraft at ridges	Attack within 90 m, previous period	+2.668****	0.0	67,488	0.7
E2 slope updraft at ridges	Stand basal area (m² ha⁻¹)	+0.260****	0.0	67,488	0.7
E2 slope updraft at ridges	Slope updraft of the terrain wind, previous summer x Stand basal area	+0.024	0.0	67,488	0.7
E2 slope updraft at ridges	Quadratic mean diameter x Slope updraft of the terrain wind, previous summer	+0.029*	0.0	67,488	0.7
E2 slope updraft at ridges	Crown closure x Slope updraft of the terrain wind, previous summer	+0.055****	0.0	67,488	0.7
E2 slope updraft at ridges	Live stems x Slope updraft of the terrain wind, previous summer	-0.055*	0.0	67,488	0.7
E2 slope updraft at ridges	Standing volume x Slope updraft of the terrain wind, previous summer	-0.030	0.0	67,488	0.7
E2 slope updraft at ridges	Topographic position index x Slope updraft of the terrain wind, previous summer	-0.029*	0.0	67,488	0.7
E2 cloud-adjusted sun	Elevation (m)	+0.307****	0.0	67,488	0.7
E2 cloud-adjusted sun	Susceptible pine basal area (m² ha⁻¹)	+0.005	0.0	67,488	0.7
E2 cloud-adjusted sun	Standing volume (m³ ha⁻¹)	+0.055*	0.0	67,488	0.7
E2 cloud-adjusted sun	Live stems (n/ha)	-0.077*	0.0	67,488	0.7
E2 cloud-adjusted sun	Quadratic mean diameter (cm)	-0.128*	0.0	67,488	0.7
E2 cloud-adjusted sun	Crown closure (%)	+0.146****	0.0	67,488	0.7
E2 cloud-adjusted sun	Terrain ruggedness index	-0.128*	0.0	67,488	0.7
E2 cloud-adjusted sun	Vector ruggedness measure	-0.014	0.0	67,488	0.7
E2 cloud-adjusted sun	Topographic position index	-0.018	0.0	67,488	0.7
E2 cloud-adjusted sun	Mid-slope position	-0.063*	0.0	67,488	0.7
E2 cloud-adjusted sun	Valley depth (m)	-0.091*	0.0	67,488	0.7
E2 cloud-adjusted sun	Profile curvature	-0.001	0.0	67,488	0.7
E2 cloud-adjusted sun	Growing-season direct radiation (kWh/m²)	-0.135*	0.0	67,488	0.7
E2 cloud-adjusted sun	Flight-window direct radiation (kWh/m²)	+0.238****	0.0	67,488	0.7
E2 cloud-adjusted sun	Windward-leeward index	+0.094****	0.0	67,488	0.7
E2 cloud-adjusted sun	Cloud-adjusted flight-hour sun, previous summer (kWh/m²)	-0.128*	0.0	67,488	0.7
E2 cloud-adjusted sun	Attack in the same cell, previous period	+0.874****	0.0	67,488	0.7
E2 cloud-adjusted sun	Attack within 90 m, previous period	+1.968****	0.0	67,488	0.7
E2 cloud-adjusted sun	Stand basal area (m² ha⁻¹)	+0.057	0.0	67,488	0.7
E2 cloud-adjusted sun	Cloud-adjusted flight-hour sun, previous summer x Stand basal area	-0.085*	0.0	67,488	0.7
E2 cloud-adjusted sun	Quadratic mean diameter x Cloud-adjusted flight-hour sun, previous summer	-0.079*	0.0	67,488	0.7
E2 cloud-adjusted sun	Crown closure x Cloud-adjusted flight-hour sun, previous summer	-0.039*	0.0	67,488	0.7
E2 cloud-adjusted sun	Live stems x Cloud-adjusted flight-hour sun, previous summer	-0.117*	0.0	67,488	0.7
E2 cloud-adjusted sun	Standing volume x Cloud-adjusted flight-hour sun, previous summer	+0.136****	0.0	67,488	0.7
E2 slope temperature	Elevation (m)	+0.313****	0.0	67,488	0.7
E2 slope temperature	Susceptible pine basal area (m² ha⁻¹)	+0.000	0.0	67,488	0.7
E2 slope temperature	Standing volume (m³ ha⁻¹)	+0.034	0.0	67,488	0.7
E2 slope temperature	Live stems (n/ha)	-0.053*	0.0	67,488	0.7

Model	Term	Estimate	SE	n	AUC
E2 slope temperature	Quadratic mean diameter (cm)	-0.111*	0.0	67,488	0.7
E2 slope temperature	Crown closure (%)	+0.152****	0.0	67,488	0.7
E2 slope temperature	Terrain ruggedness index	-0.123*	0.0	67,488	0.7
E2 slope temperature	Vector ruggedness measure	-0.010	0.0	67,488	0.7
E2 slope temperature	Topographic position index	-0.006	0.0	67,488	0.7
E2 slope temperature	Mid-slope position	-0.057*	0.0	67,488	0.7
E2 slope temperature	Valley depth (m)	-0.098*	0.0	67,488	0.7
E2 slope temperature	Profile curvature	+0.001	0.0	67,488	0.7
E2 slope temperature	Growing-season direct radiation (kWh/m²)	-0.112*	0.0	67,488	0.7
E2 slope temperature	Flight-window direct radiation (kWh/m²)	+0.361****	0.0	67,488	0.7
E2 slope temperature	Windward-leeward index	+0.092****	0.0	67,488	0.7
E2 slope temperature	Slope temperature departure, previous summer (°C)	-0.259*	0.0	67,488	0.7
E2 slope temperature	Attack in the same cell, previous period	+0.872****	0.0	67,488	0.7
E2 slope temperature	Attack within 90 m, previous period	+1.987****	0.0	67,488	0.7
E2 slope temperature	Stand basal area (m² ha⁻¹)	+0.054	0.0	67,488	0.7
E2 slope temperature	Slope temperature departure, previous summer x Stand basal area	-0.087*	0.0	67,488	0.7
E2 slope temperature	Quadratic mean diameter x Slope temperature departure, previous summer	-0.035*	0.0	67,488	0.7
E2 slope temperature	Crown closure x Slope temperature departure, previous summer	+0.009	0.0	67,488	0.7
E2 slope temperature	Live stems x Slope temperature departure, previous summer	-0.097*	0.0	67,488	0.7
E2 slope temperature	Standing volume x Slope temperature departure, previous summer	+0.102****	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Elevation (m)	+0.274****	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Susceptible pine basal area (m² ha⁻¹)	+0.029*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Standing volume (m³ ha⁻¹)	-0.163*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Live stems (n/ha)	-0.066*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Quadratic mean diameter (cm)	-0.021	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Crown closure (%)	+0.100****	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Terrain ruggedness index	-0.115*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Vector ruggedness measure	-0.013	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Topographic position index	-0.001	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Mid-slope position	-0.058*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Valley depth (m)	-0.090*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Profile curvature	-0.003	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Growing-season direct radiation (kWh/m²)	-0.170*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Flight-window direct radiation (kWh/m²)	+0.118****	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Windward-leeward index	+0.080****	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Cloud-adjusted flight-hour sun, previous summer (kWh/m²)	+0.077	0.0	67,488	0.7

Model	Term	Estimate	SE	n	AUC
E2 cloud-adjusted sun with season	Attack in the same cell, previous period	+0.899****	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Attack within 90 m, previous period	+2.676****	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Stand basal area (m² ha⁻¹)	+0.257****	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Cloud-adjusted flight-hour sun, previous summer x Stand basal area	-0.104*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Quadratic mean diameter x Cloud-adjusted flight-hour sun, previous summer	-0.087*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Crown closure x Cloud-adjusted flight-hour sun, previous summer	-0.025	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Live stems x Cloud-adjusted flight-hour sun, previous summer	-0.084*	0.0	67,488	0.7
E2 cloud-adjusted sun with season	Standing volume x Cloud-adjusted flight-hour sun, previous summer	+0.158****	0.0	67,488	0.7
E2 slope temperature with season	Elevation (m)	+0.273****	0.0	67,488	0.7
E2 slope temperature with season	Susceptible pine basal area (m² ha⁻¹)	+0.027*	0.0	67,488	0.7
E2 slope temperature with season	Standing volume (m³ ha⁻¹)	-0.178*	0.0	67,488	0.7
E2 slope temperature with season	Live stems (n/ha)	-0.056*	0.0	67,488	0.7
E2 slope temperature with season	Quadratic mean diameter (cm)	-0.009	0.0	67,488	0.7
E2 slope temperature with season	Crown closure (%)	+0.098****	0.0	67,488	0.7
E2 slope temperature with season	Terrain ruggedness index	-0.102*	0.0	67,488	0.7
E2 slope temperature with season	Vector ruggedness measure	-0.008	0.0	67,488	0.7
E2 slope temperature with season	Topographic position index	+0.008	0.0	67,488	0.7
E2 slope temperature with season	Mid-slope position	-0.053*	0.0	67,488	0.7
E2 slope temperature with season	Valley depth (m)	-0.096*	0.0	67,488	0.7
E2 slope temperature with season	Profile curvature	-0.002	0.0	67,488	0.7
E2 slope temperature with season	Growing-season direct radiation (kWh/m²)	-0.112*	0.0	67,488	0.7
E2 slope temperature with season	Flight-window direct radiation (kWh/m²)	+0.294****	0.0	67,488	0.7
E2 slope temperature with season	Windward-leeward index	+0.076****	0.0	67,488	0.7
E2 slope temperature with season	Slope temperature departure, previous summer (°C)	-0.203*	0.0	67,488	0.7
E2 slope temperature with season	Attack in the same cell, previous period	+0.898****	0.0	67,488	0.7
E2 slope temperature with season	Attack within 90 m, previous period	+2.670****	0.0	67,488	0.7
E2 slope temperature with season	Stand basal area (m² ha⁻¹)	+0.266****	0.0	67,488	0.7
E2 slope temperature with season	Slope temperature departure, previous summer x Stand basal area	-0.074*	0.0	67,488	0.7
E2 slope temperature with season	Quadratic mean diameter x Slope temperature departure, previous summer	-0.040*	0.0	67,488	0.7
E2 slope temperature with season	Crown closure x Slope temperature departure, previous summer	+0.015	0.0	67,488	0.7
E2 slope temperature with season	Live stems x Slope temperature departure, previous summer	-0.080*	0.0	67,488	0.7
E2 slope temperature with season	Standing volume x Slope temperature departure, previous summer	+0.101****	0.0	67,488	0.7
E2 slope temperature without sun	Elevation (m)	+0.266****	0.0	67,488	0.7
E2 slope temperature without sun	Susceptible pine basal area (m² ha⁻¹)	+0.024*	0.0	67,488	0.7
E2 slope temperature without sun	Standing volume (m³ ha⁻¹)	-0.201*	0.0	67,488	0.7
E2 slope temperature without sun	Live stems (n/ha)	-0.058*	0.0	67,488	0.7

Model	Term	Estimate	SE	n	AUC
E2 slope temperature without sun	Quadratic mean diameter (cm)	-0.010	0.0	67,488	0.7
E2 slope temperature without sun	Crown closure (%)	+0.111****	0.0	67,488	0.7
E2 slope temperature without sun	Terrain ruggedness index	-0.129*	0.0	67,488	0.7
E2 slope temperature without sun	Vector ruggedness measure	-0.007	0.0	67,488	0.7
E2 slope temperature without sun	Topographic position index	+0.023	0.0	67,488	0.7
E2 slope temperature without sun	Mid-slope position	-0.053*	0.0	67,488	0.7
E2 slope temperature without sun	Valley depth (m)	-0.097*	0.0	67,488	0.7
E2 slope temperature without sun	Profile curvature	-0.002	0.0	67,488	0.7
E2 slope temperature without sun	Windward-leeward index	-0.042*	0.0	67,488	0.7
E2 slope temperature without sun	Slope temperature departure, previous summer (°C)	-0.054*	0.0	67,488	0.7
E2 slope temperature without sun	Attack in the same cell, previous period	+0.893****	0.0	67,488	0.7
E2 slope temperature without sun	Attack within 90 m, previous period	+2.687****	0.0	67,488	0.7
E2 slope temperature without sun	Stand basal area (m² ha⁻¹)	+0.297****	0.0	67,488	0.7
E2 slope temperature without sun	Slope temperature departure, previous summer x Stand basal area	-0.070*	0.0	67,488	0.7
E2 slope temperature without sun	Quadratic mean diameter x Slope temperature departure, previous summer	-0.042*	0.0	67,488	0.7
E2 slope temperature without sun	Crown closure x Slope temperature departure, previous summer	+0.018	0.0	67,488	0.7
E2 slope temperature without sun	Live stems x Slope temperature departure, previous summer	-0.086*	0.0	67,488	0.7
E2 slope temperature without sun	Standing volume x Slope temperature departure, previous summer	+0.099****	0.0	67,488	0.7
E2 valley and ridge flats	Elevation (m)	+0.336****	0.0	68,813	0.7
E2 valley and ridge flats	Susceptible pine basal area (m² ha⁻¹)	-0.015	0.0	68,813	0.7
E2 valley and ridge flats	Standing volume (m³ ha⁻¹)	+0.045*	0.0	68,813	0.7
E2 valley and ridge flats	Live stems (n/ha)	-0.037*	0.0	68,813	0.7
E2 valley and ridge flats	Quadratic mean diameter (cm)	-0.115*	0.0	68,813	0.7
E2 valley and ridge flats	Crown closure (%)	+0.130****	0.0	68,813	0.7
E2 valley and ridge flats	Terrain ruggedness index	-0.117*	0.0	68,813	0.7
E2 valley and ridge flats	Vector ruggedness measure	-0.006	0.0	68,813	0.7
E2 valley and ridge flats	Topographic position index	-0.009	0.0	68,813	0.7
E2 valley and ridge flats	Mid-slope position	-0.067*	0.0	68,813	0.7
E2 valley and ridge flats	Valley depth (m)	-0.076*	0.0	68,813	0.7
E2 valley and ridge flats	Profile curvature	-0.004	0.0	68,813	0.7
E2 valley and ridge flats	Growing-season direct radiation (kWh/m²)	-0.173*	0.0	68,813	0.7
E2 valley and ridge flats	Flight-window direct radiation (kWh/m²)	+0.211****	0.0	68,813	0.7
E2 valley and ridge flats	Windward-leeward index	+0.122****	0.0	68,813	0.7
E2 valley and ridge flats	Valley bottom flatness	-0.020	0.0	68,813	0.7
E2 valley and ridge flats	Attack in the same cell, previous period	+0.882****	0.0	68,813	0.7
E2 valley and ridge flats	Attack within 90 m, previous period	+1.922****	0.0	68,813	0.7

Model	Term	Estimate	SE	n	AUC
E2 valley and ridge flats	Stand basal area (m² ha⁻¹)	+0.040	0.0	68, 813	0.7
E2 valley and ridge flats	Ridge top flatness	-0.014	0.0	68, 813	0.7
E2 valley and ridge flats	Valley bottom flatness x Stand basal area	-0.003	0.0	68, 813	0.7
E2 cold-air pooling	Elevation (m)	+0.372****	0.0	68, 813	0.7
E2 cold-air pooling	Susceptible pine basal area (m² ha⁻¹)	-0.017	0.0	68, 813	0.7
E2 cold-air pooling	Standing volume (m³ ha⁻¹)	+0.043	0.0	68, 813	0.7
E2 cold-air pooling	Live stems (n/ha)	-0.036*	0.0	68, 813	0.7
E2 cold-air pooling	Quadratic mean diameter (cm)	-0.116*	0.0	68, 813	0.7
E2 cold-air pooling	Crown closure (%)	+0.129****	0.0	68, 813	0.7
E2 cold-air pooling	Terrain ruggedness index	-0.108*	0.0	68, 813	0.7
E2 cold-air pooling	Vector ruggedness measure	-0.003	0.0	68, 813	0.7
E2 cold-air pooling	Topographic position index	-0.001	0.0	68, 813	0.7
E2 cold-air pooling	Mid-slope position	-0.064*	0.0	68, 813	0.7
E2 cold-air pooling	Valley depth (m)	-0.080*	0.0	68, 813	0.7
E2 cold-air pooling	Profile curvature	-0.005	0.0	68, 813	0.7
E2 cold-air pooling	Growing-season direct radiation (kWh/m²)	-0.172*	0.0	68, 813	0.7
E2 cold-air pooling	Flight-window direct radiation (kWh/m²)	+0.213****	0.0	68, 813	0.7
E2 cold-air pooling	Windward-leeward index	+0.118****	0.0	68, 813	0.7
E2 cold-air pooling	Cold-air pooling (°C)	-0.045	0.0	68, 813	0.7
E2 cold-air pooling	Attack in the same cell, previous period	+0.880****	0.0	68, 813	0.7
E2 cold-air pooling	Attack within 90 m, previous period	+1.918****	0.0	68, 813	0.7
E2 cold-air pooling	Stand basal area (m² ha⁻¹)	+0.048	0.0	68, 813	0.7
E2 cold-air pooling	Cold-air pooling x Stand basal area	+0.021*	0.0	68, 813	0.7
E2 canopy height	Elevation (m)	+0.277****	0.0	68, 813	0.7
E2 canopy height	Susceptible pine basal area (m² ha⁻¹)	-0.002	0.0	68, 813	0.7
E2 canopy height	Standing volume (m³ ha⁻¹)	-0.011	0.0	68, 813	0.7
E2 canopy height	Live stems (n/ha)	-0.010	0.0	68, 813	0.7
E2 canopy height	Quadratic mean diameter (cm)	-0.117*	0.0	68, 813	0.7
E2 canopy height	Crown closure (%)	+0.083****	0.0	68, 813	0.7
E2 canopy height	Terrain ruggedness index	-0.079*	0.0	68, 813	0.7
E2 canopy height	Vector ruggedness measure	-0.002	0.0	68, 813	0.7
E2 canopy height	Topographic position index	+0.011	0.0	68, 813	0.7
E2 canopy height	Mid-slope position	-0.044*	0.0	68, 813	0.7
E2 canopy height	Valley depth (m)	-0.113*	0.0	68, 813	0.7
E2 canopy height	Profile curvature	-0.014	0.0	68, 813	0.7
E2 canopy height	Growing-season direct radiation (kWh/m²)	-0.217*	0.0	68, 813	0.7

Model	Term	Estimate	SE	n	AUC
E2 canopy height	Flight-window direct radiation (kWh/m²)	+0.243****	0.0	68,813	0.7
E2 canopy height	Windward-leeward index	+0.107****	0.0	68,813	0.7
E2 canopy height	Canopy height, 2017 lidar (m)	+0.509****	0.0	68,813	0.7
E2 canopy height	Attack in the same cell, previous period	+0.857****	0.0	68,813	0.7
E2 canopy height	Attack within 90 m, previous period	+1.801****	0.0	68,813	0.7
E2 canopy height	Stand basal area (m² ha⁻¹)	+0.035	0.0	68,813	0.7
E2 canopy height	Canopy height variation, 2017 lidar (m)	+0.042***	0.0	68,813	0.7
E2 canopy height	Canopy over 5 m, 2017 lidar	-0.635*	0.0	68,813	0.7
E2 canopy height	Canopy height, 2017 lidar x Stand basal area	-0.080*	0.0	68,813	0.7
E2 winter cold	Elevation (m)	+0.359****	0.0	68,813	0.7
E2 winter cold	Susceptible pine basal area (m² ha⁻¹)	-0.013	0.0	68,813	0.7
E2 winter cold	Standing volume (m³ ha⁻¹)	-0.008	0.0	68,813	0.7
E2 winter cold	Live stems (n/ha)	-0.013	0.0	68,813	0.7
E2 winter cold	Quadratic mean diameter (cm)	-0.077*	0.0	68,813	0.7
E2 winter cold	Crown closure (%)	+0.118****	0.0	68,813	0.7
E2 winter cold	Terrain ruggedness index	-0.094*	0.0	68,813	0.7
E2 winter cold	Vector ruggedness measure	-0.003	0.0	68,813	0.7
E2 winter cold	Topographic position index	+0.007	0.0	68,813	0.7
E2 winter cold	Mid-slope position	-0.062*	0.0	68,813	0.7
E2 winter cold	Valley depth (m)	-0.079*	0.0	68,813	0.7
E2 winter cold	Profile curvature	-0.006	0.0	68,813	0.7
E2 winter cold	Growing-season direct radiation (kWh/m²)	-0.151*	0.0	68,813	0.7
E2 winter cold	Flight-window direct radiation (kWh/m²)	+0.192****	0.0	68,813	0.7
E2 winter cold	Windward-leeward index	+0.099****	0.0	68,813	0.7
E2 winter cold	Lowest winter minimum before the flight (°C)	-0.111*	0.0	68,813	0.7
E2 winter cold	Attack in the same cell, previous period	+0.875****	0.0	68,813	0.7
E2 winter cold	Attack within 90 m, previous period	+1.944****	0.0	68,813	0.7
E2 winter cold	Stand basal area (m² ha⁻¹)	+0.099***	0.0	68,813	0.7
E2 winter cold	Lowest winter minimum before the flight x Stand basal area	+0.097****	0.0	68,813	0.7
E2 annual terms	Elevation (m)	+0.329****	0.0	68,813	0.7
E2 annual terms	Susceptible pine basal area (m² ha⁻¹)	-0.019	0.0	68,813	0.7
E2 annual terms	Standing volume (m³ ha⁻¹)	+0.024	0.0	68,813	0.7
E2 annual terms	Live stems (n/ha)	-0.023	0.0	68,813	0.7
E2 annual terms	Quadratic mean diameter (cm)	-0.138*	0.0	68,813	0.7
E2 annual terms	Crown closure (%)	+0.117****	0.0	68,813	0.7
E2 annual terms	Terrain ruggedness index	+0.046	0.0	68,813	0.7

Model	Term	Estimate	SE	n	AUC
E2 annual terms	Vector ruggedness measure	+0.014	0.0	68,813	0.7
E2 annual terms	Topographic position index	-0.096***	0.0	68,813	0.7
E2 annual terms	Mid-slope position	-0.055***	0.0	68,813	0.7
E2 annual terms	Valley depth (m)	-0.064***	0.0	68,813	0.7
E2 annual terms	Profile curvature	-0.062***	0.0	68,813	0.7
E2 annual terms	Growing-season direct radiation (kWh/m²)	-0.266***	0.0	68,813	0.7
E2 annual terms	Flight-window direct radiation (kWh/m²)	+0.196***	0.0	68,813	0.7
E2 annual terms	Windward-leeward index	+0.070**	0.0	68,813	0.7
E2 annual terms	Northness	-0.097***	0.0	68,813	0.7
E2 annual terms	Attack in the same cell, previous period	+0.881***	0.0	68,813	0.7
E2 annual terms	Attack within 90 m, previous period	+1.911***	0.0	68,813	0.7
E2 annual terms	Stand basal area (m² ha⁻¹)	+0.052	0.0	68,813	0.7
E2 annual terms	Sky view factor	+0.305***	0.0	68,813	0.7
E2 annual terms	Wind shelter index	-0.023	0.0	68,813	0.7
E2 annual terms	Convergence index	-0.023***	0.0	68,813	0.7
E2 annual terms	Stand age (years)	+0.044***	0.0	68,813	0.7
E2 annual terms	Northness x Stand basal area	+0.009	0.0	68,813	0.7
E2 vertical air speed	Elevation (m)	+0.321***	0.0	68,813	0.7
E2 vertical air speed	Susceptible pine basal area (m² ha⁻¹)	-0.018	0.0	68,813	0.7
E2 vertical air speed	Standing volume (m³ ha⁻¹)	+0.033	0.0	68,813	0.7
E2 vertical air speed	Live stems (n/ha)	-0.052***	0.0	68,813	0.7
E2 vertical air speed	Quadratic mean diameter (cm)	-0.123***	0.0	68,813	0.7
E2 vertical air speed	Crown closure (%)	+0.121***	0.0	68,813	0.7
E2 vertical air speed	Terrain ruggedness index	-0.102***	0.0	68,813	0.7
E2 vertical air speed	Vector ruggedness measure	-0.003	0.0	68,813	0.7
E2 vertical air speed	Topographic position index	-0.007	0.0	68,813	0.7
E2 vertical air speed	Mid-slope position	-0.066***	0.0	68,813	0.7
E2 vertical air speed	Valley depth (m)	-0.081***	0.0	68,813	0.7
E2 vertical air speed	Profile curvature	-0.005	0.0	68,813	0.7
E2 vertical air speed	Growing-season direct radiation (kWh/m²)	-0.159***	0.0	68,813	0.7
E2 vertical air speed	Flight-window direct radiation (kWh/m²)	+0.203***	0.0	68,813	0.7
E2 vertical air speed	Windward-leeward index	+0.117***	0.0	68,813	0.7
E2 vertical air speed	Reanalysis vertical air speed, previous summer (m/s)	+0.169***	0.0	68,813	0.7
E2 vertical air speed	Attack in the same cell, previous period	+0.882***	0.0	68,813	0.7
E2 vertical air speed	Attack within 90 m, previous period	+2.051***	0.0	68,813	0.7
E2 vertical air speed	Stand basal area (m² ha⁻¹)	+0.080**	0.0	68,813	0.7

Model	Term	Estimate	SE	n	AUC
E2 vertical air speed	Reanalysis vertical air speed, previous summer x Stand basal area	-0.148*	0.0	68,813	0.7
E2 vertical air speed	Quadratic mean diameter x Reanalysis vertical air speed, previous summer	-0.032*	0.0	68,813	0.7
E2 vertical air speed	Crown closure x Reanalysis vertical air speed, previous summer	+0.031*	0.0	68,813	0.7
E2 vertical air speed	Live stems x Reanalysis vertical air speed, previous summer	+0.059****	0.0	68,813	0.7
E2 vertical air speed	Standing volume x Reanalysis vertical air speed, previous summer	+0.120****	0.0	68,813	0.7
E2 trajectory source	Elevation (m)	+0.295****	0.0	43,112	0.7
E2 trajectory source	Susceptible pine basal area (m² ha⁻¹)	-0.035*	0.0	43,112	0.7
E2 trajectory source	Standing volume (m³ ha⁻¹)	+0.048	0.0	43,112	0.7
E2 trajectory source	Live stems (n/ha)	-0.144*	0.0	43,112	0.7
E2 trajectory source	Quadratic mean diameter (cm)	-0.137*	0.0	43,112	0.7
E2 trajectory source	Crown closure (%)	+0.104****	0.0	43,112	0.7
E2 trajectory source	Terrain ruggedness index	-0.109*	0.0	43,112	0.7
E2 trajectory source	Vector ruggedness measure	+0.020	0.0	43,112	0.7
E2 trajectory source	Topographic position index	+0.016	0.0	43,112	0.7
E2 trajectory source	Mid-slope position	-0.123*	0.0	43,112	0.7
E2 trajectory source	Valley depth (m)	-0.140*	0.0	43,112	0.7
E2 trajectory source	Profile curvature	+0.003	0.0	43,112	0.7
E2 trajectory source	Growing-season direct radiation (kWh/m²)	-0.039*	0.0	43,112	0.7
E2 trajectory source	Flight-window direct radiation (kWh/m²)	+0.149****	0.0	43,112	0.7
E2 trajectory source	Windward-leeward index	+0.068**	0.0	43,112	0.7
E2 trajectory source	Trajectory endpoints over red attack, previous summer	+0.074****	0.0	43,112	0.7
E2 trajectory source	Attack in the same cell, previous period	+0.811****	0.0	43,112	0.7
E2 trajectory source	Attack within 90 m, previous period	+1.933****	0.0	43,112	0.7
E2 trajectory source	Stand basal area (m² ha⁻¹)	+0.125**	0.0	43,112	0.7
E2 trajectory source	Trajectory endpoints over red attack, previous summer x Stand basal area	-0.258*	0.0	43,112	0.7
E2 trajectory source	Quadratic mean diameter x Trajectory endpoints over red attack, previous summer	-0.015	0.0	43,112	0.7
E2 trajectory source	Crown closure x Trajectory endpoints over red attack, previous summer	-0.001	0.0	43,112	0.7
E2 trajectory source	Live stems x Trajectory endpoints over red attack, previous summer	+0.029	0.0	43,112	0.7
E2 trajectory source	Standing volume x Trajectory endpoints over red attack, previous summer	+0.143****	0.0	43,112	0.7

Table S5

Every sixteen-day period used, with its start date, the cells the imagery saw and the share classed as attacked. Periods in which the imagery saw less than a tenth of the perimeter were dropped.

Year	Period	Start	Cells seen	Attacked (%)
2006	1	2006-05-01	48,329	5.6
2006	2	2006-05-17	16,703	8.4
2006	3	2006-06-02	9,426	26.0
2006	4	2006-06-18	18,568	11.2
2006	5	2006-07-04	55,107	17.3
2006	6	2006-07-20	41,218	9.4
2006	7	2006-08-05	46,263	5.7
2006	8	2006-08-21	60,102	12.4
2007	1	2007-05-01	50,919	9.9
2007	5	2007-07-04	56,511	13.3
2007	6	2007-07-20	41,218	18.3
2007	7	2007-08-05	61,246	11.3
2007	8	2007-08-21	35,655	12.6
2008	3	2008-06-02	9,932	57.4
2008	5	2008-07-04	15,069	9.8
2008	6	2008-07-20	6,293	7.8
2008	7	2008-08-05	53,059	8.6
2008	8	2008-08-21	9,279	9.5
2009	1	2009-05-01	17,062	10.9
2009	2	2009-05-17	37,124	15.2
2009	4	2009-06-18	18,566	12.3
2009	5	2009-07-04	56,511	16.2
2009	6	2009-07-20	35,156	10.5
2009	7	2009-08-05	61,246	11.0
2009	8	2009-08-21	35,147	39.0
2010	1	2010-05-01	13,007	10.2
2010	4	2010-06-18	9,106	37.4
2010	5	2010-07-04	17,158	23.1
2010	7	2010-08-05	59,387	11.5
2010	8	2010-08-21	8,618	26.3
2011	2	2011-05-17	14,102	5.7
2011	4	2011-06-18	8,606	18.9
2011	5	2011-07-04	53,179	23.7
2011	6	2011-07-20	32,384	15.4
2011	7	2011-08-05	61,246	13.8
2011	8	2011-08-21	60,102	17.8

Year	Period	Start	Cells seen	Attacked (%)
2013	1	2013-05-01	29,392	8.5
2013	4	2013-06-18	12,760	33.1
2013	5	2013-07-04	45,303	22.6
2013	6	2013-07-20	39,873	25.8
2013	7	2013-08-05	43,605	38.8
2013	8	2013-08-21	59,452	34.2
2014	1	2014-05-01	53,035	11.1
2014	2	2014-05-17	9,048	7.4
2014	4	2014-06-18	18,286	29.2
2014	5	2014-07-04	44,890	20.2
2014	7	2014-08-05	56,249	22.9

Table S6

Accuracy of the three red-stage classifiers over repeated random splits that held out a quarter of the field plots and of the 100 m blocks of undisturbed pixels, as mean and standard deviation with the 2.5th and 97.5th percentiles of the splits. The model with the highest mean kappa set the cut used on every map.

Model	Accuracy	Accuracy 95%	Kappa	Kappa 95%	Sensitivity	Specificity	Chosen
gradient boosting	0.912 ± 0.049	0.820 to 1.000	0.810 ± 0.106	0.614 to 1.000	0.920	0.899	
random forest	0.898 ± 0.046	0.810 to 0.976	0.778 ± 0.099	0.585 to 0.948	0.925	0.854	
radial support vector machine	0.934 ± 0.037	0.859 to 1.000	0.859 ± 0.081	0.682 to 1.000	0.913	0.968	yes

Table S7

Dependence terms compared on AIC over 63,968 cell-years with a previous map. Own state was attack in the cell the year before, and the share was the proportion of cells attacked within each radius.

Terms	k	AIC	Delta AIC
intercept	1	88,680	30,456.0
intercept + own state	2	62,907	4,682.6
intercept + own state + share within 42 m	3	59,502	1,277.9

Terms	k	AIC	Delta AIC
intercept + own state + share within 90 m	3	58,264	39.4
intercept + own state + share within 150 m	3	58,224	0.0
intercept + own state + share within 210 m	3	58,380	155.3
intercept + own state + share within 510 m	3	59,448	1,223.9
intercept + own state + share within 1050 m	3	60,370	2,145.2

Table S8

Red-stage attack by quadratic mean diameter class on the balanced annual sample, with Wilson 95 per cent intervals and the mean susceptible pine and total basal area of each class. The 25 cm boundary was the source-sink threshold of the species' bionomics. The sample held equal numbers of attacked and unattacked cells in each year, so the percentages compared classes and were not landscape rates.

QMD class (cm)	n	Attacked	Attacked (%)	95% CI (%)	Pine BA (m² ha⁻¹)	BA (m² ha⁻¹)
<15	672	203	30.2	26.9 to 33.8	1.0	3.5
15-20	7,678	3,380	44.0	42.9 to 45.1	10.9	23.7
20-25	20,445	9,842	48.1	47.5 to 48.8	8.4	28.9
25-30	28,046	15,807	56.4	55.8 to 56.9	10.1	39.7
30-40	12,526	5,836	46.6	45.7 to 47.5	5.5	42.2
>40	2,633	932	35.4	33.6 to 37.2	0.4	45.4

Table S9

The annual models, each adding one mechanism, over the nine years without and the eight years with a previous map without and with the dependence terms. M0 held host size, shading and landform, M1 added basal area, M2 terrain exposure, shape and flight-window radiation, and M3 the interactions of basal area with wind shelter, July wind and radiation. Cross-validation (CV) used ten folds of 1 km blocks, and the blind test fitted the years to 2012 and predicted 2013 and 2014. The RMSE ratio was the test-fold over the training-fold root mean square error, and Theil's U bias the share of the mean squared error due to the difference of the means.

Years	Dependence	Mod	n	AIC	AUC full	AUC	AUC blind	RMSE full	RMSE CV	RMSE blind	RMSE ratio	MAE	Theil's U bias CV	Theil's U bias blind
2006 to 2014	no	M0	720	774	0.804	0.795	0.875	0.423	0.427	0.385	1.003	0.36	0e+00	0.0004
2006 to 2014	no	M1	720	772	0.807	0.800	0.879	0.422	0.425	0.383	1.000	0.36	1e-04	0.0001
2006 to 2014	no	M2	720	771	0.808	0.800	0.876	0.421	0.425	0.385	1.005	0.36	1e-04	0.0001
2006 to 2014	no	M3	720	769	0.809	0.801	0.875	0.421	0.425	0.386	1.008	0.35	1e-04	0.0026
2007 to 2014	no	M0	639	684	0.809	0.803	0.873	0.421	0.425	0.386	1.020	0.36	2e-04	0.0003
2007 to 2014	yes	M0	639	536	0.886	0.888	0.942	0.367	0.307	0.310	1.000	0.27	0e+00	0.0049
2007 to 2014	no	M1	639	680	0.813	0.804	0.878	0.418	0.422	0.384	1.007	0.35	0e+00	0.0001
2007 to 2014	yes	M1	639	534	0.887	0.888	0.943	0.367	0.307	0.309	1.009	0.27	0e+00	0.0045
2007 to 2014	no	M2	639	678	0.814	0.804	0.878	0.418	0.423	0.387	1.016	0.35	2e-04	0.0000
2007 to 2014	yes	M2	639	534	0.887	0.888	0.940	0.367	0.307	0.311	1.010	0.27	0e+00	0.0039
2007 to 2014	no	M3	639	677	0.814	0.804	0.878	0.418	0.423	0.387	1.019	0.35	1e-04	0.0016
2007 to 2014	yes	M3	639	530	0.890	0.888	0.942	0.365	0.307	0.313	0.998	0.26	0e+00	0.0264

51 **Table S10**

52 Agreement of the red-stage classifier and the terrain wind with independent measurements. The
53 classifier was scored over 100 held-out splits, and wind speed over the flight hours of June to
54 August 2005 to 2014 at four valley stations, corrected as in the models. Bias was the predicted
55 less the measured mean speed in km/h, RMSE the root mean square error, Theil's U bias the
56 share of the mean squared error due to that difference, and direction error the mean absolute
57 difference in degrees.

Variable	Measured against	n	Bias	RMS	Theil's U bias	Agreement	Direction error
Red-stage attack, classifier	Field plots and undisturbed points	18	NA	NA	NA	kappa 0.859, accuracy 0.934	NA
Wind speed (km/h), regional box, 500 m mesh	Creston station	4,23	-3	5.7	0.354	r 0.32	61.7
Wind speed (km/h), trench, 1 km mesh	Castlegar station	3,38	-8	10	0.710	r 0.34	66.1
Wind speed (km/h), trench, 1 km mesh	Creston station	4,23	-3	5.9	0.418	r 0.33	60.4
Wind speed (km/h), trench, 1 km mesh	Nelson station	2,22	-8	9.4	0.718	r 0.33	82.6
Wind speed (km/h), trench, 1 km mesh	Warfield station	4,11	-4	5.9	0.564	r 0.33	81.3

58 **Table S11**

59 Estimates of the terms tested beside M3 in the annual models, each in the specification named,
60 fitted with the dependence terms over the eight years with a previous map. β is the change in log-

odds per standard deviation, with its standard error (SE). Specifications marked with year held the summer as a factor. * p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001, **** p ≤ 0.0001.

Specification	Term	β	SE	p
lagged wind	July wind, previous summer (km/h)	-0.092***	0.011	< 0.00
lagged wind	June wind, previous summer (km/h)	+0.008	0.010	0.435
lagged wind, diameter and closure	Crown closure (%)	+0.141***	0.016	< 0.00
lagged wind, diameter and closure	July wind, previous summer (km/h)	-0.089***	0.011	< 0.00
lagged wind, diameter and closure	June wind, previous summer (km/h)	+0.007	0.011	0.512
measured heat	Surface temperature, previous flight window (°C)	+0.218***	0.015	< 0.00
measured heat	July wind, previous summer (km/h)	-0.075***	0.011	< 0.00
measured heat	June wind, previous summer (km/h)	-0.020	0.011	0.059
lidar	July wind, previous summer (km/h)	-0.092***	0.011	< 0.00
lidar	June wind, previous summer (km/h)	+0.009	0.010	0.410
lidar	Ground roughness within the cell at 1 m (m)	-0.049	0.026	0.059
lidar	Share of the cell facing north at 5 m	+0.059***	0.016	< 0.00
lidar	Spread of position at 55 m (m)	+0.010	0.014	0.502
modelled wind	Modelled wind at 90 m, previous summer (km/h)	+0.077***	0.015	< 0.00
upslope share	Upslope share of flight hours, previous summer	-0.003	0.013	0.804
shelter at the modelled wind	Shelter at the modelled wind direction, previous summer	-0.051***	0.012	< 0.00
modelled wind, diameter and closure	Crown closure (%)	+0.138***	0.016	< 0.00
modelled wind, diameter and closure	Modelled wind at 90 m, previous summer (km/h)	+0.067***	0.015	< 0.00
modelled wind, 30 m	Modelled wind at 30 m, previous summer (km/h)	+0.012	0.014	0.387
modelled wind, 10 m	Modelled wind at 10 m, previous summer (km/h)	+0.077***	0.013	< 0.00
lift	Convective lift, previous summer (m/s)	+0.180***	0.018	< 0.00
lift with year	Convective lift, previous summer (m/s)	-0.390***	0.040	< 0.00
lift, 10 m speed and upslope	Convective lift, previous summer (m/s)	-0.402***	0.040	< 0.00
lift, 10 m speed and upslope	Modelled wind at 10 m, previous summer (km/h)	+0.031*	0.014	0.033
lift, 10 m speed and upslope	Upslope share of flight hours, previous summer	+0.018	0.014	0.177
slope updraft	Slope updraft of the terrain wind, previous summer (m/s)	-0.029*	0.015	0.050
slope updraft at ridges	Slope updraft of the terrain wind, previous summer (m/s)	-0.015	0.015	0.314
cloud-adjusted sun	Cloud-adjusted flight-hour sun, previous summer (kWh/m²)	+0.063**	0.021	0.003
slope temperature	Slope temperature departure, previous summer (°C)	+0.129**	0.041	0.001
cloud-adjusted sun with year	Cloud-adjusted flight-hour sun, previous summer (kWh/m²)	+0.037	0.022	0.089
slope temperature with year	Slope temperature departure, previous summer (°C)	-0.462***	0.051	< 0.00
slope temperature without sun	Slope temperature departure, previous summer (°C)	+0.018	0.021	0.388
valley and ridge flats	Valley bottom flatness	-0.059***	0.013	< 0.00

Specification	Term	β	SE	p
valley and ridge flats	Ridge top flatness	+0.071***	0.016	< 0.001
cold-air pooling	Cold-air pooling (°C)	-0.083*	0.033	0.012
canopy height	Canopy height, 2017 lidar (m)	-0.088**	0.031	0.004
canopy height	Canopy height variation, 2017 lidar (m)	+0.171***	0.015	< 0.001
canopy height	Canopy over 5 m, 2017 lidar	-0.389***	0.032	< 0.001
winter cold	Lowest winter minimum before the flight (°C)	-0.210***	0.013	< 0.001
winter cold with year	Lowest winter minimum before the flight (°C)	-0.102	0.056	0.068
all new terrain	Valley bottom flatness	-0.059***	0.013	< 0.001
all new terrain	Ridge top flatness	+0.070***	0.016	< 0.001
all new terrain	Cold-air pooling (°C)	-0.033	0.034	0.331
all new terrain	Canopy height, 2017 lidar (m)	-0.124***	0.031	< 0.001
all new terrain	Canopy height variation, 2017 lidar (m)	+0.171***	0.015	< 0.001
all new terrain	Canopy over 5 m, 2017 lidar	-0.344***	0.033	< 0.001
all new terrain	Lowest winter minimum before the flight (°C)	-0.190***	0.013	< 0.001
sources	Surviving sources within 150 m, all directions	-0.662***	0.089	< 0.001
sources	Surviving sources 150 to 1,050 m, all directions	-0.304***	0.019	< 0.001
upwind and downwind sources	Surviving sources within 150 m, upwind	-0.316***	0.048	< 0.001
upwind and downwind sources	Surviving sources within 150 m, downwind	-0.382***	0.050	< 0.001
upwind and downwind sources	Surviving sources 150 to 1,050 m, upwind	-0.204***	0.018	< 0.001
upwind and downwind sources	Surviving sources 150 to 1,050 m, downwind	-0.124***	0.019	< 0.001
sources, unweighted	Red crowns within 150 m, all directions	+0.307	0.231	0.183
sources, unweighted	Red crowns 150 to 1,050 m, all directions	-0.312***	0.019	< 0.001
upwind and downwind sources, unweighted	Red crowns within 150 m, upwind	-0.090	0.086	0.296
upwind and downwind sources, unweighted	Red crowns within 150 m, downwind	-0.140	0.088	0.113
upwind and downwind sources, unweighted	Red crowns 150 to 1,050 m, upwind	-0.219***	0.018	< 0.001
upwind and downwind sources, unweighted	Red crowns 150 to 1,050 m, downwind	-0.119***	0.019	< 0.001
vertical air speed	Reanalysis vertical air speed, previous summer (m/s)	+0.040***	0.011	< 0.001
trajectory source	Trajectory endpoints over red attack, previous summer	+0.091***	0.012	< 0.001

Table S12

Estimates of the terms tested beside E2 in the sixteen-day models, each in the specification named, fitted with the previous period's attack in and around the cell. β is the change in log-odds per standard deviation, with its standard error (SE). Specifications marked with season held the period, or the period of each summer, as a factor. Stand basal area, entered as a control in each of these specifications, is omitted. * p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001, **** p ≤ 0.0001.

Specification	Term	β	SE	p

Table S13

Screening of every candidate variable of the annual models on the years with a previous map. β is the slope of the variable added alone to the two dependence terms, Deviance the deviance it explained with the p-value of the likelihood ratio test, VIF its variance inflation against the main effects of M3, Elastic net its coefficient in an elastic net that entered every candidate together with the dependence terms unpenalised, and W the Shapiro-Wilk statistic on 5,000 cell-years. In M3 marks the variables of the final model. * p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001, **** p ≤ 0.0001.

Variable	β	Deviance	p	VIF	Elastic net	W	In M3
Susceptible pine basal area (m² ha⁻¹)	+0.3329	886.97	< 0.001****	1.46	+0.0986	0.804	yes
Stand age (years)	+0.0923	85.93	< 0.001****	1.50	+0.0000	0.893	yes
Quadratic mean diameter (cm)	-0.014	2.19	0.139	1.95	-0.0170	0.934	yes
Sky view factor	+0.4119	1,362.54	< 0.001****	2.49	+0.1284	0.984	yes
Northness	+0.0435	18.26	< 0.001****	1.58	+0.0000	0.886	yes
Elevation (m)	+0.7528	3,987.76	< 0.001****	1.48	+0.5899	0.906	yes
Stand basal area (m² ha⁻¹)	+0.1961	371.10	< 0.001****	1.92	+0.0133	0.915	yes
Wind shelter index	+0.0650	43.04	< 0.001****	2.20	+0.0000	0.597	yes
Topographic position index	+0.1393	201.48	< 0.001****	1.74	+0.0000	0.977	yes
Convergence index	+0.0668	41.60	< 0.001****	1.29	+0.0000	0.974	yes
Profile curvature	+0.0457	21.34	< 0.001****	1.18	+0.0000	0.992	yes
Flight-window direct radiation (kWh/m²)	+0.2147	469.53	< 0.001****	2.86	+0.0000	0.993	yes
July mean wind (km/h)	+0.2349	501.58	< 0.001****	1.01	+0.1862	0.916	yes
June mean wind (km/h)	+0.0526	17.97	< 0.001****	1.01	+0.0000	0.948	yes
July wind, previous summer (km/h)	-0.107	117.15	< 0.001****	1.40	-0.0276	0.939	
June wind, previous summer (km/h)	+0.0020	0.04	0.838	1.10	-0.0521	0.948	
Surface temperature, previous flight window (°C)	-0.055	21.42	< 0.001****	1.63	+0.0000	0.970	
Ground roughness within the cell at 1 m (m)	-0.398	1,314.02	< 0.001****	4.53	-0.0648	0.981	
Share of the cell facing north at 5 m	-0.009	0.88	0.349	2.20	+0.0116	0.624	
Spread of position at 55 m (m)	-0.288	696.43	< 0.001****	1.29	+0.0000	0.822	
Crown closure (%)	+0.1823	325.49	< 0.001****	2.06	+0.1309	0.799	
Modelled wind at 90 m, previous summer (km/h)	+0.1808	357.43	< 0.001****	2.02	+0.0000	0.919	
Upslope share of flight hours, previous summer	-0.046	21.70	< 0.001****	1.47	+0.0000	0.916	
Shelter at the modelled wind direction, previous summer	-0.151	220.28	< 0.001****	1.18	-0.0342	0.728	
Modelled wind at 30 m, previous summer (km/h)	+0.1178	148.21	< 0.001****	1.81	+0.0000	0.941	
Modelled wind at 10 m, previous summer (km/h)	+0.1484	235.31	< 0.001****	1.70	+0.0222	0.943	

Variable	β	Deviance	p	VIF	Elastic net	W	In M3
Convective lift, previous summer (m/s)	+0.2220	530.01	< 0.001****	4.87	+0.0151	0.999	
Slope updraft of the terrain wind, previous summer (m/s)	-0.090	83.20	< 0.001****	1.94	-0.0028	0.907	
Cloud-adjusted flight-hour sun, previous summer (kWh/m²)	+0.1904	378.54	< 0.001****	34.67	+0.0000	0.996	
Slope temperature departure, previous summer (°C)	+0.2407	587.93	< 0.001****	19.23	+0.0000	0.997	
Trajectory endpoints over red attack, previous summer	+0.0982	79.14	< 0.001****	1.31	+0.0749	0.832	

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Methods S1

The wind component of the MicroMet model of Liston & Elder (2006) is seven equations, implemented directly from the source paper with its equation numbers given below.

Terrain slope β and slope azimuth ξ came from the elevation model, equations 12 and 13 of Liston & Elder (2006). Curvature Ω_c is a cell's elevation minus the mean of the two opposite cells one curvature length scale away, taken on four direction lines and averaged, their equation 14, and the length scale was estimated as the first lag at which elevation autocorrelation fell below 0.5, which was 600 m here. Assigning θ to wind bearing, slope in the wind direction was computed as follows

$$\Omega_s = \beta\cos(\theta - \xi) \qquad (1)$$

with Ω_s and Ω_c each scaled to $[-0.5, 0.5]$. The terrain weighting factor applied to the observed speed W was derived using

$$W_w = 1 + \gamma_s\Omega_s + \gamma_c\Omega_c, \qquad \gamma_s = \gamma_c = 0.5 \qquad (2)$$

This provided the basis for calculation of the terrain-modified speed using

$$W_t = W_w W \qquad (3)$$

and a diversion of wind direction computed as

$$\theta_d = -1/2\ \Omega_s\sin\left[2(\xi - \theta)\right]. \qquad (4)$$

By *Equation 2*, W_w depends on direction and not on speed, so it was computed once for each of 16 wind direction sectors of 22.5 degrees, the sixteen points of the compass, and each hourly observation was multiplied by the surface for its own sector, so that nothing was averaged before the terrain acted on it, and over all sectors W_w ran from 0.66 to 1.32.

Methods S2

The wind and heat terms of section 2.3 of the main text rested on these sources. Convective lift was the velocity scale of Deardorff (1970) from the ERA5-Land sensible heat flux (Muñoz-Sabater et al., 2021) and the ERA5 boundary layer height (Hersbach et al., 2020). Back-trajectories were computed with HYSPLIT (Stein et al., 2015) on the North American Regional Reanalysis

(Mesinger et al., 2006). Sun on each slope under each hour's cloud came from r.sun in GRASS GIS (Šúri & Hofierka, 2004), and afternoon air temperature from microclima (Maclean et al., 2019).

Three groups of terms that the earlier models lacked were added beside M3 and E2. Valley bottoms and ridge tops were measured by the multiresolution indices of valley bottom and ridge top flatness (Gallant & Dowling, 2003), computed in SAGA GIS on the 30 m elevation model. Canopy height was the 2017 lidar surface model less its bare-earth model at 1 m, summarised to each 30 m cell as its mean, its standard deviation and the share of the cell above 5 m, and because the lidar was flown after the outbreak it measured the canopy that remained. Winter cold was the lowest daily minimum temperature and the number of days at or below -30 °C in the winter, October to April, that ended before the flight behind each map, from the ERA5-Land daily minimum (Muñoz-Sabater et al., 2021) moved to each cell's elevation at the lapse rate the reanalysis cells gave on that day, with the cooling that cold air draining into the cell would add, from the basins, flow accumulation and drainage model of microclima (Maclean et al., 2019), on every day whose night met that package's conditions for cold-air drainage at the site. Over the ten winters the temperature rose with elevation, an inversion, on 0.01 of days, the conditions for drainage held on 0.32 of nights, drainage cooled the coldest cells by up to 9.3 °C, and the lowest winter minimum across the study area averaged -26.8 °C and fell no lower than -35.4 °C, so that the days at or below -30 °C, 2.0 on average in the coldest winter, did not enter the models. The sixteen-day model was also refitted with the terms of the annual model it lacked, northness, sky view, the wind shelter index, convergence and stand age, and with the station pressure of each period, the flight-hour departure from each station's May to September mean and its three-hour change, for the period and for the same period one year earlier.

The source of each year's attack was the red crowns of the year before, whose brood had come through the winter in those trees. Winter survival was computed for every 30 m cell with the cold tolerance model of Régnière & Bentz (2007), in the R code of Goodsman (2024), from ERA5-Land daily temperature (Muñoz-Sabater et al., 2021) moved to each cell's elevation and converted to phloem temperature (Bolstad et al., 1997). Around each cell the share of cells attacked the previous year, weighted by survival, was computed within 150 m and from 150 to 1,050 m in eight compass sectors. Each sector was weighted by the share of flight hours the wind blew from it at the Akokli Creek station, and the upwind and downwind shares were compared by a Wald test of the difference between their coefficients.

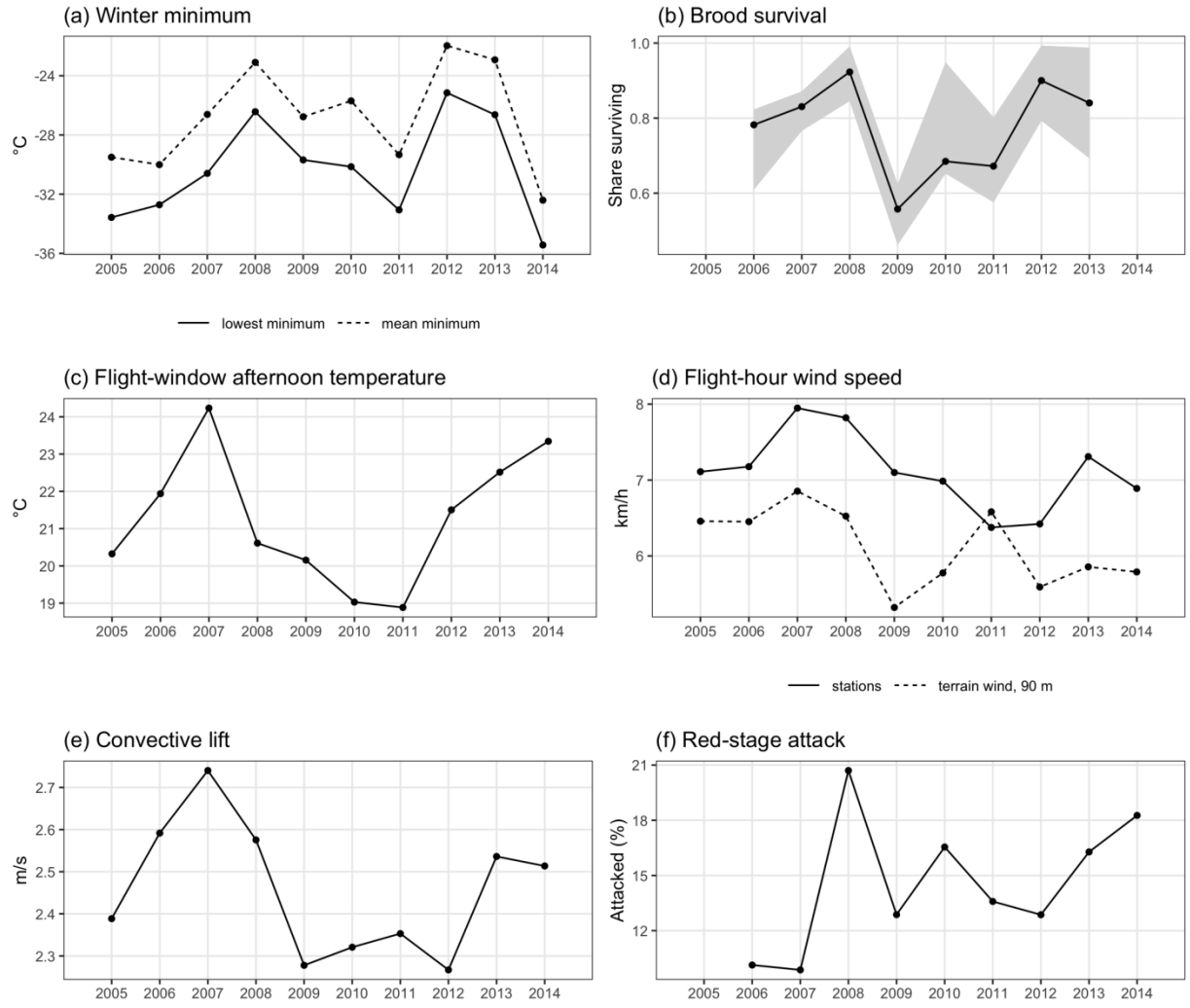
Results S1

With the previous year's attack and the dependence terms in the annual model, valley bottom flatness entered at -0.059 ($p < 0.001$) and ridge top flatness at +0.071 ($p < 0.001$), cold-air pooling at -0.083 ($p = 0.012$), mean canopy height at -0.088 ($p = 0.004$), its variation at +0.171 ($p < 0.001$) and the share above 5 m at -0.389 ($p < 0.001$). The lowest minimum of the winter before the flight entered at -0.210 ($p < 0.001$), and with the summer held at -0.102 ($p = 0.068$). In the sixteen-day models with the previous period's attack, valley bottom flatness entered at -0.020 ($p = 0.179$), ridge top flatness at -0.014 ($p = 0.174$), cold-air pooling at -0.045 ($p = 0.058$), mean canopy height at +0.509 ($p < 0.001$) and the winter minimum at -0.111 ($p < 0.001$). With the terms of the annual model added to the sixteen-day model, northness entered at -0.097 ($p < 0.001$), sky view at +0.305 ($p < 0.001$), the wind shelter index at -0.023 ($p = 0.233$), convergence at -0.023 ($p = 0.030$) and stand age at +0.044 ($p < 0.001$). The pressure departure of the previous summer's period entered the lagged model with dependence terms at +0.011 ($p = 0.318$) and its three-hour change at +0.149 ($p < 0.001$).

In the flight window the late-morning surface temperature of north-facing ground was 21.4 °C against 24.8 °C on south-facing ground, averaged over the ten summers and 108 scenes, and it correlated -0.42 with northness and 0.02 with modelled flight-window radiation (Figure S14). With the measured temperature of the previous summer in place of modelled radiation, northness entered at +0.258 ($p < 0.001$) against +0.102 ($p < 0.001$) beside radiation, and temperature itself at +0.560 ($p < 0.001$). With the three terms from the 1 m terrain added to the lagged model, ground roughness within the cell entered at -0.086 ($p < 0.001$), the share of the cell facing north at +0.028 ($p = 0.036$) and the spread of crown-scale position at +0.007 ($p = 0.584$), and the AIC moved from 77087 to 77074.

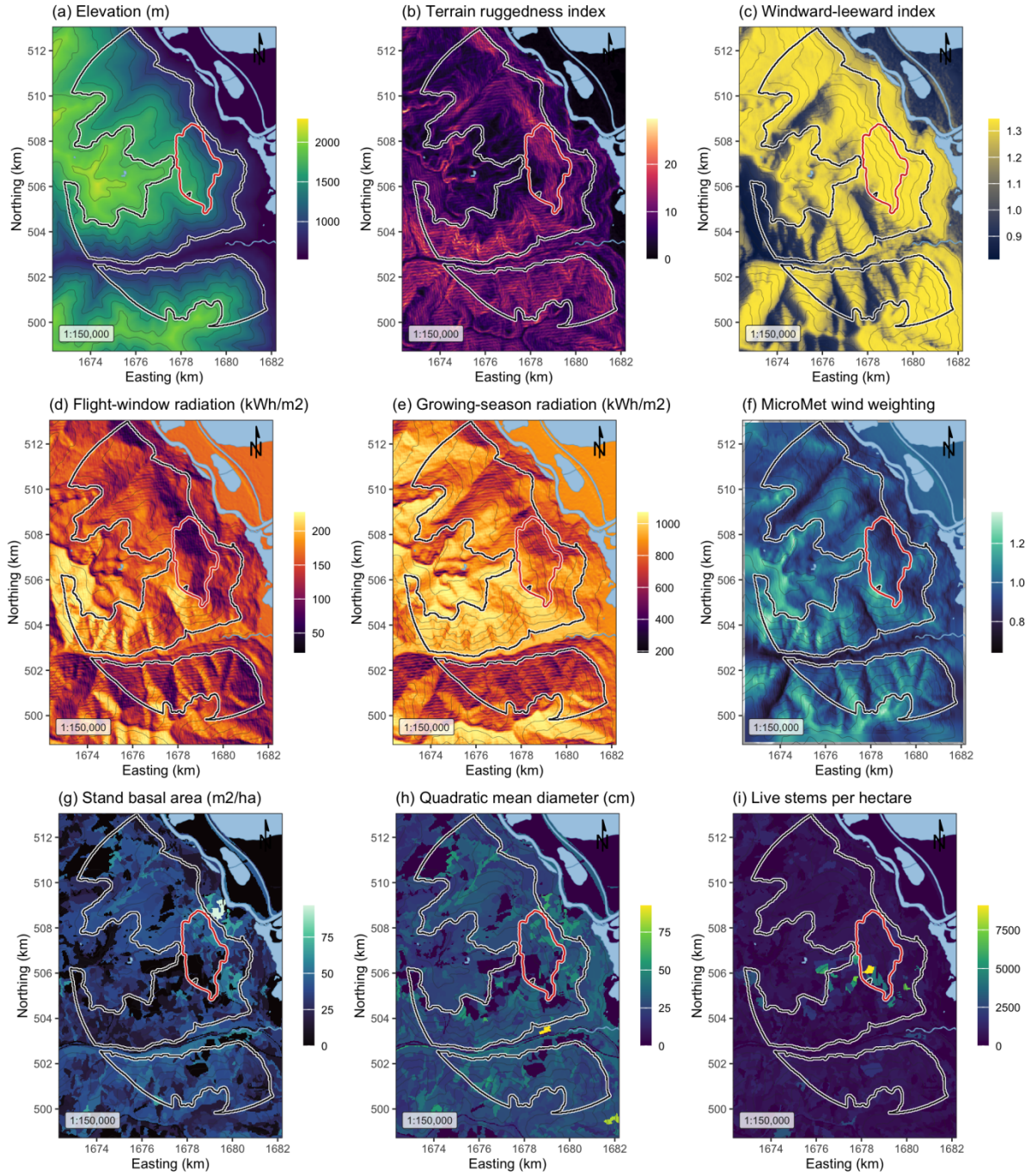
Attack upwind of a cell did not raise its attack more than attack downwind. In the survival-weighted model the two shares within 150 m did not differ ($p = 0.453$), from 150 to 1,050 m the upwind share entered at -0.235 against -0.151 downwind, a difference of -0.084 [Wald $z = -2.67$, $p = 0.007$], and splitting the shares by wind direction raised the AIC from 52,907 to 52,950. Every specification is listed with its estimates in Tables S11 and S12.

Figure S1



Weather of the study years at the site. (a) The mean and the lowest daily minimum temperature of each October to April winter, ERA5-Land moved to each cell's elevation, by the year the winter ended. (b) Modelled overwinter survival of the brood, mean with the range across cells. (c) Afternoon air temperature of the flight window, 1 July to 15 August, from microclima. (d) Flight-hour wind speed, the station mean and the terrain wind at 90 m, corrected. (e) Convective lift of the flight window. (f) The share of the perimeter attacked on each annual map, which records the flight of the summer before.

174 **Figure S2**

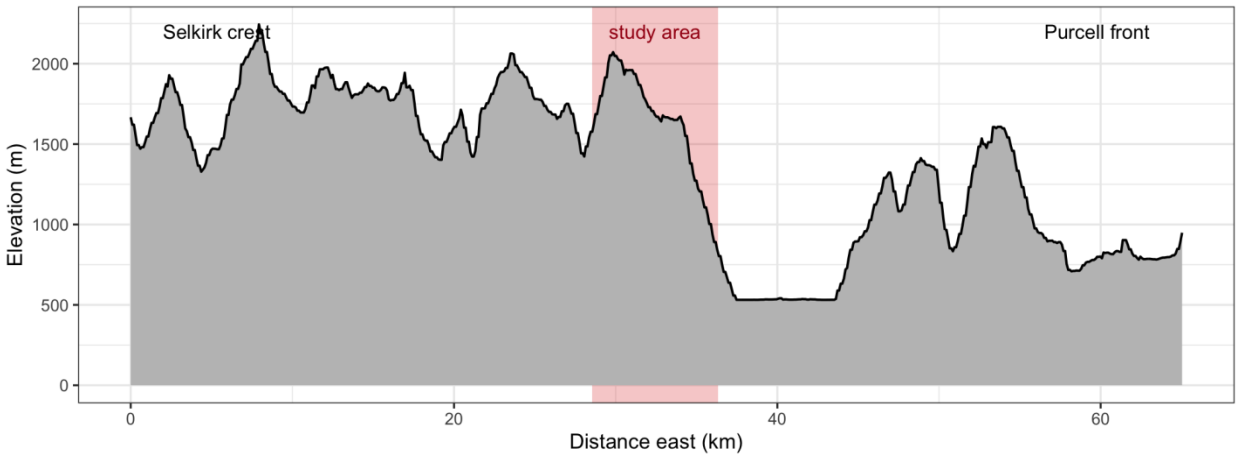


175

176 Landscape, terrain and stand surfaces across the study area, all EPSG:3153 at 30 m over Esri
177 World Shaded Relief. (a) elevation; (b) terrain ruggedness index; (c) windward-leeward index at
178 the prevailing bearing; (d) flight-window direct radiation; (e) growing-season direct radiation; (f)
179 the MicroMet wind weighting factor at the prevailing bearing; (g) stand basal area; (h) quadratic

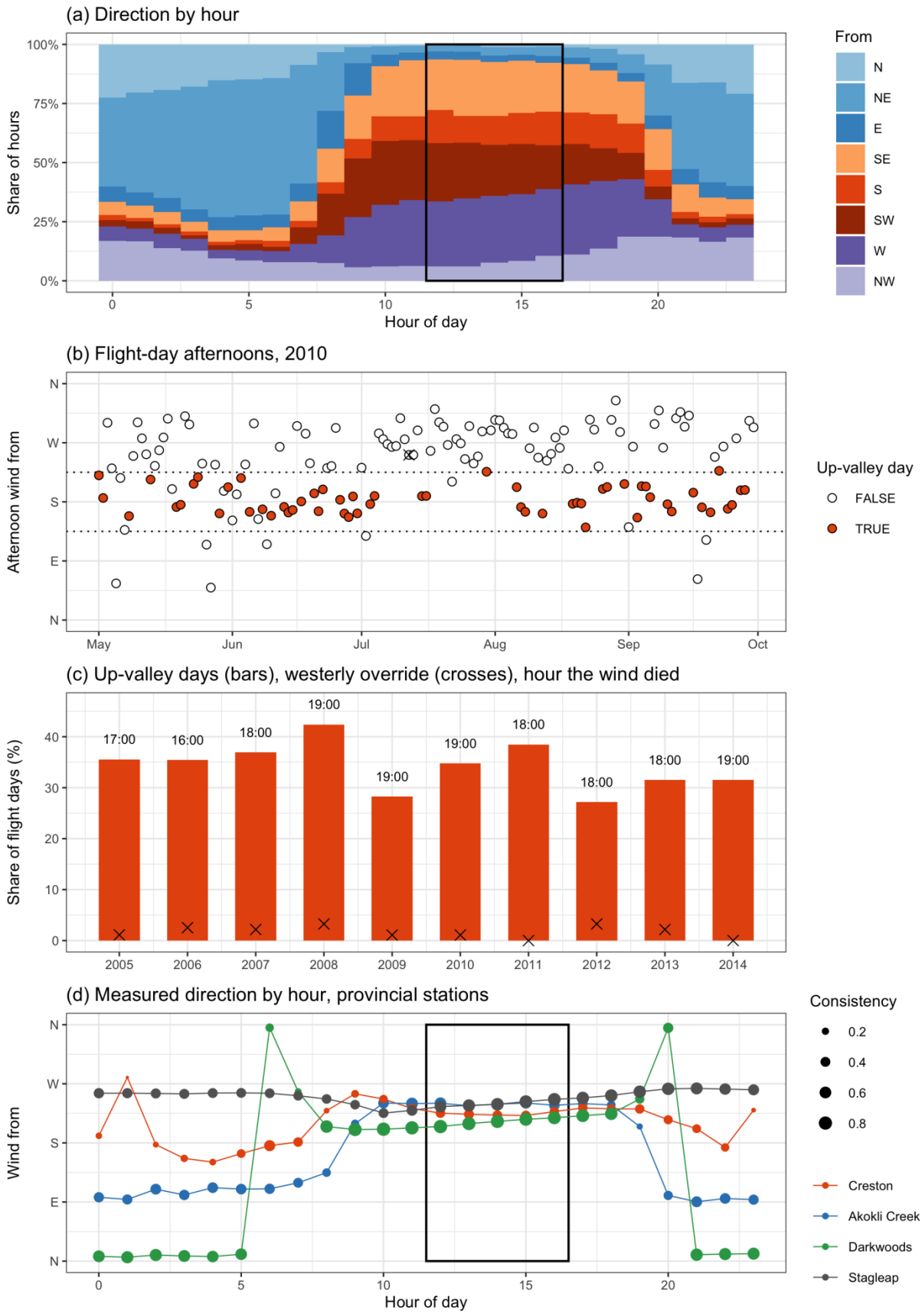
mean diameter; (i) live stems per hectare. The white outline marked the study perimeter and the red outline the 2015 Mt Midgeley burn, with contours at 200 m.

Figure S3



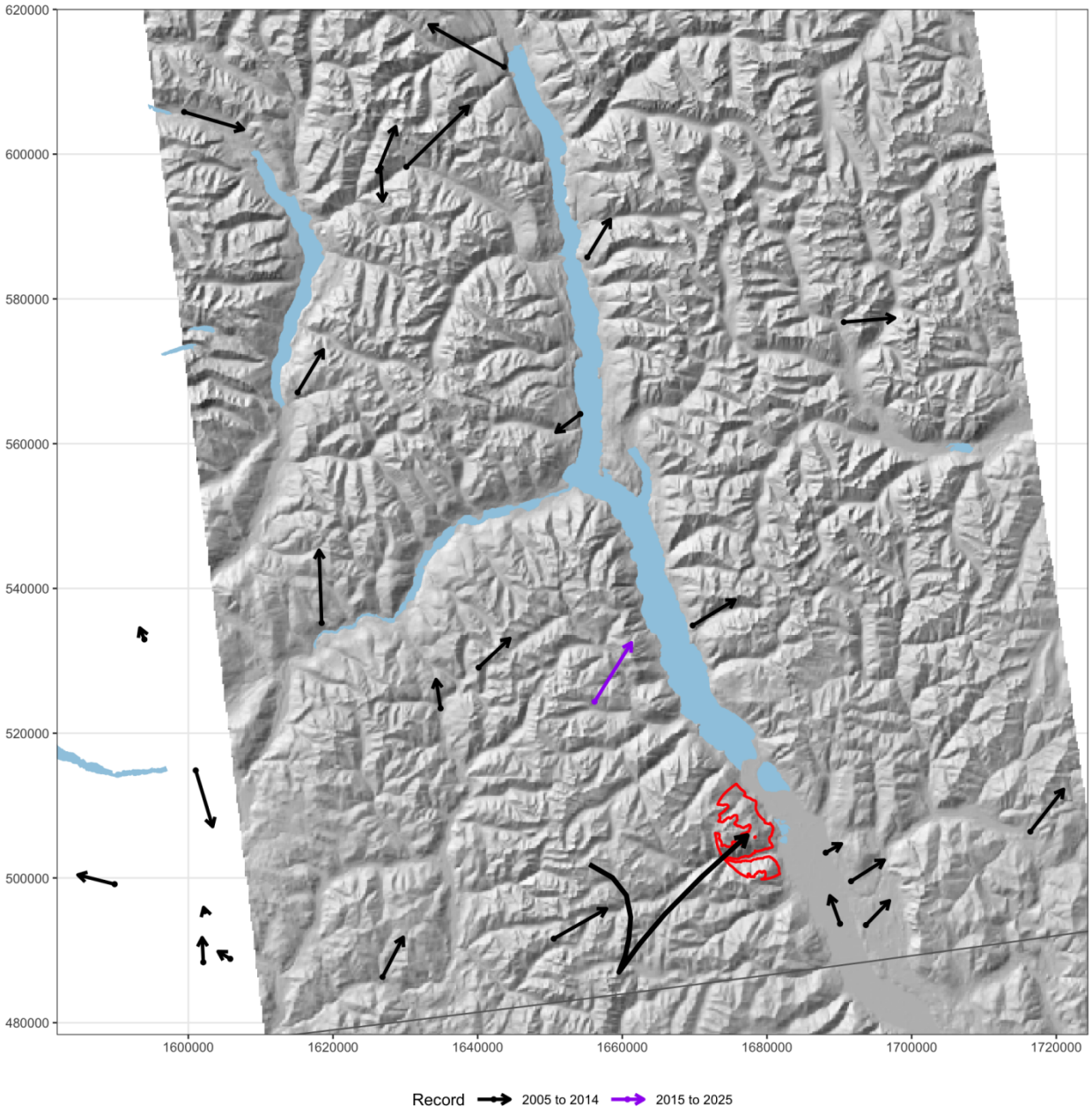
Relief from the Selkirk crest east across the study area to the Purcell front, along the latitude of the centre of the perimeter, from the Copernicus 30 m elevation model sampled every 100 m. The red band is the span of the study perimeter. The level floor at 531 m is the Creston flats and the south arm of Kootenay Lake, which the ground reaches within 7.6 km of the crest of the study area at 2,072 m.

189 **Figure S4**


(a) Direction by hour
Share of hours
100%
75%
50%
25%
0%
0
5
10
15
20
Hour of day
From
N
NE
E
SE
S
SW
W
NW
(b) Flight-day afternoons, 2010
Afternoon wind from
N
W
S
E
N
May
Jun
Jul
Aug
Sep
Oct
Up-valley day
FALSE
TRUE
(c) Up-valley days (bars), westerly override (crosses), hour the wind died
Share of flight days (%)
40
30
20
10
0
17:00
16:00
18:00
19:00
19:00
19:00
18:00
18:00
18:00
19:00
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
(d) Measured direction by hour, provincial stations
Wind from
N
W
S
E
N
0
5
10
15
20
Hour of day
Consistency
0.2
0.4
0.6
0.8
Creston
Akokli Creek
Darkwoods
Stagleap

The valley wind at the Creston station, May to September 2005 to 2014. (a) The share of hours in each eight-point sector by hour of the day, where S and SW are the up-valley sectors. (b) The afternoon wind of each flight day of 2010 as its direction, with up-valley days filled and days on which a westerly above 15 km/h overrode the valley wind marked with a cross. (c) By year over June to August, the share of flight days that were up-valley days and the share overridden by a westerly, with the median hour at which the afternoon wind fell below 5 km/h written above each year. (d) The measured direction by hour of day over June to August 2005 to 2014 at three stations of the provincial networks, Creston on the valley side, Akokli Creek on the east shore of the south arm of Kootenay Lake, the Stagleap ridge, and Darkwoods on the study area's high ground, whose record begins in October 2014 and is shown for June to August 2015 to 2025, the vector mean of the hours above 5 km/h with point size the consistency of direction; the box marks the flight hours.

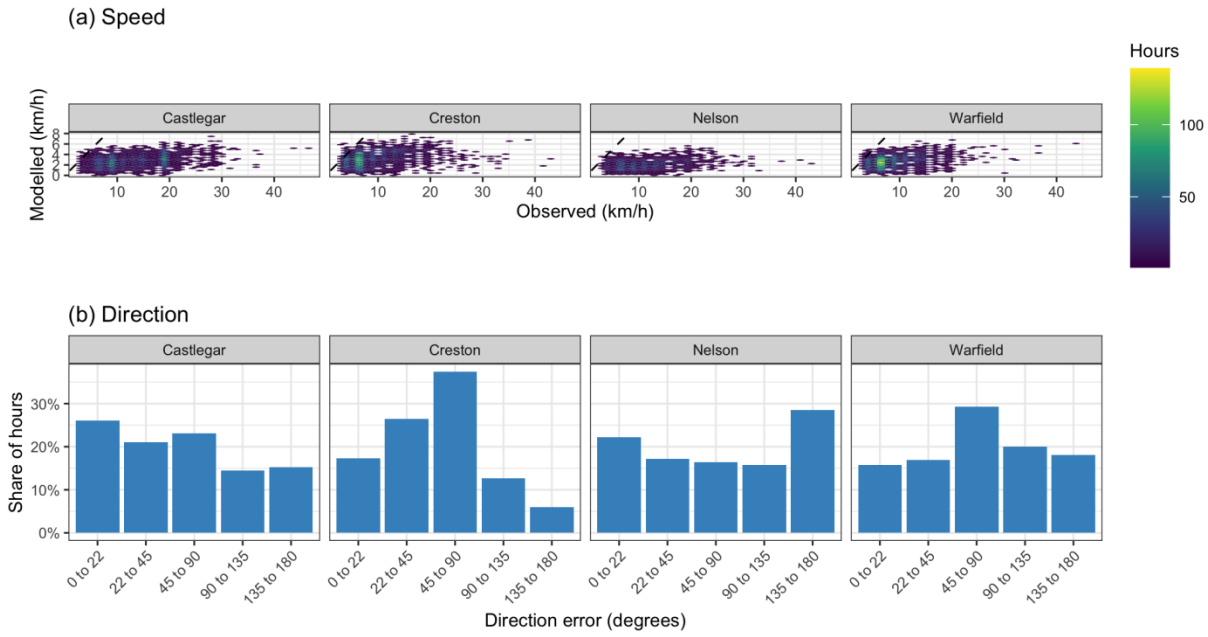
Figure S5



The measured afternoon wind along the Purcell Trench and the Creston valley in the flight hours, 12:00 to 16:59 of June to August. Each arrow points where the wind went at one hourly station of the provincial networks, its direction the mean of the hourly directions weighted by the windy

208 hours, and its length the consistency of that direction; black arrows are 2005 to 2014 and the
209 purple arrow is the Darkwoods fire weather station, 2015 to 2025. The thick black line is the mean
210 path of the air over the 12 h before 14:00 on the flight days of 2005 to 2013, from HYSPLIT,
211 ending over the site. Relief from the 150 m Copernicus model, lakes from the Freshwater Atlas in
212 blue and the international border in grey; the red outline is the study perimeter.

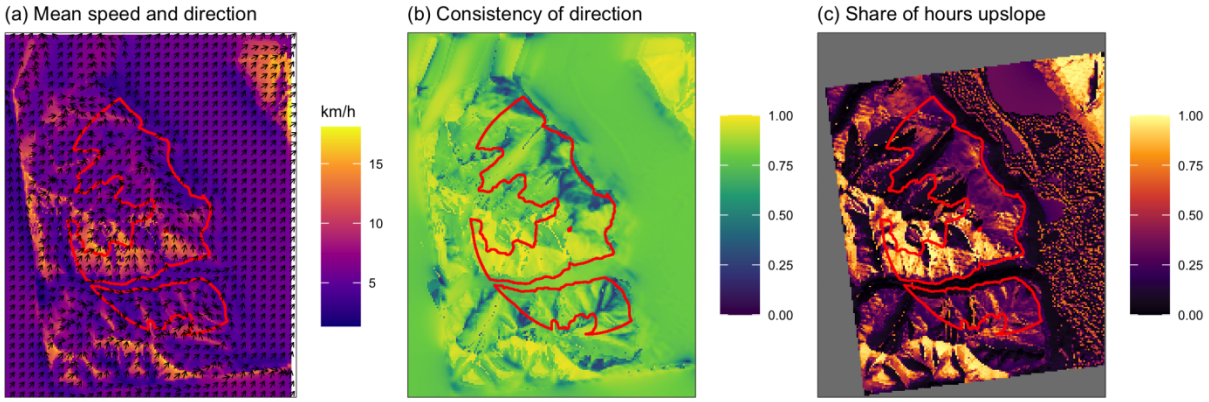
213 **Figure S6**



214

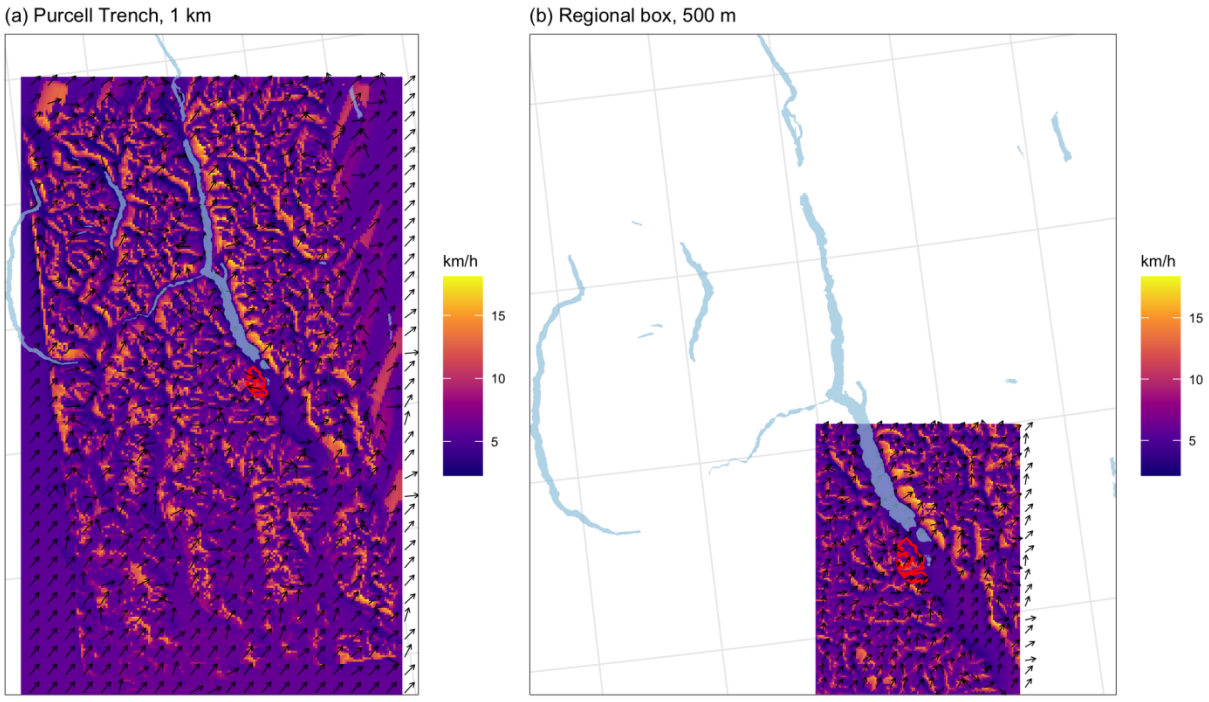
215 The modelled wind against the stations over every flight hour of June to August 2005 to 2014, on
216 the trench run at 1 km. (a) Modelled against observed speed at each station, with the one-to-one
217 line. (b) The error of direction on hours above 5 km/h, as the share of hours within each band of
218 degrees, by station.

Figure S7



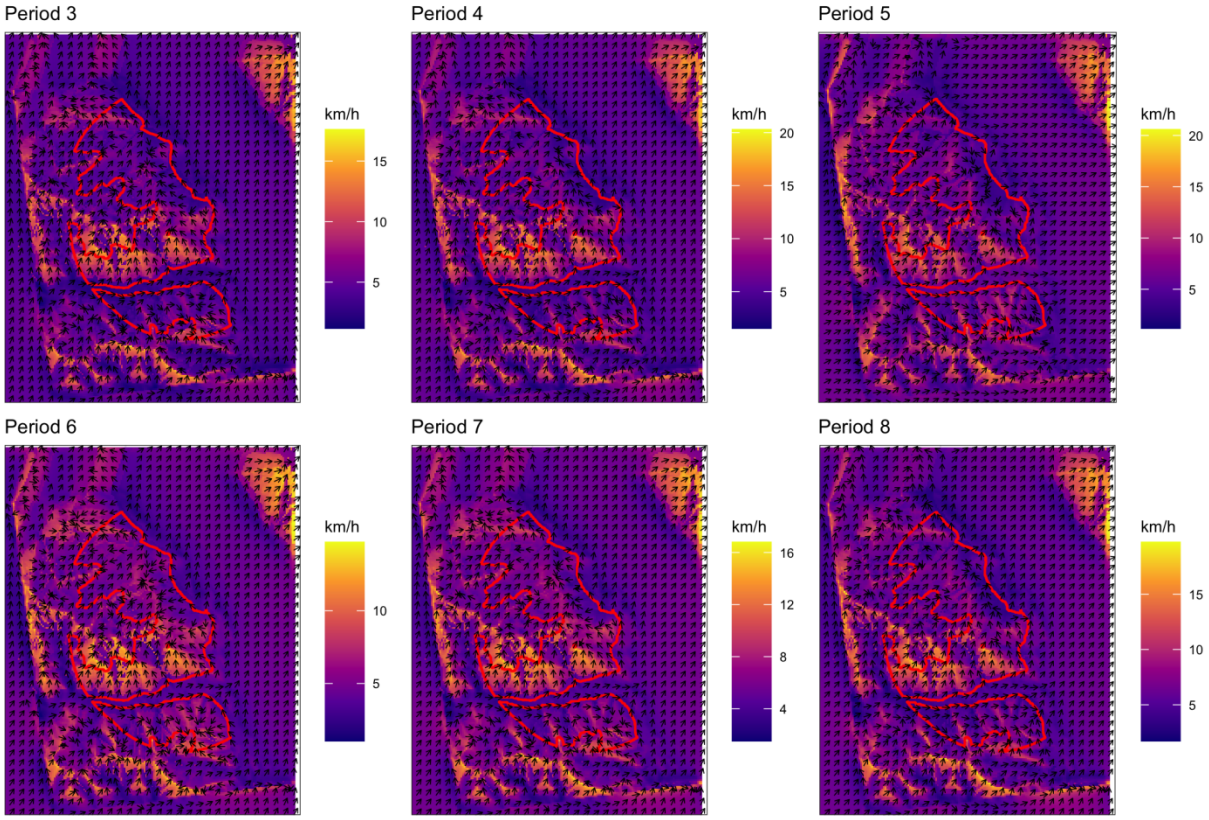
The terrain wind over the site from WindNinja at a 90 m mesh, averaged over the flight windows, 1 July to 15 August, of 10 summers, every flight hour 12:00 to 16:59. (a) Mean speed with the mean direction drawn as arrows, one per six cells. (b) The consistency of direction, the length of the mean wind vector over the mean speed, 1 for a wind that never changed direction. (c) The share of flight hours in which the air moved upslope. Speeds are multiplied by the model's flat-ground factor at this mesh, 1.63. The red outline is the study perimeter.

Figure S8



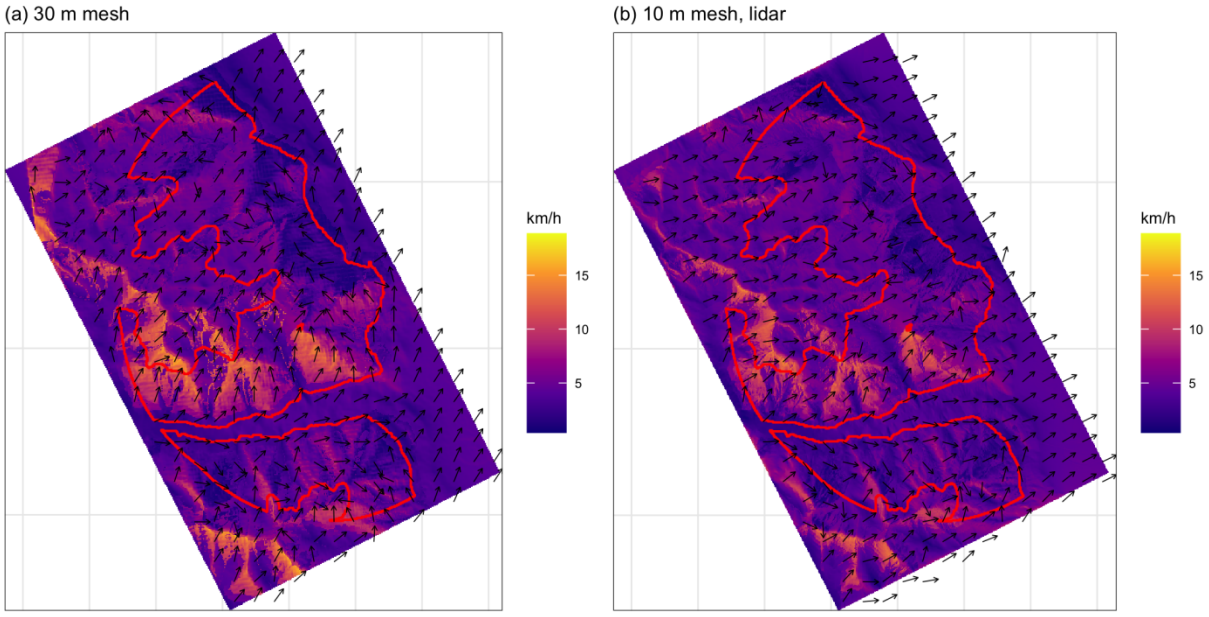
The terrain wind at the two regional scales, averaged over the flight windows of the summers run, every flight hour. (a) The Purcell Trench from Pend Oreille to the north end of Kootenay Lake at a 1 km mesh, with the 60 by 80 km box outlined. (b) The 60 by 80 km box around the site at 500 m. Colour is mean speed, arrows the mean direction, the red outline the study perimeter, and lakes from the Freshwater Atlas in blue. Speeds are multiplied by the model's flat-ground factor at each mesh, 1.65 at 1 km and 1.64 at 500 m.

Figure S9



The terrain wind over the site by sixteen-day period through the 2010 season at a 90 m mesh, every flight hour, periods 3 to 8 from 2 June to 5 September. Colour is mean speed and arrows the mean direction, one per six cells; the red outline is the study perimeter. Speeds are multiplied by the model's flat-ground factor at this mesh, as in the previous figure.

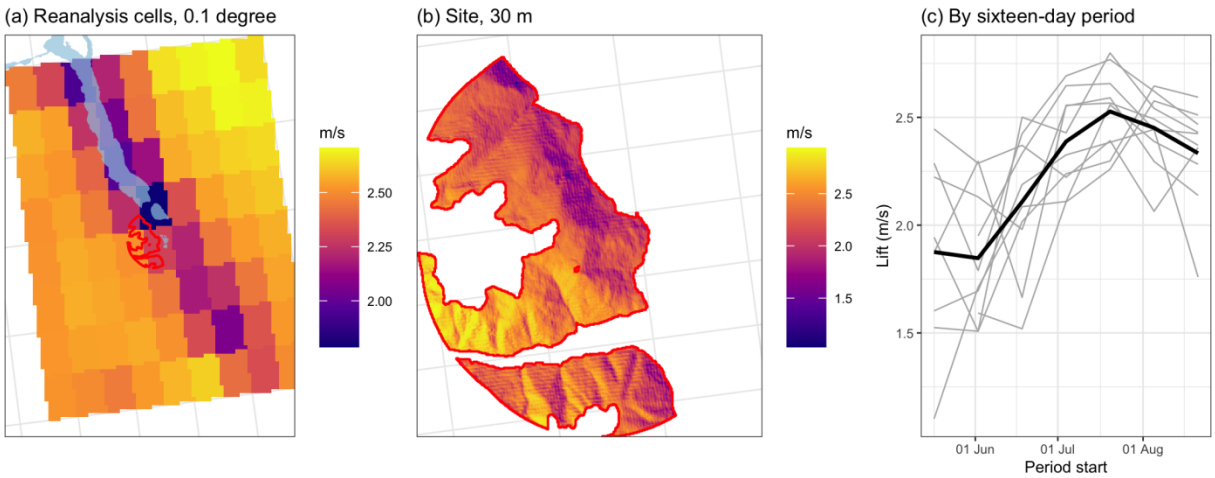
241 **Figure S10**



242

243 The terrain wind over the microsite window, the extent of the 1 m lidar, at two meshes, each the
244 mean of the flight-window runs of 10 summers, one run per window at its mean flight-hour
245 condition at 14:00. (a) A 30 m mesh on the 30 m elevation model. (b) A 10 m mesh on the lidar
246 ground model, run as nine tiles whose cores were joined. Colour is mean speed and arrows the
247 mean direction, one every 600 m. Speeds are multiplied by the model's flat-ground factor, 1.62 at
248 30 m and 1.62 at 10 m. The red outline is the study perimeter.

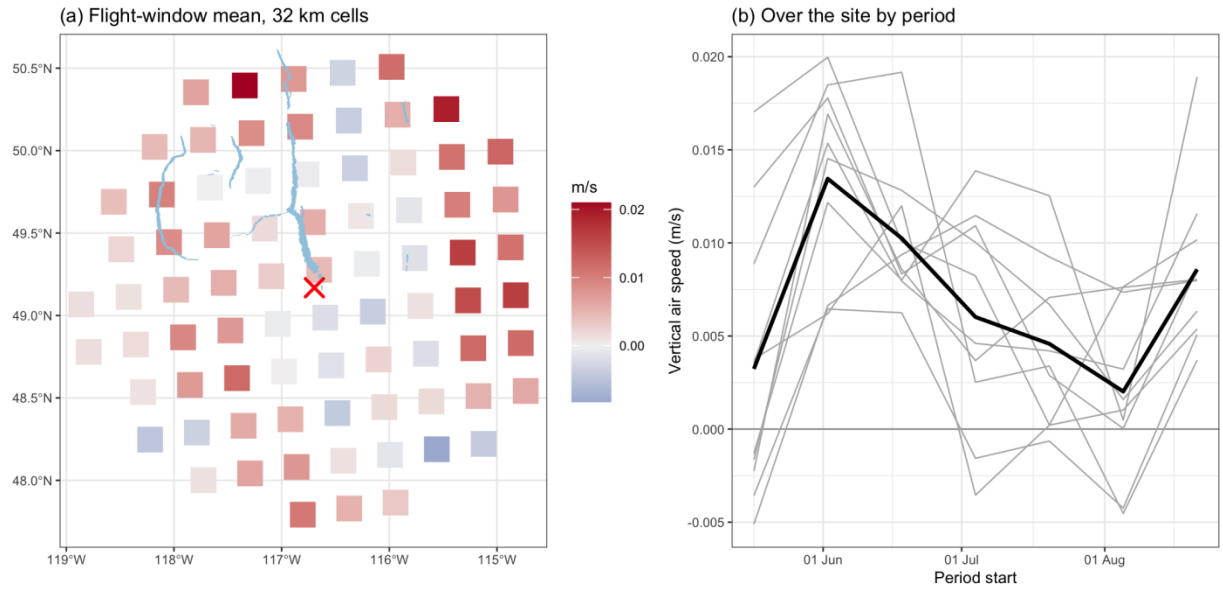
249 **Figure S11**



250

251 Convective lift, the characteristic speed of rising air in the afternoon mixed layer, from the
252 reanalysis heat flux and the ERA5 boundary layer height, over the flight hours 12:00 to 16:59 of
253 2005 to 2014. (a) The ERA5-Land cells of the 60 by 80 km box, the mean over the flight windows,
254 1 July to 15 August, before any slope scaling; lakes from the Freshwater Atlas in blue. (b) The
255 site at 30 m, the same mean with each cell scaled by the cube root of its share of the flight-window
256 sun. (c) The mean over the box of each sixteen-day period, one grey line per summer and the
257 mean of the ten summers in black. The red outline is the study perimeter.

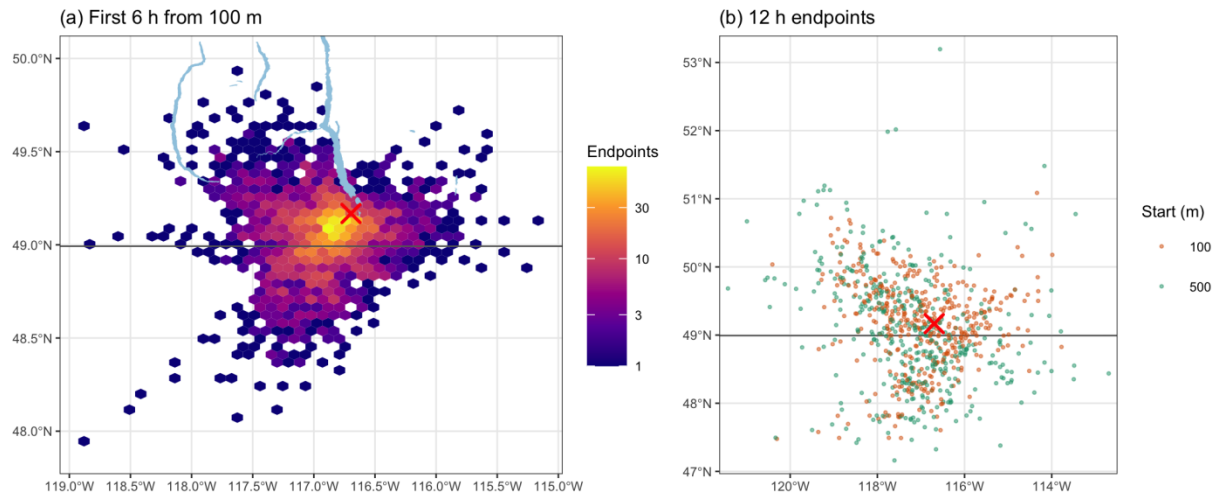
258 **Figure S12**



259

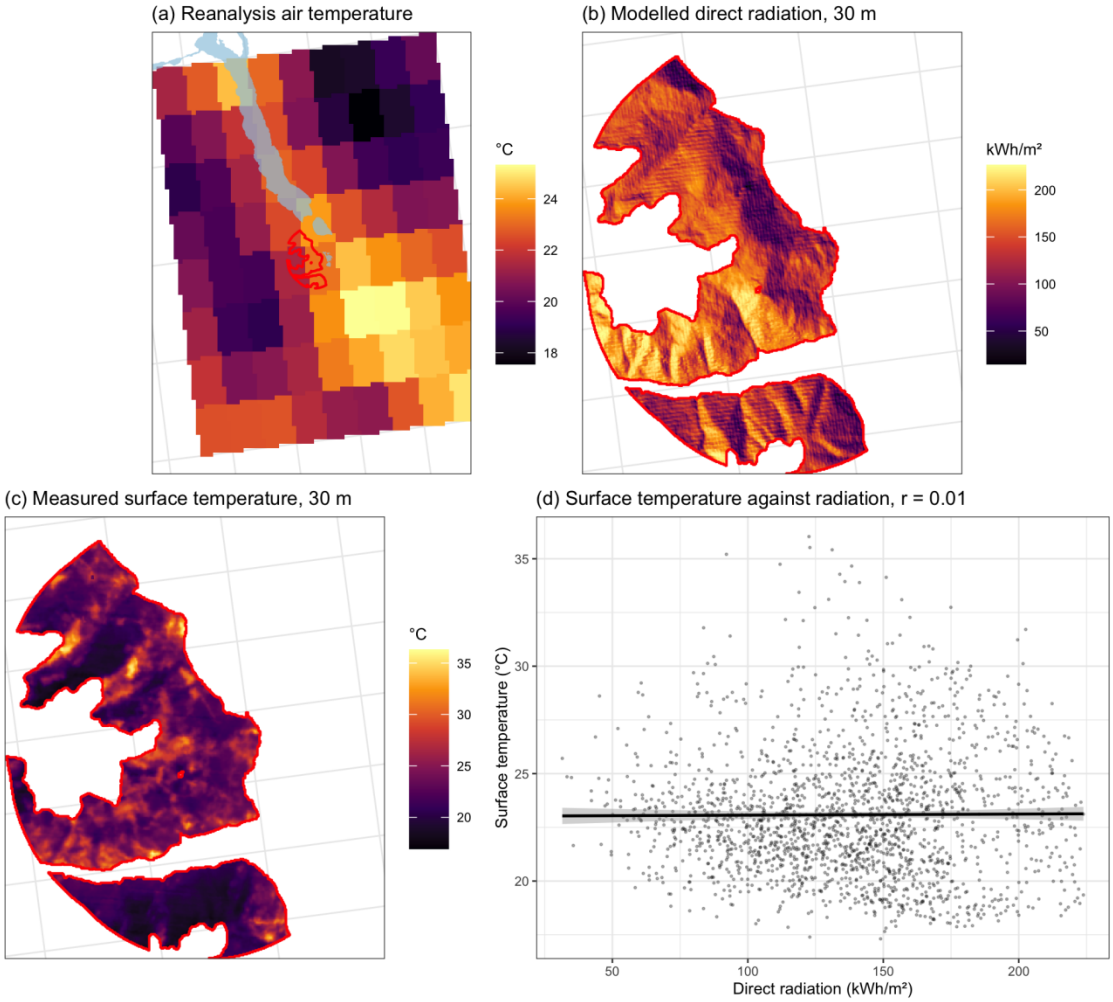
260 Vertical air speed from the North American Regional Reanalysis between 850 and 650 hPa at
261 14:00 and 17:00, positive for rising air. (a) The mean over the flight windows, 1 July to 15 August
262 of 2005 to 2014, at each 32 km cell within 160 km of the site, the red cross; lakes from the
263 Freshwater Atlas in blue. (b) The mean at the cell over the site by sixteen-day period, one grey
264 line per summer and the mean of the ten summers in black.

Figure S13



Back-trajectories of the air over the site at 14:00 on every flight day, 1 July to 15 August of 2005 to 2013, from HYSPLIT on the North American Regional Reanalysis. (a) The hourly endpoints of the first 6 h from 100 m above ground, counted in hexagons, over the Purcell Trench and Kootenay Lake. (b) The 12 h endpoints from 100 m and 500 m above ground. The site is the red cross, lakes from the Freshwater Atlas are blue and the international border is grey.

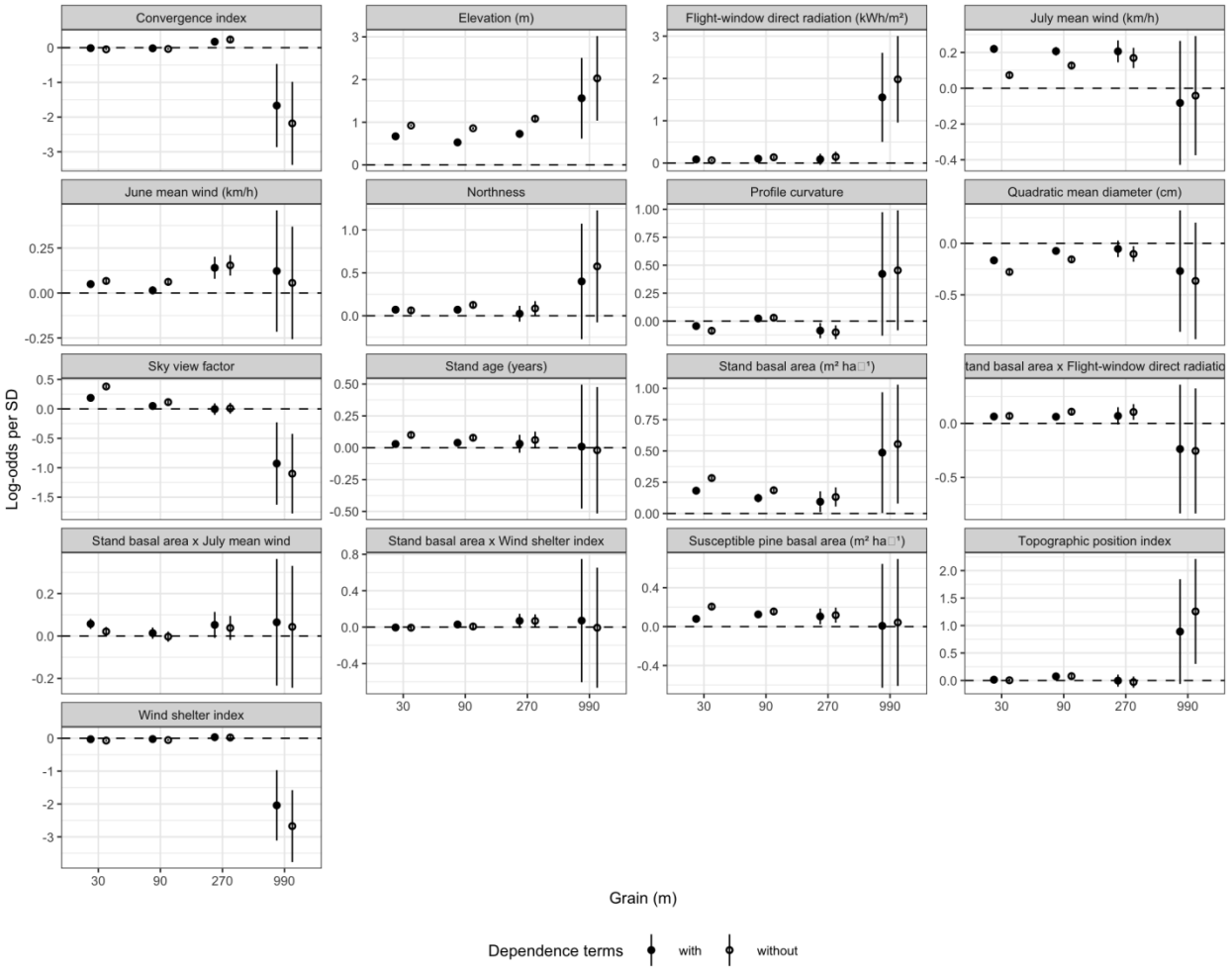
272 **Figure S14**



273

274 Heat over the flight windows, 1 July to 15 August of 2005 to 2014, at the two scales at which it
275 was measured. (a) ERA5-Land air temperature at 2 m over the 60 by 80 km box, the mean of the
276 flight hours 12:00 to 16:59. (b) Direct radiation over the site in the flight window, computed in
277 SAGA GIS from the 30 m elevation model. (c) Landsat land surface temperature over the site at
278 the late-morning overpass, the median of the clear flight-window scenes of each summer
279 averaged over the summers. (d) Measured surface temperature against modelled radiation over
280 the site, one point per cell of a 2,000-cell sample, with the least-squares line. The red outline is
281 the study perimeter.

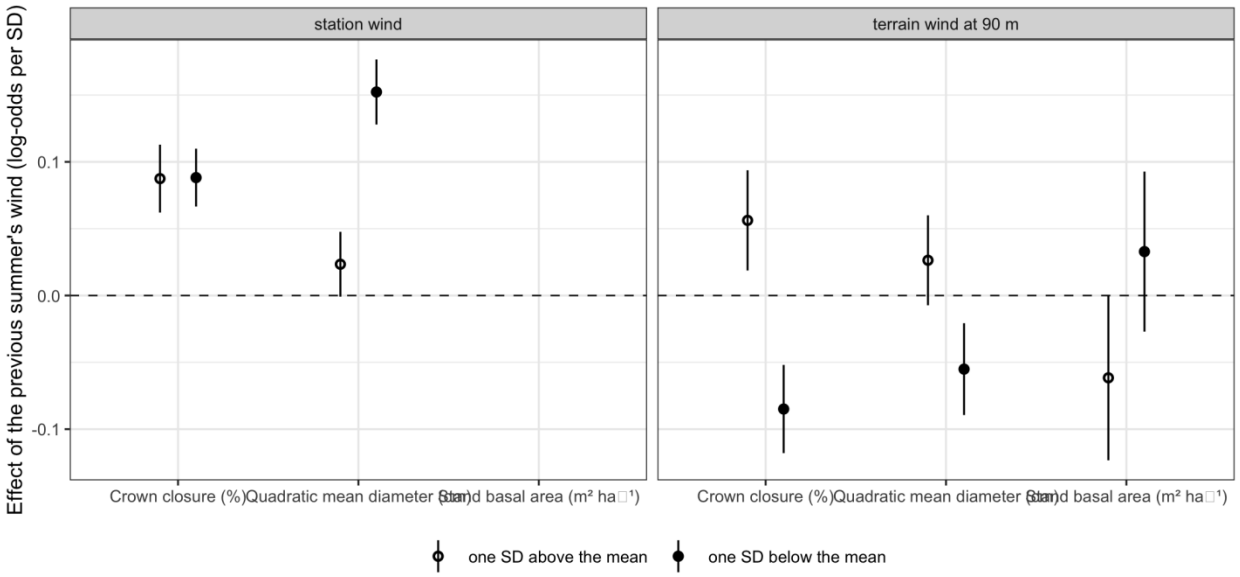
282 **Figure S15**



283

284 The coefficients of M3 refitted at four grains, with their 95 per cent confidence intervals, without
285 the dependence terms, open circles, and with them rebuilt at each grain, filled circles.

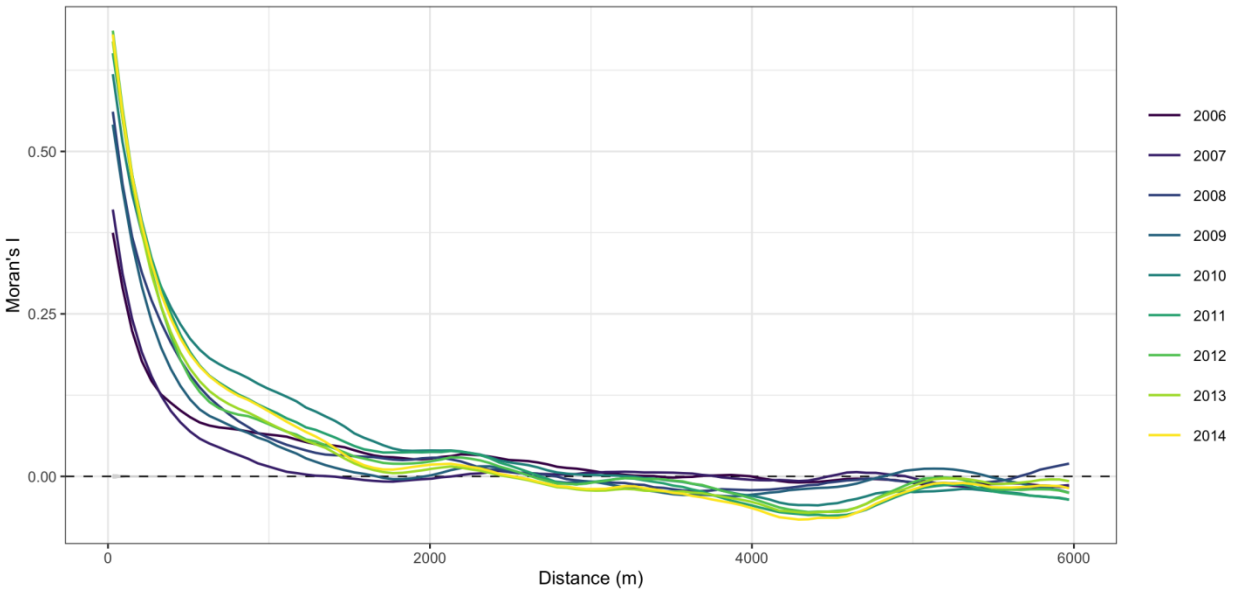
286 **Figure S16**



287

288 Effect of the previous summer's station and terrain wind on the log-odds of attack, per standard
289 deviation, in stands one standard deviation below and above the mean crown closure, quadratic
290 mean diameter and basal area, from the sixteen-day models with the previous period's attack,
291 with 95 per cent confidence intervals.

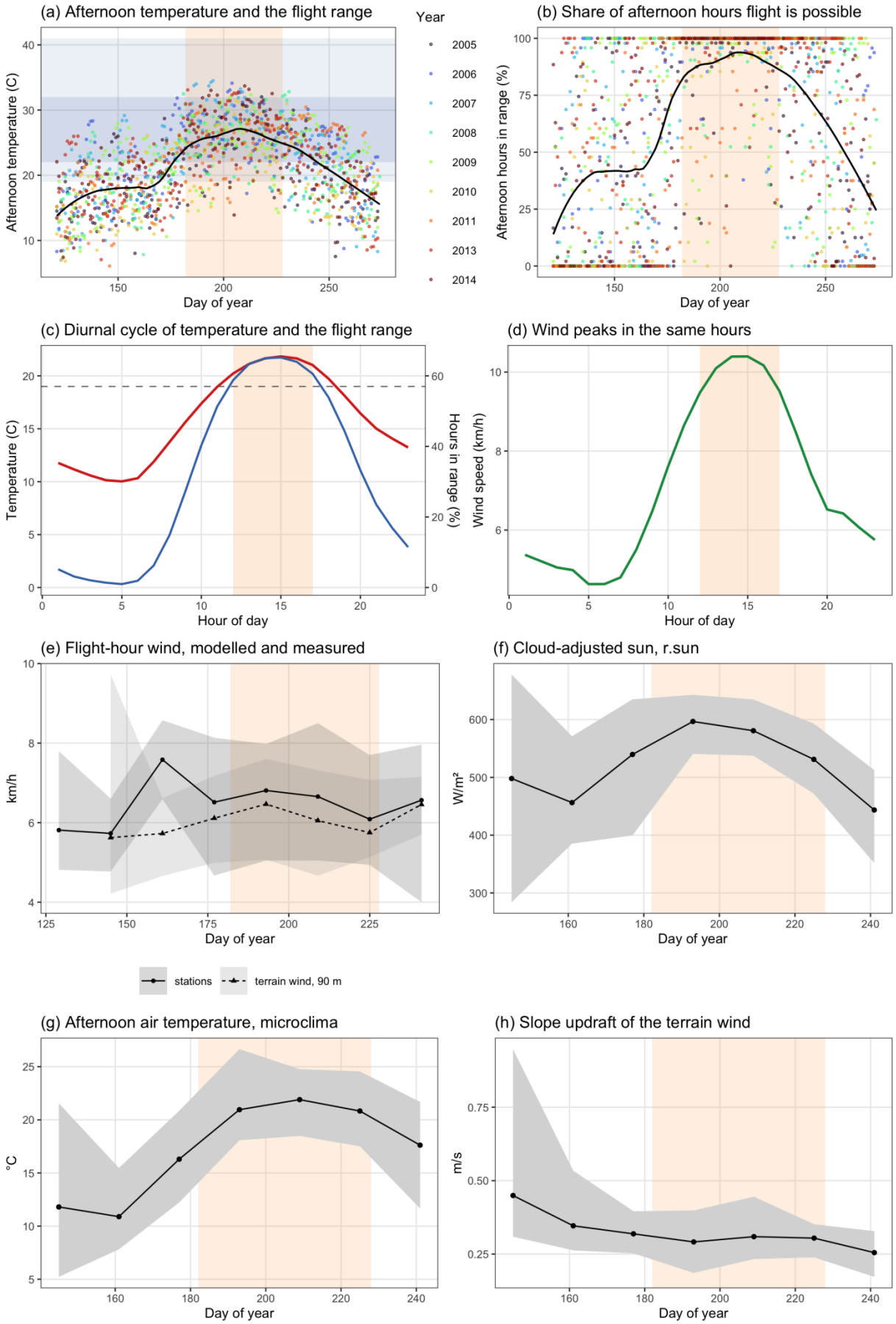
292 **Figure S17**



293

294 Spatial pattern of attack on the annual maps. Moran’s I by 60 m distance class, one line a year,
295 against 999 random relabellings of the attacked cells, grey.

296 **Figure S18**


(a) Afternoon temperature and the flight range
Afternoon temperature (C)
Day of year
Year
2005
2006
2007
2008
2009
2010
2011
2013
2014
(b) Share of afternoon hours flight is possible
Afternoon hours in range (%)
Day of year
(c) Diurnal cycle of temperature and the flight range
Temperature (C)
Hours in range (%)
Hour of day
(d) Wind peaks in the same hours
Wind speed (km/h)
Hour of day
(e) Flight-hour wind, modelled and measured
km/h
Day of year
stations
terrain wind, 90 m
(f) Cloud-adjusted sun, r.sun
W/m²
Day of year
(g) Afternoon air temperature, microclima
°C
Day of year
(h) Slope updraft of the terrain wind
m/s
Day of year

The flight window against the landscape's own climate, from hourly Environment and Climate Change Canada station records for May to September of the study years. The shaded band marks the window, 1 July to 15 August, taken from the bionomics and not fitted to these data. (a) Mean afternoon temperature by day of year, with the 19 to 41 °C flight range and the 22 to 32 °C peak band marked. (b) The share of afternoon hours inside the flight range, day by day. (c) Mean temperature and the share of all hours inside the flight range, by hour, with the 12:00 to 17:00 restriction shaded. (d) Mean wind speed by hour, showing that the hours in which the beetle could fly were also the windiest of the day. (e) to (h) The modelled weather of the flight hours by sixteen-day period, the mean over the summers with the range of the summer means shaded, for (e) wind speed from WindNinja at 90 m, corrected, against the stations, (f) cloud-adjusted sun from r.sun, (g) afternoon air temperature from microclima, and (h) the slope updraft of the terrain wind, each the mean over the study area.

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Author Response to Reviewers

AFE(2026)5804, resubmitted as a new manuscript

Dear Dr Ranger and Dr MacQuarrie,

Thank you for the decision of 22 September 2026 and for the invitation to resubmit. Each comment is quoted below as it appeared in the decision email, followed by what was done and where in the revised manuscript it can be found. Section numbers refer to the revised manuscript. Two defects in the submitted data were repaired during the revision, the wind terms were rebuilt, and several findings changed as a result. The repairs, and which findings held and which changed, are set out after the responses so that nothing in the new manuscript arrives unannounced.

Editor-in-Chief

Comment 1. “I agree with the Associate Editor that the manuscript is within the scope of AFE, but the Introduction and Discussion need to be thoroughly revised to include existing literature.”

Response. The literature was gathered in two searches under written protocols, on 23 and 24 September 2026, which screened 1,257 and 4,430 unique records, and the reference list of the main text now holds 61 works against 30 in the submitted manuscript, with the sources of the wind and heat models moved to the Supporting Information. The Introduction (section 1) was rewritten on that literature, one study to a sentence. It opens with the beetle as a native disturbance agent of pine forests and the history of its outbreaks (Macias Fauria & Johnson, 2009; Raffa et al., 2008; Safranyik & Carroll, 2006), defines disturbance refugia and the three mechanisms proposed for them (Cartwright, 2018; Krawchuk et al., 2020), and then sets out the biology each mechanism rests on, how a mass attack overcomes the defences of a tree and the threshold at which it does so (Boone et al., 2011; Raffa & Berryman, 1983), why large trees are the better hosts (Carroll & Safranyik, 2004; Safranyik et al., 2010), how beetles fly within a stand and how attack spreads from the year before (Aukema et al., 2008; Kunegel-Lion & Lewis, 2020; Robertson et al., 2007; Safranyik et al., 1989; Safranyik et al., 1992), how cold and drought act (Chapman et al., 2012; Logan & Powell, 2001; Preisler et al., 2012), how terrain entered earlier models (Nelson et al., 2007; Walter & Platt, 2013; Wulder et al., 2006), what thinning trials found and why (Amman et al., 1988; Bartos & Amman, 1989; Fettig et al., 2007; Mitchell et al., 1983;

Shore et al., 2006), and the two ways wind can act (Ainslie & Jackson, 2010; de la Giroday et al., 2011; Hynum & Berryman, 1980; Jackson et al., 2008). The Discussion (section 4) sets each finding against that literature, as described under comments 4 and 5.

Comment 2. “In addition, the novelty of the study needs to be explicitly stated since numerous modelling papers are submitted to AFE and we have limited room.”

Response. The novelty is stated in three places, as what the study did. The first statement of the abstract names the gap, how little is known about how the three mechanisms act together once attack nearby and earlier is taken into account. The last paragraph of the Introduction states that few studies had tested these mechanisms together while accounting for attack nearby, and sets the three questions. The opening paragraph of the Discussion states that the novelty of the study lay in testing the three mechanisms of Krawchuk et al. (2020) together on one landscape at 30 m with attack in and around each cell the year before in every model, and in testing the two ways wind could act, breaking up the pheromone plume and depositing beetles on sheltered ground, with a terrain wind model over the afternoon flight hours of each sixteen-day period.

Associate Editor

Comment 3. “I am, however, concerned that the arguments in the introduction and discussion are not well supported or linked to the (admittedly vast) literature on MPB.”

Response. Every argument in the Introduction now rests on named studies of this beetle, as set out under comment 1, and each of the three questions is derived from them. The first question follows the mechanisms proposed by Krawchuk et al. (2020) and the refugia Cartwright (2018) mapped in southern Oregon, with the thinning trials behind stand density and the 25 cm diameter at which Carroll & Safranyik (2004) placed the change from beetle sink to beetle source behind host size. The second follows Bartos & Amman (1989), who measured more light, warmth and wind in a thinned stand than in the unthinned stand beside it. The third follows the finding of Hynum & Berryman (1980) that pioneer beetles could not tell hosts from non-hosts while landing, and the settlement of beetles where wind slows (de la Giroday et al., 2011). The terms for attack nearby and in the previous year, entered in every model, follow Aukema et al. (2008), Robertson et al. (2007) and Kunegel-Lion & Lewis (2020). The Discussion rests on the same literature, section by section.

Comment 4. “My initial impression is that the Discussion reiterates what has already been stated in the results.”

Response. The Discussion (section 4) was rewritten in four sections that mirror the three questions of the Results, and coefficients are repeated only where an argument needs them. Its opening paragraph reads the dominance of attack nearby and the year before against Aukema et al. (2008) on the Chilcotin Plateau, the 140 m over which previous attack raised attack in Colorado (Walter & Platt, 2013) and the late phase of the outbreak (Meddens & Hicke, 2014). Section 4.1 reads the density result against the stand susceptibility rating used in British Columbia (Safranyik et al., 2010; Shore et al., 2000), the thinning trials and the history of the province’s pine (Taylor & Carroll, 2003), reads the shade result against the observation that the heaviest attacks on a bole are found on its northern side (Safranyik & Carroll, 2006) and that attacks fall at midday with the highest temperatures (Schmid et al., 1991), and reads host size against the dependence of brood production on phloem thickness (Amman, 1972), the 25 cm boundary between beetle sinks and sources (Carroll & Safranyik, 2004) and the depletion of large pine by epidemics (Shore et al., 2006). Section 4.2 reads the wind result against the stand measurements of Bartos & Amman (1989) and Thistle et al. (2004), wind above the canopy carrying beetles into new stands (Chen & Jackson, 2017; Jackson et al., 2008) and pheromones from adjacent stands overriding the benefit of spacing (Schmid & Mata, 2005). Section 4.3 reads the shelter result against the windward attack de la Giroday et al. (2011) found in the Peace River region, the crosswind and downwind flight that Safranyik et al. (1989) recorded, the flight limit of Carroll & Safranyik (2004), the north-facing onset in the Cypress Hills (Kunegel-Lion & Lewis, 2020), deposition within the stand (Byers, 2000) and the scale of inference that Aukema et al. (2008) raised. Section 4.4 sets out the limits of the study and what the findings mean for management.

Comment 5. “Their revised discussion should link the author’s findings to the biology, ecology or management of the MPB system.”

Response. Each section of the Discussion reads a result through the beetle rather than through the model. The biology appears in sections 4.1 and 4.3, through the brood that thicker phloem feeds, the side of the bole that is attacked, the fall in attack at midday heat, flight crosswind and downwind of an attacked tree and the wind speed above which the beetle cannot fly. The ecology appears in sections 4.1 and 4.3, through the age structure of the province’s pine, the pine that grew on the higher ground of this range, the phase of the outbreak and the difference between a refugium defined at 30 m and one defined at 1 km. Management appears in sections 4.2 and 4.4. Section 4.2 states that a stand opened by thinning was protected by the wind only during flight

and least where attacked trees stood close, and section 4.4 states that the refugia this landscape showed were stands with little host, the structure that thinning and a younger age structure produce, which terrain maps alone could not locate, and sets that against the rating under which partial cutting lowers the susceptibility of a stand (Mata et al., 2003; Shore et al., 2000), the advice to manage on a landscape basis because small cut parcels surrounded by unmanaged stands may not be protected (Schmid & Mata, 2005), and the conclusion of Six et al. (2014) that the field evidence for thinning during an outbreak was weaker than policy had assumed.

Analysis repairs

Two defects in the submitted data were repaired, the wind terms were rebuilt, and the spatial statistics of Aukema et al. (2008) were added. The design was kept, with annual maps for the first and third questions and sixteen-day maps for the second.

1. The attack maps were rebuilt with a new classifier, trained on the 112 Landsat pixels of the field plots, four to a plot, against undisturbed forest digitised for the companion study, and tested over 100 random splits that each held out a quarter of the plots and of the 100 m blocks of undisturbed pixels. The radial support vector machine reached kappa 0.859 ± 0.081, and its cut was applied to the annual maps and to the 47 sixteen-day maps alike (section 2.2, Table S6). The table of attack by diameter class was rebuilt on the repaired maps (Table S8). The annual maps in the submitted manuscript came from an earlier classifier trained on 38 plots, with a leave-one-out kappa of 0.475, while its Methods described the classification of Murphy et al. (2026). The summer of 2012, which the submitted manuscript left out, was mapped from Landsat 7 alone and entered the annual models (section 2.2).
2. The terrain-adjusted station wind was recomputed on an elevation model extending beyond the study area. In the submitted analysis its curvature term was computed on an elevation model cut to the study area, so the field covered 18,388 of the 61,923 cells, every wind term rested on that partial field, and the models were fitted on the cells it covered. It now covers all of them (section 2.2).
3. A terrain wind model, WindNinja, was run from the reanalysis over the site at 90 m for every flight hour of ten summers, over the surrounding 60 by 80 km box at 500 m, over the whole Purcell Trench at 1 km with hourly values kept at four valley stations, and over the site at 30 m and the lidar terrain at 10 m once per flight window. It was checked against the stations hour by hour. At Creston the station recorded 9.1 km/h over the flight hours, the reanalysis cell over it 5.8 and the model 3.2, with direction errors of 61 and 59 degrees, so the direction is the reanalysis's and the speed carries a constant step inside the model. The step was measured on flat ground at every mesh and removed, the direction limit is stated in the Methods (section 2.3, Table S10), and the fields entered the models beside the station wind, with the wind of the same period of the previous summer as well as the same period's, because the red crowns of a period record the flight of the summer before.

The spatial statistics were added beside the submitted models. Join counts and a Moran's I correlogram described the pattern of attack on each annual map before any model was fitted. The models were then built in the order of Aukema et al. (2008), with the terms for attack in and around each cell the year before chosen first, and the final annual model was refitted with a latent spatial

field and at grains of 90, 270 and 990 m (section 2.4, equations 1 to 3). Every candidate variable was screened beside the dependence terms by likelihood ratio test, variance inflation and an elastic net (Table 1), and the sixteen-day models and every further specification are reported in full in the Supporting Information (Tables S4, S11 and S12).

Findings

The Results (section 3) now answer the three questions in turn, each section naming the variables that bear on the question with their estimates, and the Discussion follows the same order. Three findings held. Stand density acted through the amount of host, with basal area entering at +0.430 log-odds per standard deviation in the submitted manuscript and at +0.276 in the revised final model, +0.167 with the terms for previous attack and +0.200 under the spatial field, so the thinnest stands had the least attack. Attack still peaked in stands of 25 to 30 cm quadratic mean diameter and fell in stands of larger trees, which held little pine. Ground facing the afternoon wind still had the least attack, with the wind shelter index entering at -0.267 before and at -0.073 now, -0.032 with the terms for previous attack ($p = 0.043$), and with no interaction with basal area ($p = 0.971$), as deposition predicts; the shelter index at each cell's own modelled wind direction, the share of flight hours blowing upslope and the slope updraft all pointed the same way (section 3.3).

Three findings changed. First, the submitted manuscript read shading as unsupported because north-facing ground had more attack, +0.384 before and +0.073 now. The revised manuscript reads the shade mechanism through the terms that measure shade by terrain, sky view, +0.285 before and +0.170 now with the terms for previous attack, and the direct sun of the flight window, +0.085, so terrain-shaded ground had less attack as the hypothesis predicted while north faces remained the exception, and it states that the effect did not survive the spatial field, under which sky view fell to +0.037 ($p = 0.285$). Second, the submitted manuscript tested plume disruption with the station wind and standing volume, an interaction of -0.017, which could no longer be distinguished from zero once the previous period's attack entered ($p = 0.296$). With the terrain wind of the previous summer and crown closure as the measure of openness, the interaction was +0.071 ($p < 0.001$), so that the wind effect was -0.085 in stands one standard deviation below the mean crown closure and +0.056 one above it. Wind lowered attack where the canopy was open and raised it where it was closed, which supports plume disruption in the sixteen-day models, while in the annual models the station wind raised attack at every density, +0.224 (section 3.2). Third, the shelter effect, although it held in sign, was small and changed sign under the latent spatial field, to +0.096 ($p < 0.001$), and lost significance at a 90 m grain, so the revised manuscript

reports it as weak and dependent on grain and place. The submitted abstract concluded, “Refugia are therefore proposed to lie on warm, windward ground in thin stands during windy flight periods.” The revised abstract concludes that refugia on this landscape were stands with little host, and that terrain and wind added small, scale-dependent modifiers to an outbreak whose spread was mostly contagion.

The added spatial statistics produced the result the revised manuscript leads with. Attack was clustered in every year, with 3.2 to 5.8 times as many pairs of neighbouring attacked cells as random permutations produced, and with positive autocorrelation to a median of 2,640 m. Attack in each cell and within 150 m the year before dominated every model it entered, raising the area under the receiver operating characteristic curve of the final annual model from 0.814 to 0.890. Residual Moran’s I remained significant in every year, at a median of 0.345 even under the spatial field, so the p-values of every model were optimistic, and section 4.3 says so.

Every number in the manuscript is computed by code in the manuscript file itself, and the file, the derived data and the analysis code are public at *https://github.com/seamusrobertmurphy/beetle-topography-and-wind-study*.

Yours sincerely,

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